

ARM®-based 32-bit Cortex®-M4 MCU with 64 to 256 KB Flash, sLib, USB OTG, 11 timers, 1 ADC, 2 COMPs, 12 communication interfaces

Feature

- **Core: ARM® 32-bit Cortex®-M4 CPU**
 - 150 MHz maximum frequency, with a memory protection unit (MPU)
 - Single-cycle multiplication and hardware division
 - DSP instructions
- **Memories**
 - 64 to 256 Kbytes of main Flash instruction/data memory
 - 18 Kbytes of system memory used as a Bootloader or as a general instruction/data memory (one-time-configured)
 - 32 Kbytes of SRAM
 - sLib: configurable part of main Flash set as a library area with code executable but secured, non-readable
- **Clock, reset, and supply management**
 - 2.6 to 3.6 V application supply and I/Os
 - POR, PDR, and programmable voltage detector (PVD)
 - 4 to 25 MHz crystal oscillator
 - Internal 48 MHz factory-trimmed RC (accuracy 1 % at $T_A = 25\text{ }^{\circ}\text{C}$, 2.5 % at $T_A = -40$ to $+105\text{ }^{\circ}\text{C}$)
 - Internal 40 kHz RC with calibration
 - 32 kHz oscillator for ERTC with calibration
- **Low power**
 - Sleep, Stop, and Standby modes
 - V_{BAT} supply for ERTC and twenty 32-bit backup registers
- **One 12-bit, 0.5 μs A/D converter (up to 16 channels)**
 - Conversion range: 0 to 3.6V
 - One sample-and-hold capability
 - Temperature sensor
- **Two analog comparators**
- **DMA: 14-channel DMA controller**
 - Supported peripherals: timers, ADC, SDIO, I²Ss, SPIs, I²Cs, and USARTs
- **Debug mode**
 - Serial wire debug (SWD) and JTAG interfaces
- **Up to 55 fast I/Os**
 - 27/39/55 multi-functional bi-directional I/Os, all mappable on 16 external interrupt vectors and almost all 5V-tolerant
 - All fast I/Os, control registers accessible with f_{AHB} speed
- **Up to 11 timers**
 - Up to 5 x 16-bit timers + 2 x 32-bit timers, each with 4 IC/OC/PWM or pulse counter and quadrature (incremental) encoder input
 - 1 x 16-bit motor control PWM advanced timers with dead-time generator and emergency stop
 - 2 x watchdog timers (Independent and Window)
 - SysTick timer: a 24-bit downcounter
- **ERTC: enhanced RTC with subsecond accuracy and hardware calendar**
- **Up to 12 communication interfaces**
 - 2 x I²C interfaces (SMBus/PMBus)
 - Up to 5 x USARTs (ISO7816 interface, LIN, IrDA capability, modem control)
 - 2 x SPIs (50 Mbit/s), both with I²S interface multiplexed
 - CAN interface (2.0B Active) with dedicated 256 bytes SRAM
 - USB 2.0 full-speed device/host/OTG controller with dedicated 1280 bytes SRAM, device mode supporting crystal-less
 - SDIO interface
- **CRC calculation unit, 96-bit unique ID**
- **Packages**
 - LQFP64 10 x 10 mm
 - LQFP64 7 x 7 mm
 - LQFP48 7 x 7 mm
 - QFN48 6 x 6 mm
 - QFN32 4 x 4 mm

Table 1. Device summary

Flash	Part number
256 KBytes	AT32F415RCT7, AT32F415RCT7-7, AT32F415CCT7, AT32F415CCU7, AT32F415KCU7-4
128 KBytes	AT32F415RBT7, AT32F415RBT7-7, AT32F415CBT7, AT32F415CBU7, AT32F415KBU7-4
64 KBytes	AT32F415R8T7, AT32F415R8T7-7, AT32F415C8T7, AT32F415K8U7-4

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1 Introduction

This datasheet provides the ordering information and mechanical device characteristics of the AT32F415 microcontrollers.

The AT32F415 datasheet should be read in conjunction with the [AT32F415 reference manual](#).

For information on programming, erasing, and protection of the internal Flash memory please also refer to the [AT32F415 reference manual](#).

For information on the Cortex®-M4 core please refer to the Cortex®-M4 Technical Reference Manual, available from the www.arm.com website at the following address:

<http://infocenter.arm.com>

2 Description

The AT32F415 incorporates the high-performance ARM® Cortex®-M4 32-bit RISC core operating at 150 MHz. The Cortex®-M4 core features a full set of DSP instructions and a memory protection unit (MPU) which enhances application security.

The AT32F415 incorporates high-speed embedded memories (up to 256 Kbytes of Flash memory and 32 Kbytes of SRAM), enhanced I/Os and peripherals connected to two APB buses. Any block of the Flash memory can be protected by the sLib, functioning as a security area with code-executable only.

The AT32F415 offers one 12-bit ADC, two analog comparators, five general-purpose 16-bit timers plus two general-purpose 32-bit timers, and one PWM timers for motor control, as well as standard and advanced communication interfaces, up to two I²Cs, two SPIs (both multiplexed as I²Ss), one SDIO, five USARTs, an USB OTG full-speed interface, and a CAN.

The AT32F415 operates in the -40 to +105 °C temperature range, from a 2.6 to 3.6 V power supply. A comprehensive set of power-saving mode allows the design of low-power application.

These features make the AT32F415 suitable for a wide range of applications such as:

- Consumer
 - Camera holder/stabilizer
 - Micro printer
 - Bar-code scanner
 - USB Hub
 - Smart card reader
 - E-Sports accessories (keyboard, mouse, joystick...)
- IoT
 - Smart home applications
 - IoT sensor node/gateway
- Industrial automation
 - LED asynchronous controller/display
 - BMS
 - Robot controller
 - Electrical controller
- Motor control
 - Electric Vehicle
 - BLDC/PMSM motor control
 - Servo motor control

2.1 Device overview

The AT32F415 offers devices in five different package types: from 32 pins to 64 pins. Depending on the different packages, the pin-to-pin is completely compatible among devices, and also the software and functionality. Only different sets of peripherals are included. The description below gives an overview of the complete range of peripherals proposed in different devices.

Table 2. AT32F415 features and peripheral counts

Part Number		AT32F415 xxU7-4			AT32F415 xxU7		AT32F415 xxT7			AT32F415 xxT7-7			AT32F415 xxT7		
		K8	KB	KC	CB	CC	C8	CB	CC	R8	RB	RC	R8	RB	RC
CPU frequency (MHz)		150													
Flash (KBytes)		64	128	256	128	256	64	128	256	64	128	256	64	128	256
SRAM (KBytes)		32													
Timers	Advanced-control	1			1		1			1			1		
	32-bit general-purpose	2			2		2			2			2		
	16-bit general-purpose	5			5		5			5			5		
	SysTick	1			1		1			1			1		
	IWDG	1			1		1			1			1		
	WWDG	1			1		1			1			1		
	Enhanced RTC	1			1		1			1			1		
Communication	I ² C	2			2		2			2			2		
	SPI/I ² S	2/2 ⁽¹⁾			2/2 ⁽¹⁾		2/2 ⁽¹⁾			2/2			2/2		
	USART+UART	2+0			3+0		3+0			3+2			3+2		
	SDIO	1 ⁽²⁾			1 ⁽²⁾		1 ⁽²⁾			1			1		
	USB OTG FS	1			1		1			1			1		
	CAN	1			1		1			1			1		
Analog	12-bit ADC numbers/ channels	1													
		10			10		10			16			16		
	Comparator	2													
GPIO		27			39		39			55			55		
Operating temperatures		-40 to +105 °C													
Packages		QFN32 4 x 4 mm			QFP48 6 x 6 mm		LQFP48 7 x 7 mm			LQFP64 7 x 7 mm			LQFP64 10 x 10 mm		

(1) Only I²S1 exists MCK pin on LQFP48, QFN48, and QFN32 packages.

(2) SDIO supports maximum 4-bit (D0~D3) mode on LQFP48, QFN48, and QFN32 packages.

2.2 Overview

2.2.1 ARM® Cortex®-M4 core and DSP instruction set

The ARM Cortex®-M4 is the latest generation of ARM processors for embedded systems. It was developed to provide a low-cost platform that meets the needs of MCU implementation, with a reduced pin count and low-power consumption, while delivering outstanding computational performance and an advanced response to interrupts.

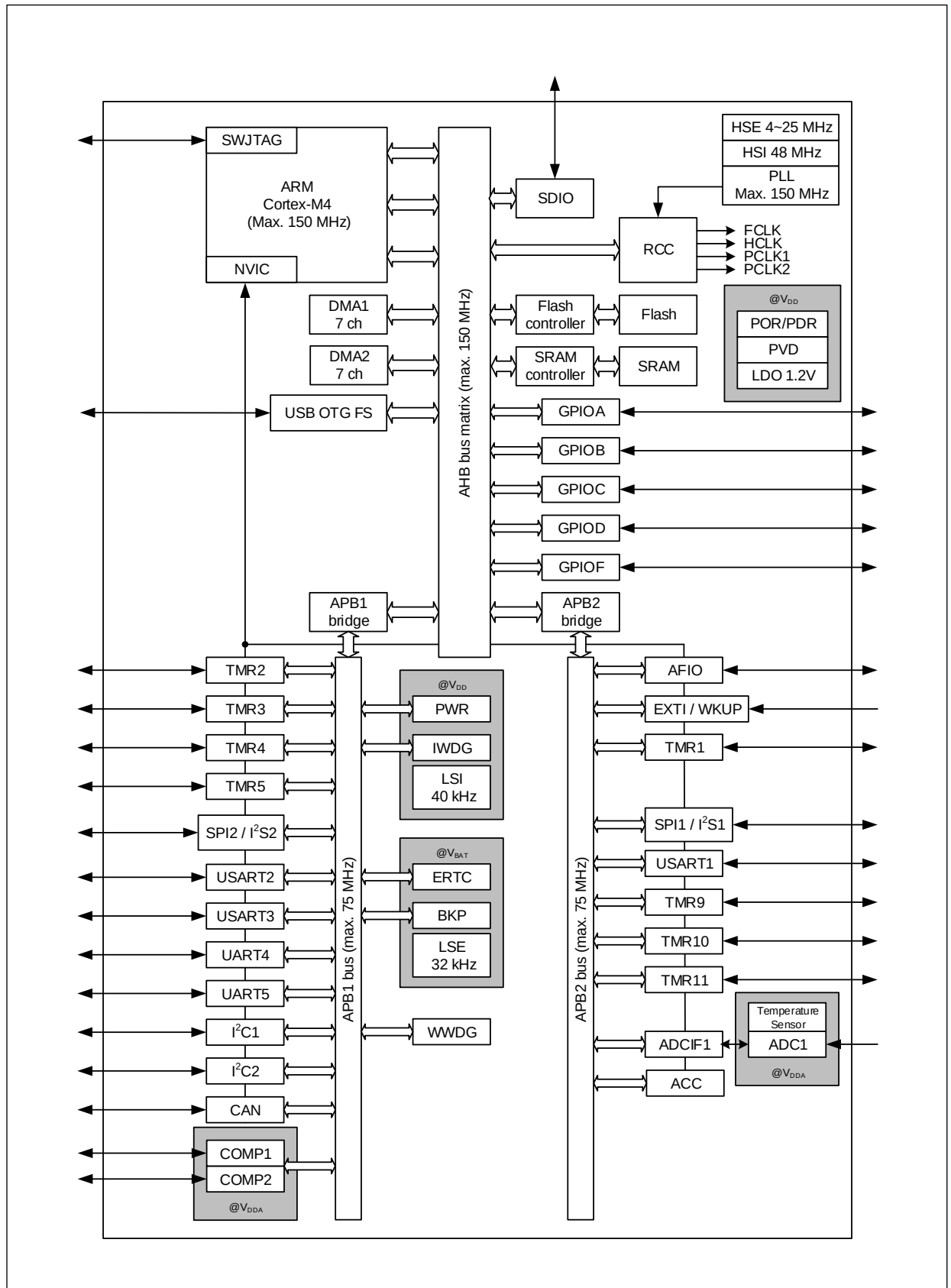
The ARM Cortex®-M4 32-bit RISC processor features exceptional code efficiency, delivering the high-performance expected from an ARM core in the memory size usually associated with 8- and 16-bit devices. The processor supports a set of DSP instructions which allow efficient signal processing and complex algorithm execution.

With its embedded ARM core, the AT32F415 is compatible with all ARM tools and software.

[Figure 1](#) shows the general block diagram of the AT32F415.

Note: Cortex®-M4 is binary compatible with Cortex®-M3.

Figure 1. AT32F415 block diagram



(1) Operating temperatures: -40 to +105 °C. Junction temperature reaches 125 °C.

2.2.2 Memory protection unit (MPU)

The memory protection unit (MPU) is used to manage the CPU accesses to memory to prevent one task to accidentally corrupt the memory or resources used by any other active task. This memory area is organized into up to 8 protected areas that can in turn be divided up into 8 subareas. The protection area sizes are between 32 bytes and the whole 4 gigabytes of addressable memory.

The MPU is especially helpful for applications where some critical or certified code has to be protected against the misbehavior of other tasks. It is usually managed by an RTOS (real-time operating system). If a program accesses a memory location that is prohibited by the MPU, the RTOS can detect it and take action. In an RTOS environment, the kernel can dynamically update the MPU area setting, based on the process to be executed.

The MPU is optional and can be bypassed for applications that do not need it.

2.2.3 Embeded Flash memory

Up to 256 Kbytes of embedded Flash is available for storing programs and data. User can configure any part of the Flash memory protected by the sLib, functioning as a security area with code-executable only but non-readable. sLib is a mechanism that protects the intelligence of solution vendors and facilitates the second-level development by customers.

There are 18 Kbytes of system memory embedded on the AT32F415, in which the Bootloader is resided. If the Bootloader is not used, this block can be one-time configured as a general perpose instruction/data area.

2.2.4 CRC (cyclic redundancy check) calculation unit

The CRC (cyclic redundancy check) calculation unit is used to get a CRC code from a 32-bit data word and a fixed generator polynomial.

Among other applications, CRC-based techniques are used to verify data transmission or storage integrity. In the scope of the EN/IEC 60335-1 standard, they offer a means of verifying the Flash memory integrity. The CRC calculation unit helps compute a signature of the software during runtime, to be compared with a reference signature generated at link time and stored at a given memory location.

2.2.5 Embedded SRAM

32 Kbytes of SRAM is embedded and it can be accessed (read/write) at CPU clock speed with 0 wait states.

2.2.6 Nested vectored interrupt controller (NVIC)

The AT32F415 embed a nested vectored interrupt controller able to manage 16 priority levels and handle up to 55 maskable interrupt channels plus the 16 interrupt lines of the Cortex®-M4.

- Closely coupled NVIC gives low-latency interrupt processing
- Interrupt entry vector table address passed directly to the core
- Closely coupled NVIC core interface
- Allows early processing of interrupts
- Processing of late arriving higher priority interrupts
- Support for tail-chaining

- Processor state automatically saved
- Interrupt entry restored on interrupt exit with no instruction overhead

This hardware block provides flexible interrupt management features with minimal interrupt latency.

2.2.7 External interrupt/event controller (EXTI)

The external interrupt/event controller consists of 23 edge detector lines used to generate interrupt/event requests. Each line can be independently configured to select the trigger event (rising edge, falling edge, both) and can be masked independently. A pending register maintains the status of the interrupt requests. The EXTI can detect an external line with a pulse width shorter than the Internal AHB clock period. Up to 55 GPIOs can be connected to the 16 external interrupt lines.

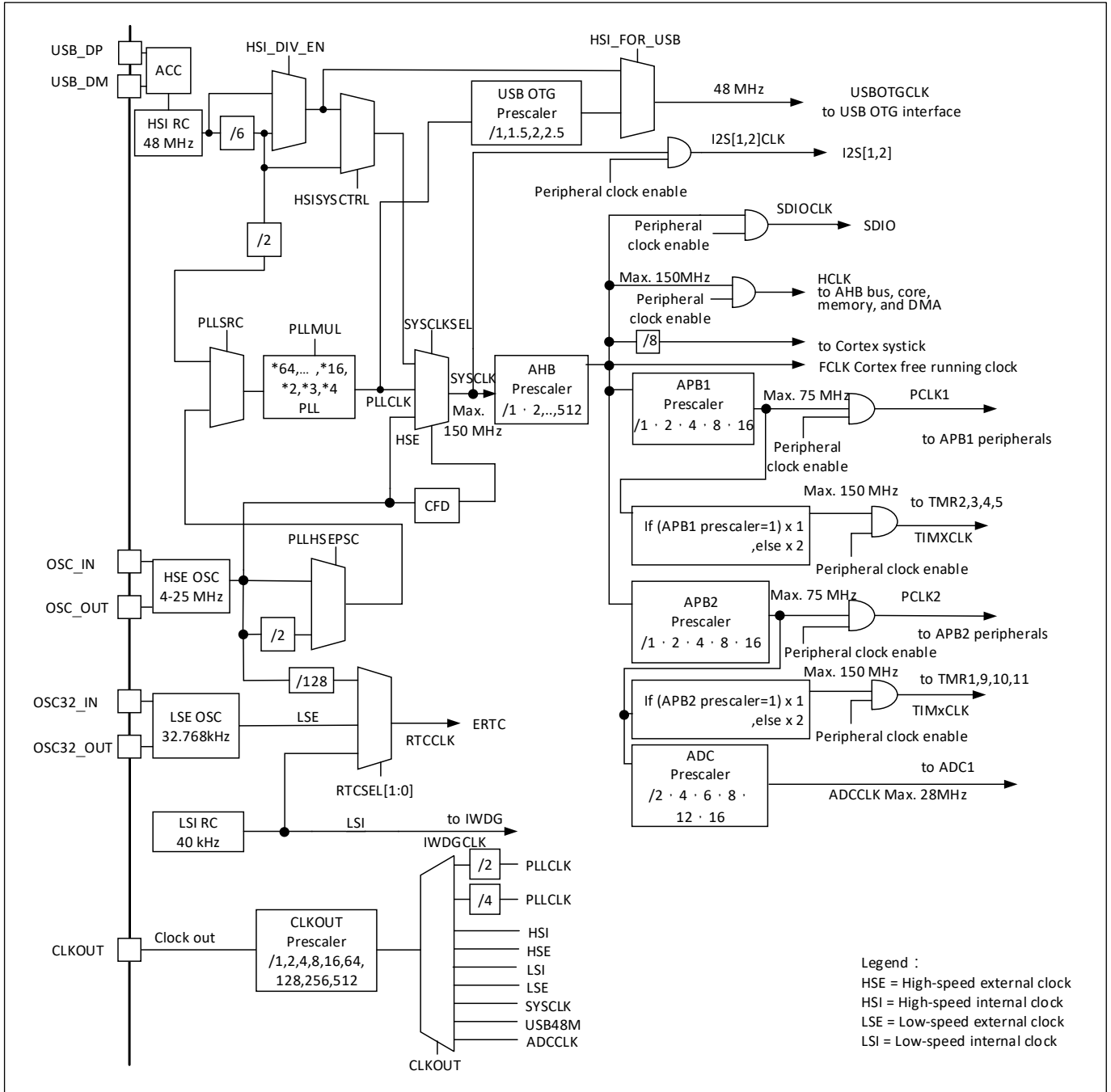
2.2.8 Clocks and startup

System clock selection is performed on startup, however the internal RC 48 MHz oscillator (HSI) through a divided-by-6 divider (8 MHz) is selected as default CPU clock on reset. An external 4 to 25 MHz clock (HSE) can be selected, in which case it is monitored for failure. If failure is detected, the system automatically switches back to the internal RC oscillator. A software interrupt is generated. Similarly, full interrupt management of the PLL clock entry is available when necessary (for example with failure of an indirectly used external oscillator).

Several prescalers allow the configuration of the AHB and the APB (APB1 and APB2) frequency. The maximum frequency of the AHB domain is 150 MHz. The maximum allowed frequency of the APB domains are 75 MHz. See [Figure 2](#) for details on the clock tree.

The AT32F415 embeds an automatic clock calibration (ACC) block, which calibrates the internal RC 48 MHz oscillator. This assures the most precise accuracy of the HSI in the full range of the operating temperatures.

Figure 2. Clock tree



(1) When using USB OTG function and its clock source is from PLL, CPU frequency must be 48 MHz, 72 MHz, 96 MHz, 120 MHz, 144 MHz, or 192 MHz; when its clock source is direct from HSI 48 MHz, CPU frequency can be any frequency from 48 MHz to 96 MHz.

2.2.9 Boot modes

At startup, boot pins are used to select one of three boot options:

- Boot from user Flash
- Boot from system memory
- Boot from embedded SRAM

The bootloader is stored in system memory. It is used to reprogram the Flash memory through USART1, USART2, or the device mode of USB OTG (DFU: device firmware update). [Table 3](#) provides the supporting interfaces of the Bootloader to different AT32F415 part numbers and pin configurations.

Table 3. The Bootloader supporting pin configurations

Interface	Pin
USART1	PA9: USART1_TX PA10: USART1_RX
USART2	PA2: USART2_TX ⁽¹⁾ PA3: USART2_RX ⁽¹⁾
USB OTG FS	PA11: OTG_FS_DM PA12: OTG_FS_DP

(1) Note that pins used are not 5 V tolerant.

2.2.10 Power supply schemes

- $V_{DD} = 2.6 \sim 3.6$ V: external power supply for I/Os and the internal regulator provided externally through V_{DD} pins.
- $V_{DDA} = 2.6 \sim 3.6$ V: external analog power supplies for ADC and DAC. V_{DDA} and V_{SSA} must be connected to V_{DD} and V_{SS} , respectively.
- $V_{BAT} = 1.8 \sim 3.6$ V: power supply for ERTC, external clock 32 kHz oscillator and backup registers (through power switch) when V_{DD} is not present.

For more detail on how to connect power pins, refer to [Figure 10](#).

2.2.11 Power supply supervisor

The device has an integrated power-on reset (POR)/power-down reset (PDR) circuitry. It is always active, and ensures proper operation starting from/down to 2.6 V. The device remains in reset mode when V_{DD} is below a specified threshold, $V_{POR/PDR}$, without the need for an external reset circuit.

The device features an embedded programmable voltage detector (PVD) that monitors the V_{DD} power supply and compares it to the V_{PVD} threshold. An interrupt can be generated when V_{DD} drops below the V_{PVD} threshold and/or when V_{DD} is higher than the V_{PVD} threshold. The interrupt service routine can then generate a warning message and/or put the MCU into a safe state. The PVD is enabled by software. Refer to [Table 11](#) for the characteristic values of $V_{POR/PDR}$ and V_{PVD} .

2.2.12 Voltage regulator

The regulator has three operation modes: main (MR), low-power (LPR), and power down.

- Main mode (MR) is used in the nominal regulation mode (Run) or in the Stop mode
- Low-power mode (LPR) can be used in the Stop mode
- Power down mode is used in Standby mode: the regulator output is in high impedance and the kernel circuitry is powered down, inducing zero consumption of the regulator (but the contents of the registers and SRAM are lost)

This regulator is always enabled after reset. It is disabled in Standby mode.

2.2.13 Low-power modes

The AT32F415 supports three low-power modes to achieve the best compromise between low-power consumption, short startup time and available wakeup sources:

- **Sleep mode**

In Sleep mode, only the CPU is stopped. All peripherals continue to operate and can wake up the CPU when an interrupt/event occurs.

- **Stop mode**

Stop mode achieves the lowest power consumption while retaining the content of SRAM and registers. All clocks in the 1.2 V domain are stopped, the PLL, the HSI RC and the HSE crystal oscillators are disabled. The voltage regulator is put in normal mode.

The device can be woken up from Stop mode by any of the EXTI line. The EXTI line source can be one of the 16 external lines, the PVD output, the ERTC alarm, the USB OTG or the COMP wakeup.

- **Standby mode**

The Standby mode is used to achieve the lowest power consumption. The internal voltage regulator is switched off so that the entire 1.2 V domain is powered off. The PLL, the HSI RC and the HSE crystal oscillators are also switched off. After entering Standby mode, SRAM and register contents are lost except for registers in the Backup domain and Standby circuitry.

The device exits Standby mode when an external reset (NRST pin), an IWDG reset, a rising edge on the WKUP pin, or an ERTC alarm occurs.

Note: The ERTC, the IWDG, and the corresponding clock sources are not stopped by entering Stop or Standby mode.

2.2.14 Direct memory access (DMA)

The flexible 14-channel general-purpose DMAs (7 channels for DMA1 and 7 channels for DMA2) are able to manage memory-to-memory, peripheral-to-memory, and memory-to-peripheral transfers. The two DMA controllers support circular buffer management, removing the need for user code intervention when the controller reaches the end of the buffer.

Each channel is connected to dedicated hardware DMA requests, with support for software trigger on each channel. Configuration is made by software and transfer sizes between source and destination are independent.

The DMA can be used with the main peripherals: SPIs, I²Cs, USARTs, general-purpose and advanced-control timers TMRx, I²Ss, SDIO, and ADC.

2.2.15 ERTC (enhanced real-time clock) and backup registers

The backup domain includes:

- The enhanced real-time clock (ERTC)
- Twenty 32-bit backup registers

The enhanced real-time clock (ERTC) is an independent BCD timer/counter. It supports the following features:

- Calendar with second, minute, hour (12 or 24 format), week day, date, month, year, in BCD (binary-coded decimal) format.
- The sub-seconds value is also available in binary format.
- Automatic correction for 28, 29 (leap year), 30, and 31 days of the month.
- Two programmable alarms and one auto-reload wakeup timer (WUT) for periodic events with wake up from Stop or Standby mode
- Calibrated using an external 512 Hz output to compensate for any natural quartz deviation.

Two alarm registers are used to generate an alarm at a specific time and calendar fields can be independently masked for alarm comparison. To generate a periodic interrupt, a 16-bit programmable binary auto-reload downcounter with programmable resolution is available and allows automatic wakeup and periodic alarms from every 120 μ s to every 36 hours.

A 20-bit prescaler is used for the time base clock. It is by default configured to generate a time base of 1 second from a clock at 32.768 kHz.

The backup registers are 32-bit registers used to store 80 bytes of user application data when V_{DD} power is not present. Backup registers are not reset by a system, a power reset, or when the device wakes up from the Standby mode.

Additional 32-bit registers contain the programmable alarm subseconds, seconds, minutes, hours, day, and date.

The ERTC and twenty backup registers are supplied through a switch that is powered either from the V_{DD} supply when present or from the V_{BAT} pin.

The ERTC clock sources can be:

- A 32.768 kHz external crystal, external resonator, or oscillator (LSE)
- The internal low power RC oscillator (LSI, with typical frequency of 40 kHz)
- The high-speed external clock (HSE) divided by 128

2.2.16 Timers and watchdogs

The AT32F415 devices include one advanced-control timers, seven general-purpose timers, two watchdog timers, and a SysTick timer.

The table below compares the features of the advanced-control and general-purpose timers.

Table 4. Timer feature comparison

Timer	Counter resolution	Counter type	Prescaler factor	DMA request generation	Capture/compare channels	Complementary outputs
TMR1	16-bit	Up, down, up/down	Any integer between 1 and 65536	Yes	4	Yes
TMR2 TMR5	32-bit	Up, down, up/down	Any integer between 1 and 65536	Yes	4	No
TMR3 TMR4	16-bit	Up, down, up/down	Any integer between 1 and 65536	Yes	4	No
TMR9	16-bit	Up	Any integer between 1 and 65536	No	2	No
TMR10 TMR11	16-bit	Up	Any integer between 1 and 65536	No	1	No

Advanced-control timers (TMR1)

An advanced-control timers (TMR1) can each be seen a three-phase PWM multiplexed on six channels. They have complementary PWM outputs with programmable inserted dead-times. They can also be seen as a complete general-purpose timer. The four independent channels can be used for:

- Input capture
- Output compare
- PWM generation (edge or center-aligned modes)
- One-pulse mode output

If configured as a standard 16-bit timer, it has the same features as the TMRx timer. If configured as the 16-bit PWM generator, it has full modulation capability (0-100%).

In debug mode, the advanced-control timer counter can be frozen and the PWM outputs disabled to turn off any power switch driven by these outputs.

Many features are shared with those of the general-purpose TMR timers which have the same architecture. The advanced-control timer can therefore work together with the TMR timers via the link feature for synchronization or event chaining.

General-purpose timers (TMRx)

There are seven synchronizable general-purpose timers embedded in the AT32F415.

- **TMR2, TMR3, TMR4, and TMR5**

The AT32F415 has 4 full-featured general-purpose timers: TMR2, TMR3, TMR4, and TMR5. The TMR2 and TMR5 timers are based on a 32-bit auto-reload up/down counter and a 16-bit prescaler. The TMR3 and TMR4 timers are based on a 16-bit auto-reload up/down counter and a 16-bit prescaler. They all feature four independent channels for input capture/output compare, PWM or one-pulse mode output. This gives up to 16 input capture/output compare/PWMs.

The TMR2, TMR3, TMR4, and TMR5 general-purpose timers can work together, or with the other general-purpose timers and the advanced-control timers via the link feature for synchronization or event chaining. In debug mode, their counter can be frozen. Any of these general-purpose timers can be used to generate PWM outputs.

The TMR2, TMR3, TMR4, and TMR5 are capable of handling quadrature (incremental) encoder signals and the digital outputs from one to three hall-effect sensors.

- **TMR9**

TMR9 is based on a 16-bit auto-reload upcounter, a 16-bit prescaler, and two independent channels for input capture/output compare, PWM, or one-pulse mode output. It can be synchronized with the TMR2, TMR3, TMR4, and TMR5 full-featured general-purpose timers. It can also be used as simple time bases.

- **TMR10 and TMR11**

These timers are based on a 16-bit auto-reload upcounter, a 16-bit prescaler, and one independent channels for input capture/output compare, PWM, or one-pulse mode output. They can be synchronized with the TMR2, TMR3, TMR4, and TMR5 full-featured general-purpose timers. They can also be used as simple time bases.

Independent watchdog (IWDG)

The independent watchdog is based on a 12-bit downcounter and 8-bit prescaler. It is clocked from an independent 40 kHz internal RC and as it operates independently from the main clock, it can operate in Stop and Standby modes. It can be used either as a watchdog to reset the device when a problem occurs, or as a free running timer for application timeout management. It is hardware or software configurable through the option bytes. The counter can be frozen in debug mode.

Window watchdog (WWDG)

The window watchdog is based on a 7-bit downcounter that can be set as free running. It can be used as a watchdog to reset the device when a problem occurs. It is clocked from the main clock. It has an early warning interrupt capability and the counter can be frozen in debug mode.

SysTick timer

This timer is dedicated to real-time operating systems, but could also be used as a standard down counter. It features:

- A 24-bit down counter
- Autoreload capability
- Maskable system interrupt generation when the counter reaches 0
- Programmable clock source

2.2.17 Inter-integrated circuit interface (I²C)

Two I²C bus interfaces can operate in multi-master and slave modes. They can support standard and fast modes.

They support 7/10-bit addressing mode and 7-bit dual addressing mode (as slave). A hardware CRC generation/verification is included.

They can be served by DMA and they support SMBus 2.0/PMBus.

2.2.18 Universal synchronous/asynchronous receiver transmitters (USART)

The AT32F415 embeds 3 universal synchronous/asynchronous receiver transmitters (USART1, USART2, and USART3) and 2 universal asynchronous receiver transmitters (UART4 and UART5).

These five interfaces provide asynchronous communication, IrDA SIR ENDEC support, multiprocessor communication mode, single-wire half-duplex communication mode, and have LIN Master/Slave capability.

3 USARTs and 2 UARTs are able to communicate at speeds of up to 4.6875 Mbit/s.

USART1, USART2, and USART3 also provide hardware management of the CTS and RTS signals, Smart Card mode (ISO 7816 compliant) and SPI-like communication capability. All interfaces can be served by the DMA controller except for UART5.

2.2.19 Serial peripheral interface (SPI)

Two SPIs are able to communicate up to 50 Mbits/s in slave and master modes in full-duplex and simplex communication modes. The 3-bit prescaler gives 8 master mode frequencies and the frame is configurable to 8 bits or 16 bits. The hardware CRC generation/verification supports basic SD Card/MMC/SDHC modes.

Both SPIs can be served by the DMA controller.

2.2.20 Inter-integrated sound interface (I²S)

Two standard I²S interfaces (multiplexed with SPI) are available, that can be operated in master or slave mode. These interfaces can be configured to operate with 16/32 bit resolution, as input or output channels. Audio sampling frequencies from 8 kHz up to 96 kHz are supported. When any of the I²S interfaces is/are configured in master mode, the master clock can be output to the external DAC/CODEC at 256 times the sampling frequency.

2.2.21 Secure digital input/output interface (SDIO)

One SD/SDIO/MMC host interface is available, that supports MultiMediaCard System Specification Version 4.2 in three different data bus modes: 1-bit (default), 4-bit and 8-bit. The interface allows data transfer at up to 50 MHz in 8-bit mode, and is compliant with SD Memory Card Specifications Version 2.0.

The SDIO Card Specification Version 2.0 is also supported with two different data bus modes: 1-bit (default) and 4-bit.

The current version supports only one SD/SDIO/MMC4.2 card at any one time and a stack of MMC4.1 or previous.

In addition to SD/SDIO/MMC, this interface is also fully compliant with the CE-ATA digital protocol Rev1.1.

2.2.22 Controller area network (CAN)

One CAN is compliant with specifications 2.0A and B (active) with a bit rate up to 1 Mbit/s. It can receive and transmit standard frames with 11-bit identifiers as well as extended frames with 29-bit identifiers. It has three transmit mailboxes, two receive FIFOs with 3 stages and 14 scalable filter banks. CAN controller has dedicated 256 bytes SRAM, which is not shared with other peripherals.

2.2.23 Universal serial bus on-the-go full-speed (USB OTG FS)

The AT32F415 embed a USB OTG full-speed (12 Mb/s) device/host/OTG peripheral with integrated transceivers. The USB OTG FS peripheral is compliant with the USB 2.0 specification and with the OTG 1.3 specification. It has software-configurable endpoint setting and supports suspend/resume. The USB OTG FS controller requires a dedicated 48 MHz clock that is generated by a PLL; as a device peripheral, the HSI 48 MHz clock source can be used as the USB OTG FS clock directly.

The major features are:

- 1280 KBytes of SRAM used exclusively by the endpoints (not shared with any other peripherals)
- 4 bidirectional endpoints
- 8 host channels with periodic OUT support
- The SOF output can be used to synchronize the external audio DAC clock in isochronous mode
- In accordance with the USB 2.0 Specification, the supported transfer speeds are:
 - In Host mode: full-speed and low speed
 - In Device mode: full-speed

For OTG/Host modes, a power switch is needed in case bus-powered devices are connected

2.2.24 General-purpose inputs/outputs (GPIO)

Each of the GPIO pins can be configured by software as output (push-pull or open-drain), as input (with or without pull-up or pull-down), or as peripheral alternate function. Most of the GPIO pins are shared with digital or analog alternate functions.

The I/O's alternate function configuration can be locked, if needed, in order to avoid spurious writing to the I/Os registers by following a specific sequence.

2.2.25 Remapping capability

This feature allows the use of a maximum number of peripherals in a given application. Indeed, alternate functions are available not only on the default pins but also on other specific pins onto which they are remappable. This has the advantage of making board design and port usage much more flexible.

For details refer to [Table 5](#), it shows the list of remappable alternate functions and the pins onto which they can be remapped. See the AT32F415 reference manual for software considerations.

2.2.26 Analog to digital converter (ADC)

One 12-bit analog-to-digital converters are embedded into AT32F415 devices and it has up to 16 external channels, performing conversions in single-shot or scan modes. In scan mode, automatic conversion is performed on a selected group of analog inputs.

The ADC can be served by the DMA controller.

An analog watchdog feature allows very precise monitoring of the converted voltage of one, some or all selected channels. An interrupt is generated when the converted voltage is outside the programmed thresholds.

The events generated by the general-purpose timers (TMRx) and the advanced-control timer (TMR1) can be internally connected to the ADC start trigger and injection trigger, respectively, to allow the application to synchronize A/D conversion and timers.

2.2.27 Temperature sensor

The temperature sensor has to generate a voltage that varies linearly with temperature. The conversion range is between $2.6\text{ V} \leq V_{\text{DDA}} \leq 3.6\text{ V}$. The temperature sensor is internally connected to the ADC1_IN16 input channel which is used to convert the sensor output voltage into a digital value.

2.2.28 Comparator (COMP)

The AT32F415 embeds two rail-to-rail comparators with programmable reference voltage (internal or external), hysteresis, speed, selectable output polarity, and noise filter.

The reference voltage can be one of the following:

- External I/O
- Internal reference voltage or submultiple (1/4, 1/2, 3/4). Refer to [Table 12](#) for the value and precision of the internal reference voltage.

Both comparators can wake up from Stop mode, generate interrupts and breaks for the timers and can be also combined into a window comparator.

2.2.29 Serial wire JTAG debug port (SWJ-DP)

The ARM SWJ-DP Interface is embedded, and is a combined JTAG and serial wire debug port that enables either a serial wire debug or a JTAG probe to be connected to the target. The JTAG TMS and TCK pins are shared respectively with SWDIO and SWCLK and a specific sequence on the TMS pin is used to switch between JTAG-DP and SW-DP.

3 Pinouts and pin descriptions

Figure 3. AT32F415 LQFP64 pinout

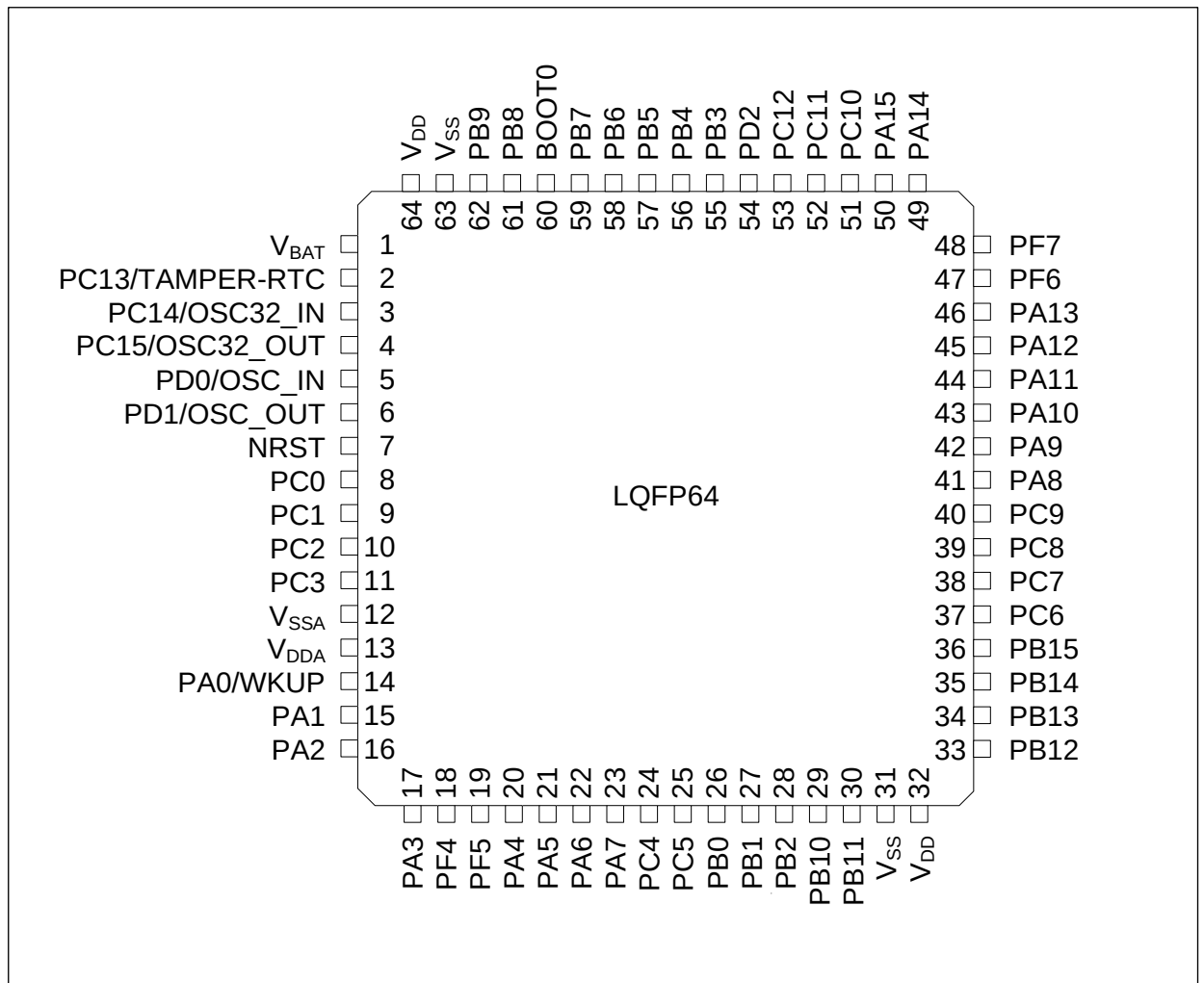


Figure 4. AT32F415 LQFP48 pinout

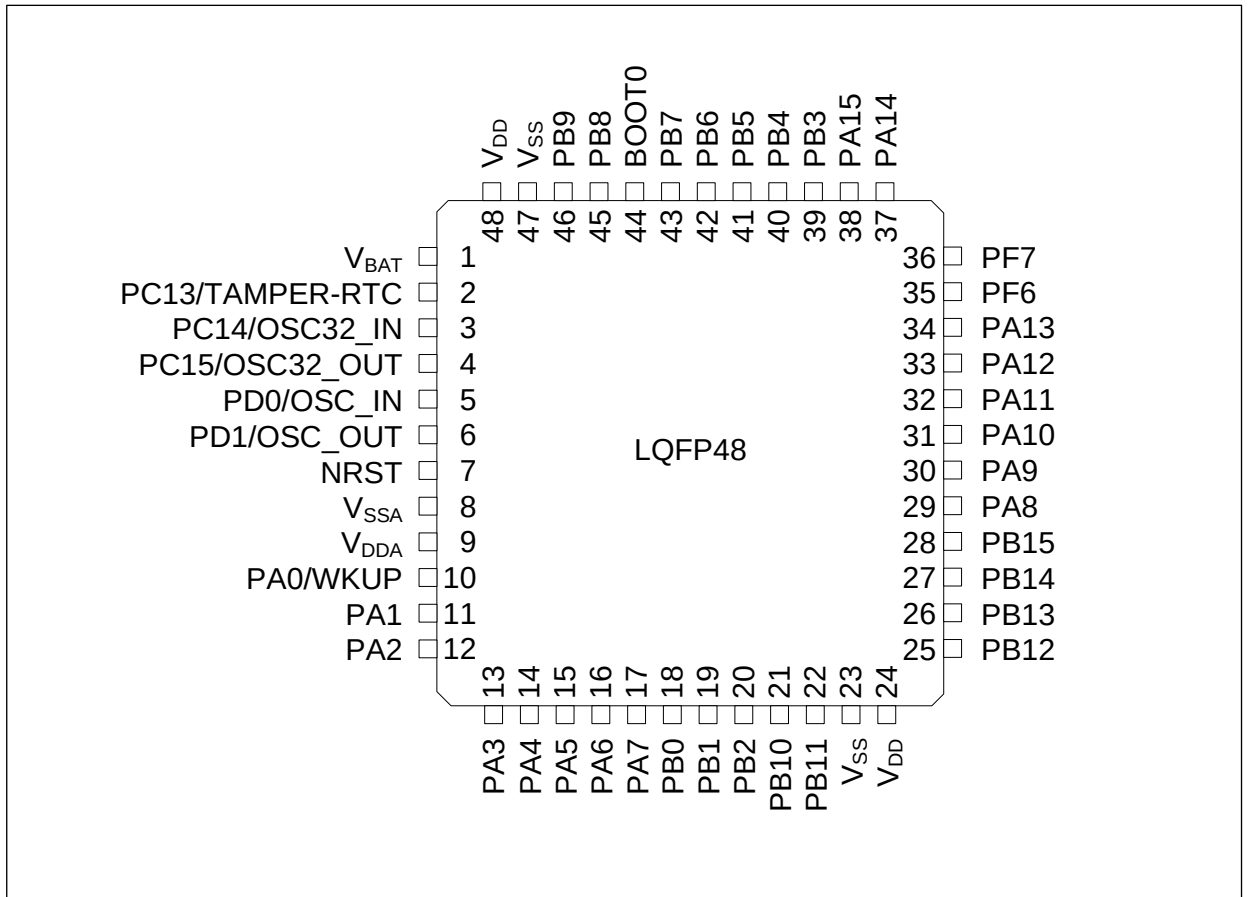


Figure 5. AT32F415 QFN48 pinout

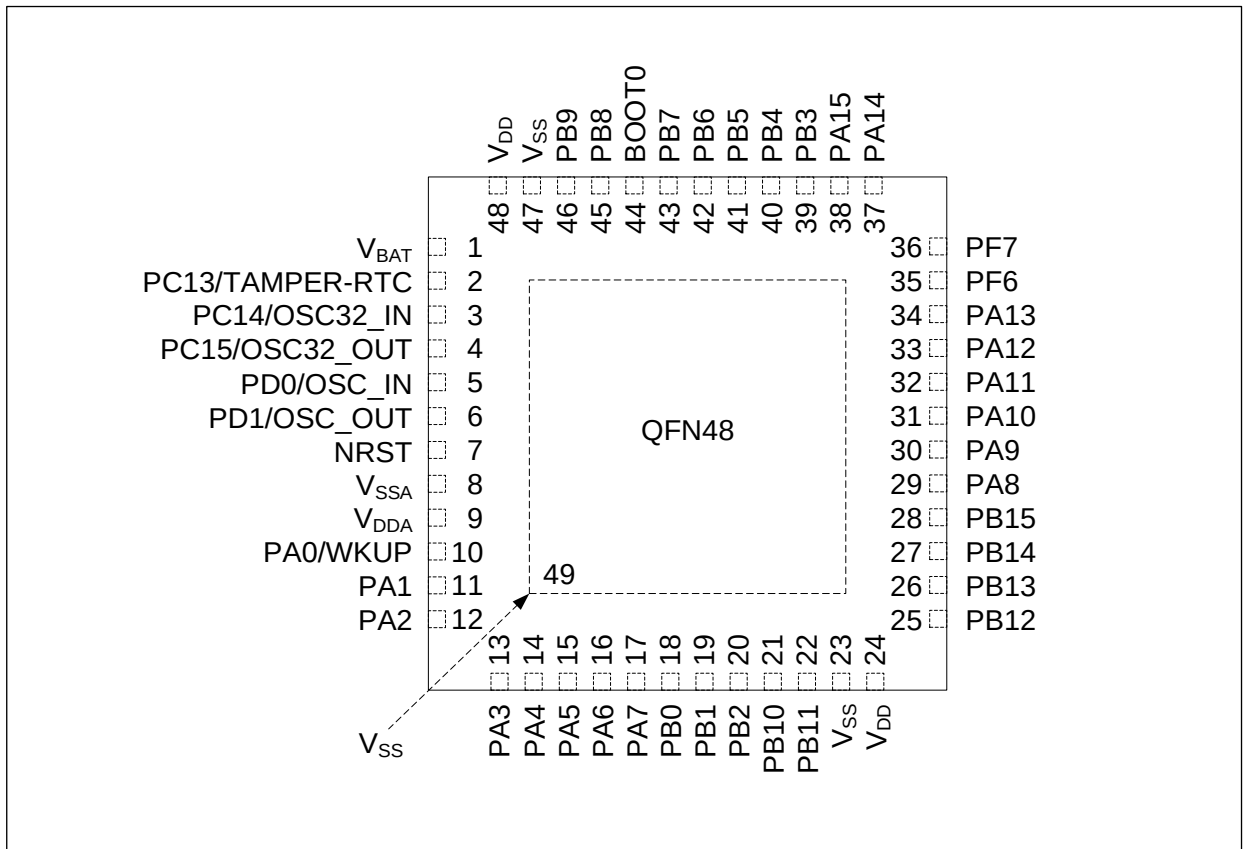
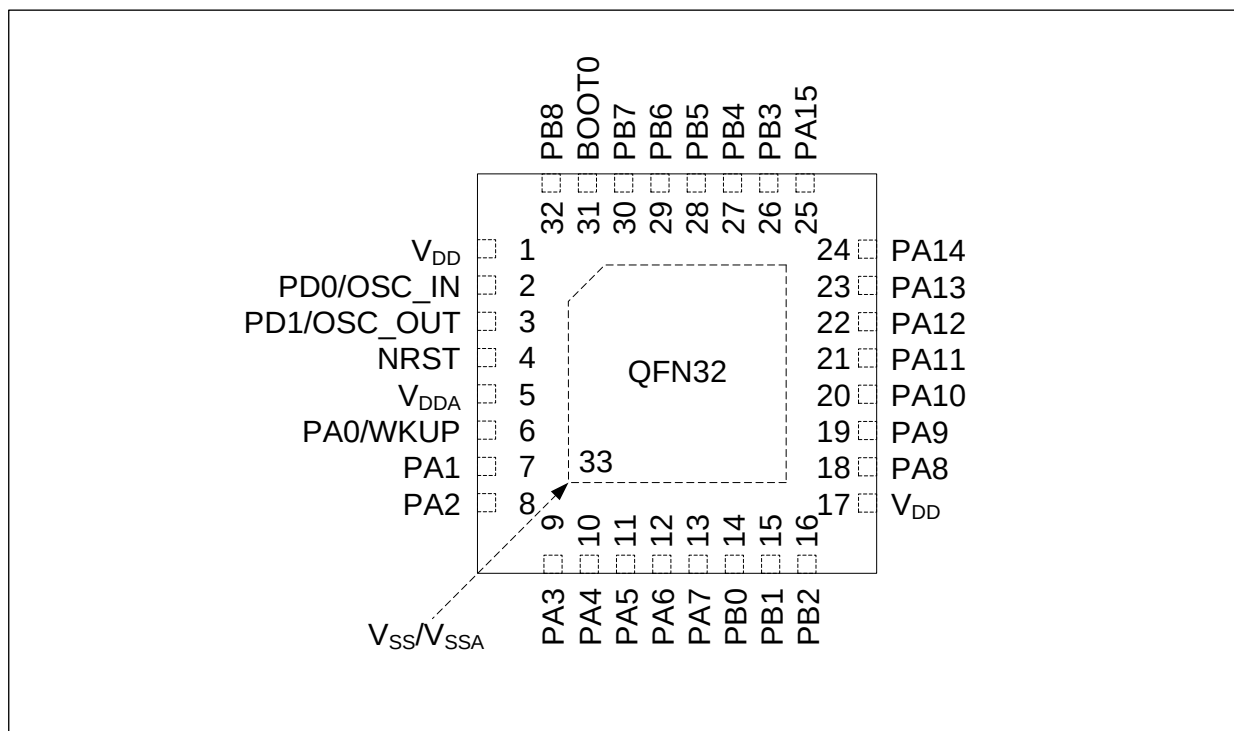


Figure 6. AT32F415 QFN32 pinout



The table below is the pin definition of the AT32F415. "-" presents there is no such pinout on the related package. The multi-functions list follows priority from high to low. In principle, the analog signals have higher priority than the digital signals, and the digital output signals have higher priority than the digital input signals.

Table 5. AT32F415 series pin definitions

Pin number				Pin name	Type ⁽¹⁾	IO level ⁽²⁾	Main function	Alternate functions ⁽³⁾	
QFN32	LQFP48	QFN48	LQFP64					Default	Remap
-	1	1		V _{BAT}	S	-	V _{BAT}	-	-
-	2	2		PC13 ⁽⁴⁾	I/O	-	PC13	TAMPER-RTC ⁽⁵⁾	-
-	3	3		PC14 ⁽⁴⁾	I/O	-	PC14	OSC32_IN ⁽⁵⁾	-
-	4	4		PC15 ⁽⁴⁾	I/O	-	PC15	OSC32_OUT ⁽⁵⁾	-
2	5	5		PD0 ⁽⁶⁾	I/O	-	OSC_IN	OSC_IN	PD0
3	6	6		PD1 ⁽⁶⁾	I/O	-	OSC_OUT	OSC_OUT	PD1
4	7	7		NRST	I/O	-	NRST	-	-
-	-	8		PC0	I/O	-	PC0	ADC1_IN10	SDIO_D0
-	-	9		PC1	I/O	-	PC1	ADC1_IN11	SDIO_D1
-	-	10		PC2	I/O	-	PC2	ADC1_IN12	SDIO_D2
-	-	11		PC3	I/O	-	PC3	ADC1_IN13	SDIO_D3
-	8	12		V _{SSA}	S	-	V _{SSA}	-	-
5	9	13		V _{DDA}	S	-	V _{DDA}	-	-
6	10	14		PA0-WKUP	I/O	-	PA0	ADC1_IN0 / WKUP / COMP1_OUT ⁽⁷⁾ / COMP1_INP2 / COMP1_INM6 / USART2_CTS / TMR2_CH1 ⁽⁷⁾ / TMR2_ETR ⁽⁷⁾ / TMR5_CH1 ⁽⁷⁾	TMR1_ETR
7	11	15		PA1	I/O	-	PA1	ADC1_IN1 / COMP1_INP1 / USART2_RTS / TMR2_CH2 ⁽⁷⁾ / TMR5_CH2 ⁽⁷⁾	-
8	12	16		PA2	I/O	-	PA2	ADC1_IN2 / COMP2_OUT ⁽⁷⁾ / COMP2_INP2 / COMP2_INM6 / USART2_TX / TMR2_CH3 ⁽⁷⁾ / TMR5_CH3 / TMR9_CH1 ⁽⁷⁾	SDIO_CK
9	13	17		PA3	I/O	-	PA3	ADC1_IN3 / COMP2_INP1 / USART2_RX / TMR2_CH4 ⁽⁷⁾ / TMR5_CH4 / TMR9_CH2 ⁽⁷⁾	SDIO_CMD
-	-	18		PF4	I/O	FT	PF4	-	UART4_TX / TMR5_CH1
-	-	19		PF5	I/O	FT	PF5	-	UART4_RX / TMR5_CH2
10	14	20		PA4	I/O	-	PA4	ADC1_IN4 / COMP1_INM4 / COMP2_INM4 / USART2_CK / SPI1_NSS ⁽⁷⁾ / I2S1_WS ⁽⁷⁾	SDIO_D4 / SDIO_D0
11	15	21		PA5	I/O	-	PA5	ADC1_IN5 / COMP1_INP0 / COMP1_INM5 / COMP2_INM5 / SPI1_SCK ⁽⁷⁾ / I2S1_CK ⁽⁷⁾	USART3_CK / SDIO_D5 / SDIO_D1
12	16	22		PA6	I/O	-	PA6	ADC1_IN6 / SPI1_MISO ⁽⁷⁾ / TMR3_CH1 ⁽⁷⁾	COMP1_OUT / USART3_RX / SDIO_D6 / SDIO_D2 / TMR1_BKIN / TMR10_CH1

Pin number				Pin name	Type ⁽¹⁾	IO level ⁽²⁾	Main function	Alternate functions ⁽³⁾	
QFN32	LQFP48	QFN48	LQFP64					Default	Remap
13	17	23		PA7	I/O	-	PA7	ADC1_IN7 / COMP2_INP0 / SPI1_MOSI ⁽⁷⁾ / I2S1_SD ⁽⁷⁾ / TMR3_CH2 ⁽⁷⁾	COMP2_OUT / USART3_TX / SDIO_D7 / SDIO_D3 / TMR1_CH1N / TMR11_CH1
-	-	24		PC4	I/O	-	PC4	ADC1_IN14	SDIO_CK
-	-	25		PC5	I/O	-	PC5	ADC1_IN15	SDIO_CMD
14	18	26		PB0	I/O	-	PB0	ADC1_IN8 / I2S1_MCK ⁽⁷⁾ / TMR3_CH3 ⁽⁷⁾	USART3_RTS / TMR1_CH2N
15	19	27		PB1	I/O	-	PB1	ADC1_IN9 / TMR3_CH4 ⁽⁷⁾	USART3_CTS / TMR1_CH3N
16	20	28		PB2	I/O	FT	PB2/ BOOT1	-	-
-	21	29		PB10	I/O	FT	PB10	I2C2_SCL ⁽⁷⁾ / USART3_TX ⁽⁷⁾	TMR2_CH3
-	22	30		PB11	I/O	FT	PB11	I2C2_SDA ⁽⁷⁾ / USART3_RX ⁽⁷⁾	TMR2_CH4
-	23	31		V _{SS}	S	-	V _{SS}	-	-
17	24	32		V _{DD}	S	-	V _{DD}	-	-
-	25	33		PB12	I/O	FT	PB12	USART3_CK ⁽⁷⁾ / I2C2_SMBA ⁽⁷⁾ / SPI2_NSS ⁽⁷⁾ / I2S2_WS ⁽⁷⁾ / TMR1_BKIN ⁽⁷⁾	-
-	26	34		PB13	I/O	FT	PB13	TMR1_CH1N ⁽⁷⁾ / USART3_CTS ⁽⁷⁾ / SPI2_SCK ⁽⁷⁾ / I2S2_CK ⁽⁷⁾	-
-	27	35		PB14	I/O	FT	PB14	TMR1_CH2N ⁽⁷⁾ / USART3_RTS ⁽⁷⁾ / SPI2_MISO ⁽⁷⁾	TMR9_CH1
-	28	36		PB15	I/O	FT	PB15	TMR1_CH3N ⁽⁷⁾ / RTC_REFIN / SPI2_MOSI ⁽⁷⁾ / I2S2_SD ⁽⁷⁾	TMR9_CH2
-	-	37		PC6	I/O	FT	PC6	I2S2_MCK ⁽⁷⁾ / SDIO_D6 ⁽⁷⁾	TMR1_CH1 / TMR3_CH1
-	-	38		PC7	I/O	FT	PC7	SDIO_D7 ⁽⁷⁾	I2S2_MCK / TMR1_CH2 / TMR3_CH2
-	-	39		PC8	I/O	FT	PC8	SDIO_D0 ⁽⁷⁾	TMR1_CH3 / TMR3_CH3
-	-	40		PC9	I/O	FT	PC9	SDIO_D1 ⁽⁷⁾	I2C2_SDA / TMR1_CH4 / TMR3_CH4
18	29	41		PA8	I/O	FT	PA8	OTG_FS_SOF / CLKOUT / USART1_CK / TMR1_CH1	I2C2_SCL
19	30	42		PA9	I/O	FT	PA9	OTG_FS_VBUS ⁽⁸⁾ / USART1_TX ⁽⁷⁾ / TMR1_CH2	I2C2_SMBA
20	31	43		PA10	I/O	-	PA10	OTG_FS_ID / USART1_RX ⁽⁷⁾ / TMR1_CH3	-
21	32	44		PA11	I/O	-	PA11	OTG_FS_DM / USART1_CTS / CAN_RX ⁽⁷⁾ / TMR1_CH4	COMP1_OUT
22	33	45		PA12	I/O	-	PA12	OTG_FS_DP / USART1_RTS / CAN_TX ⁽⁷⁾ / TMR1_ETR	COMP2_OUT
23	34	46		PA13	I/O	FT	JTMS-SWDIO	-	PA13
-	35	47		PF6	I/O	FT	PF6	-	I2C1_SCL / I2C2_SCL
-	36	48		PF7	I/O	FT	PF7	-	I2C1_SDA / I2C2_SDA
24	37	49		PA14	I/O	FT	JTCK-SWCLK	-	PA14

Pin number				Pin name	Type ⁽¹⁾	IO level ⁽²⁾	Main function	Alternate functions ⁽³⁾	
QFN32	LQFP48	QFN48	LQFP64					Default	Remap
25	38	50		PA15	I/O	FT	JTDI	-	PA15 / SPI1_NSS / I2S1_WS / SPI2_NSS / I2S2_WS / TMR2_CH1 / TMR2_ETR
-	-	51		PC10	I/O	FT	PC10	UART4_TX ⁽⁷⁾ / SDIO_D2 ⁽⁷⁾	USART3_TX
-	-	52		PC11	I/O	FT	PC11	UART4_RX ⁽⁷⁾ / SDIO_D3 ⁽⁷⁾	USART3_RX
-	-	53		PC12	I/O	FT	PC12	UART5_TX / SDIO_CK ⁽⁷⁾	USART3_CK
-	-	54		PD2	I/O	FT	PD2	UART5_RX / SDIO_CMD ⁽⁷⁾ / TMR3_ETR	-
26	39	55		PB3	I/O	FT	JTDO	-	PB3 / TRACESWO / SPI1_SCK / I2S1_CK / SPI2_SCK / I2S2_CK / TMR2_CH2
27	40	56		PB4	I/O	FT	NJTRST	-	PB4 / SPI1_MISO / SPI2_MISO / I2C2_SDA / TMR3_CH1
28	41	57		PB5	I/O	FT	PB5	I2C1_SMBA	SPI1_MOSI / I2S1_SD / SPI2_MOSI / I2S2_SD / TMR3_CH2
29	42	58		PB6	I/O	FT	PB6	I2C1_SCL ⁽⁷⁾ / TMR4_CH1	USART1_TX / I2S1_MCK
30	43	59		PB7	I/O	FT	PB7	I2C1_SDA ⁽⁷⁾ / TMR4_CH2	USART1_RX
31	44	60		BOOT0	I	-	BOOT0	-	-
32	45	61		PB8	I/O	FT	PB8	SDIO_D4 ⁽⁷⁾ / TMR4_CH3 / TMR10_CH1 ⁽⁷⁾	I2C1_SCL / CAN_RX
-	46	62		PB9	I/O	FT	PB9	SDIO_D5 ⁽⁷⁾ / TMR4_CH4 / TMR11_CH1 ⁽⁷⁾	I2C1_SDA / CAN_TX
-	47	63		V _{SS}	S	-	V _{SS}	-	-
1	48	64		V _{DD}	S	-	V _{DD}	-	-
-	-/49	-		V _{SS}	S	-	V _{SS}	-	-
33	-	-		V _{SS} /V _{SSA}	S	-	V _{SS} /V _{SSA}	-	-

(1) I = input, O = output, S = supply.

(2) FT = 5 V tolerant.

(3) If several peripherals share the same I/O pin, to avoid conflict between these alternate functions only one peripheral should be enabled at a time through the peripheral clock enable bit (in the corresponding RCC peripheral clock enable register).

(4) PC13, PC14, and PC15 are supplied through the power switch. Since the switch only sinks a limited amount of current (3 mA), the use of GPIOs PC13 to PC15 in output mode is limited: the normal sourcing/sinking strength should be used with a maximum load of 30 pF and these IOs must not be used as a current source (e.g. to drive an LED).

(5) Main function after the first backup domain power-up. Later on, it depends on the contents of the Backup registers even after reset (because these registers are not reset by the main reset). For details on how to manage these IOs, refer to the Battery backup domain and BKP register description sections in the AT32F415 reference manual.

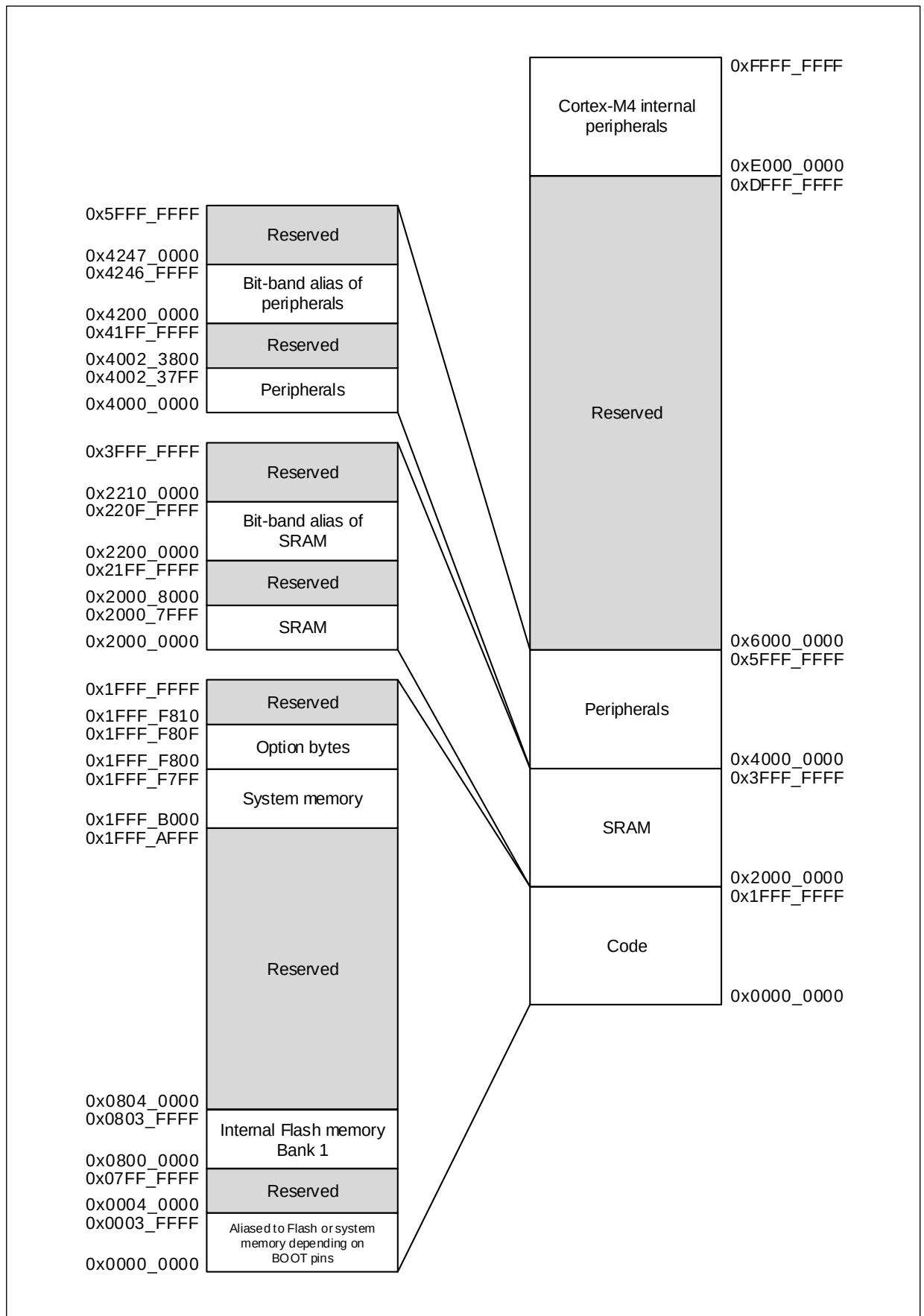
(6) The pins number 5 and 6 of the LQFP64, LQFP48, and QFN48 packages and the pins number 2 and 3 of the QFN32 packages are configured as OSC_IN/OSC_OUT after reset, the functionality of PD0 and PD1 can be remapped by software on these pins. For more details, refer to Alternate function I/O and debug configuration section in the AT32F415 reference manual.

(7) This alternate function can be remapped by software to some other port pins (if available on the used package). For more details, refer to the Alternate function I/O and debug configuration section in the AT32F415 reference manual.

(8) When USB OTG FS is used and configured as a device, PA9 should keep high level. Its GPIO and other alternative functions could not be used.

4 Memory mapping

Figure 7. Memory map



5 Electrical characteristics

5.1 Parameter conditions

Unless otherwise specified, all voltages are referenced to V_{SS} .

5.1.1 Minimum and maximum values

Unless otherwise specified the minimum and maximum values are guaranteed in the worst conditions of ambient temperature, supply voltage and frequencies by tests in production with an ambient temperature at $T_A = 25\text{ }^{\circ}\text{C}$ and $T_A = T_A \text{ max.}$

Data based on characterization results, design simulation and/or technology characteristics are indicated in the table footnotes and are not tested in production.

5.1.2 Typical values

Unless otherwise specified, typical data are based on $T_A = 25\text{ }^{\circ}\text{C}$, $V_{DD} = 3.3\text{ V}$. They are given only as design guidelines and are not tested.

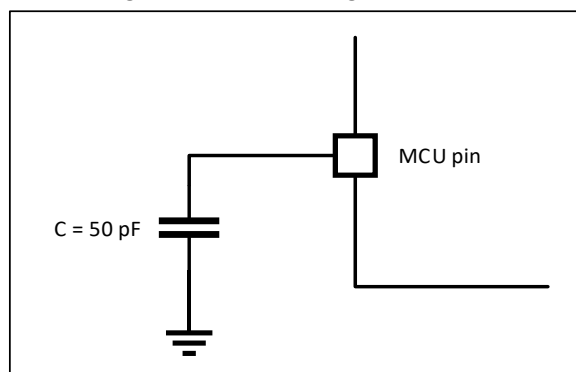
5.1.3 Typical curves

Unless otherwise specified, all typical curves are given only as design guidelines and are not tested.

5.1.4 Loading capacitor

The loading conditions used for pin parameter measurement are shown in [Figure 8](#).

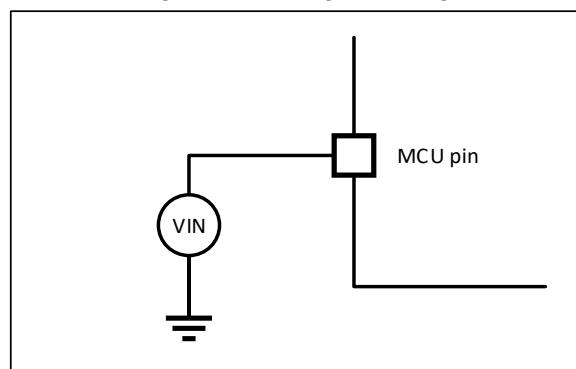
Figure 8. Pin loading conditions



5.1.5 Pin input voltage

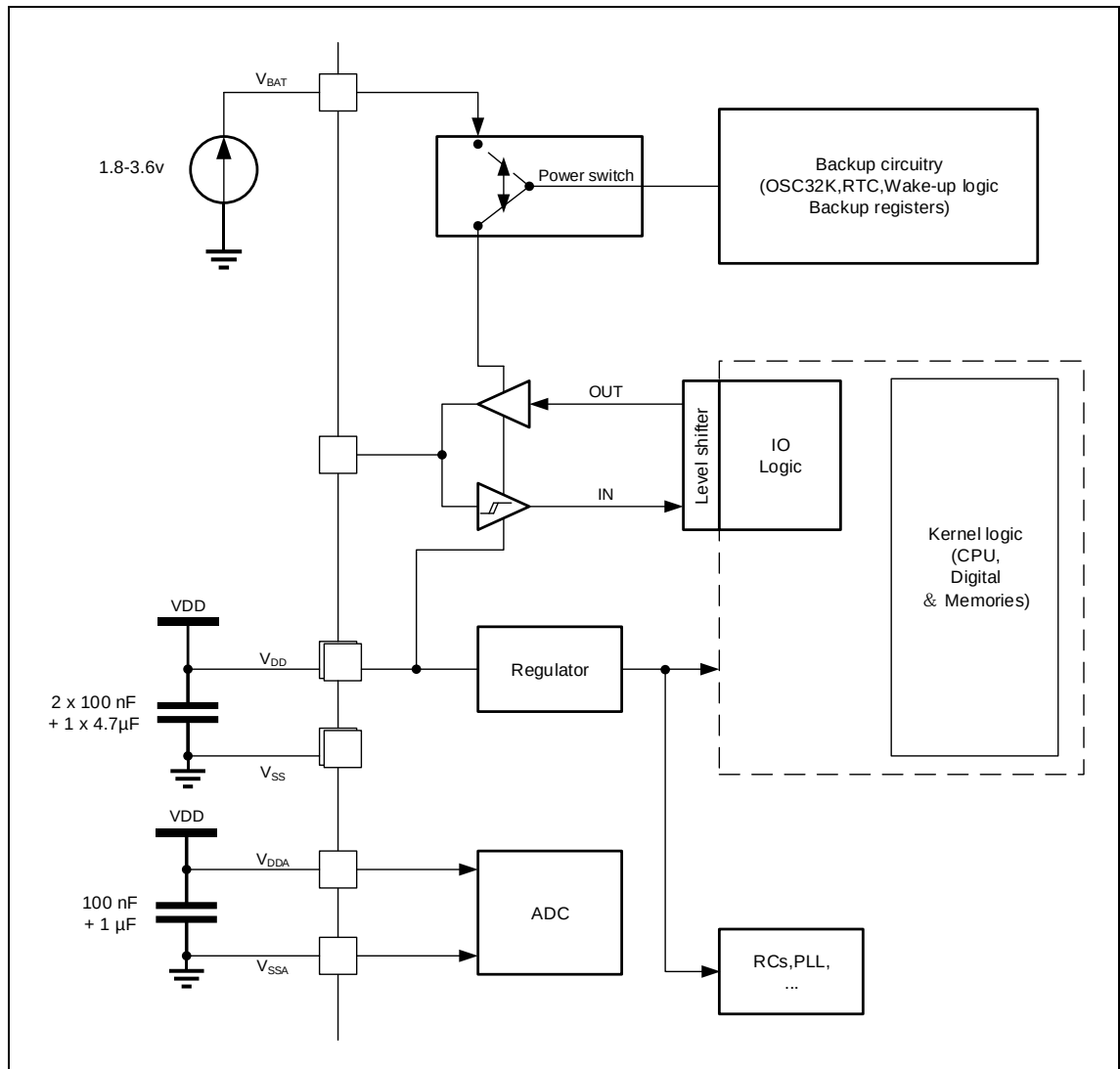
The input voltage measurement on a pin of the device is described in [Figure 9](#).

Figure 9. Pin input voltage



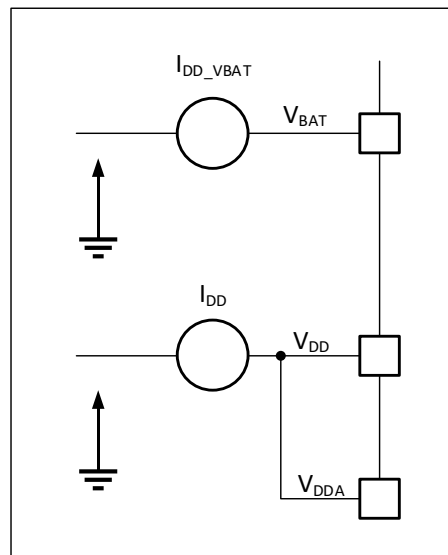
5.1.6 Power supply scheme

Figure 10. Power supply scheme



5.1.7 Current consumption measurement

Figure 11. Current consumption measurement scheme



5.2 Absolute maximum ratings

Stresses above the absolute maximum ratings listed in [Table 6](#), [Table 7](#), and [Table 8](#) may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

Table 6. Voltage characteristics

Symbol	Ratings	Min	Max	Unit
$V_{DD}-V_{SS}$	External main supply voltage (including V_{DDA} and V_{DD}) ⁽¹⁾	-0.3	4.0	V
V_{IN}	Input voltage on five volt tolerant pin ⁽²⁾	$V_{SS}-0.3$	6.0	
	Input voltage on any other pin	$V_{SS}-0.3$	4.0	
$ \Delta V_{DDx} $	Variations between different V_{DD} power pins	-	50	mV
$ V_{SSx}-V_{SS} $	Variations between all the different ground pins ⁽²⁾	-	50	

(1) All main power (V_{DD} , V_{DDA}) and ground (V_{SS} , V_{SSA}) pins must always be connected to the external power supply, in the permitted range.

Table 7. Current characteristics

Symbol	Ratings	Max	Unit
I_{VDD}	Total current into V_{DD}/V_{DDA} power lines (source) ⁽¹⁾	150	mA
I_{VSS}	Total current out of V_{SS} ground lines (sink) ⁽¹⁾	150	
I_{IO}	Output current sunk by any I/O and control pin	25	
	Output current source by any I/Os and control pin	-25	

(1) All main power (V_{DD} , V_{DDA}) and ground (V_{SS} , V_{SSA}) pins must always be connected to the external power supply, in the permitted range.

Table 8. Thermal characteristics

Symbol	Ratings	Value	Unit
T_{STG}	Storage temperature range	-60 ~ +150	°C
T_J	Maximum junction temperature	125	

5.3 Operating conditions

5.3.1 General operating conditions

Table 9. General operating conditions

Symbol	Parameter	Conditions	Min	Max	Unit
f _{HCLK}	Internal AHB clock frequency	-	0	150	MHz
f _{PCLK1}	Internal APB1 clock frequency	-	0	75	
f _{PCLK2}	Internal APB2 clock frequency	-	0	75	
V _{DD}	Standard operating voltage	-	2.6	3.6	V
V _{DDA} ⁽¹⁾	Analog operating voltage	Must be the same potential as V _{DD} ⁽¹⁾	2.6	3.6	V
V _{BAT}	Backup operating voltage	-	1.8	3.6	V
P _D	Power dissipation: T _A = 105 °C	LQFP64 (10 x 10 mm)	-	266	mW
		LQFP64 (7 x 7 mm)	-	249	
		LQFP48	-	260	
		QFN48	-	515	
		QFN32	-	335	
T _A	Ambient temperature	-	-40	105	°C

(1) It is recommended to power V_{DD} and V_{DDA} from the same source. A maximum difference of 300 mV between V_{DD} and V_{DDA} can be tolerated during power-up and operation.

5.3.2 Operating conditions at power-up / power-down

The parameters given in the table below are derived from tests performed under the ambient temperature condition summarized in [Table 9](#).

Table 10. Operating conditions at power-up / power-down

Symbol	Parameter	Conditions	Min	Max	Unit
t _{VDD}	V _{DD} rise time rate	-	0	∞ ⁽¹⁾	ms/V
	V _{DD} fall time rate		20	∞	μs/V

(1) If V_{DD} rising time rate is slower than 6 ms/V, the code should access the backup registers after V_{DD} higher than V_{POR} + 0.1V.

5.3.3 Embedded reset and power control block characteristics

The parameters given in the table below are derived from tests performed under ambient temperature and V_{DD} supply voltage conditions summarized in [Table 9](#).

Table 11. Embedded reset and power control block characteristics

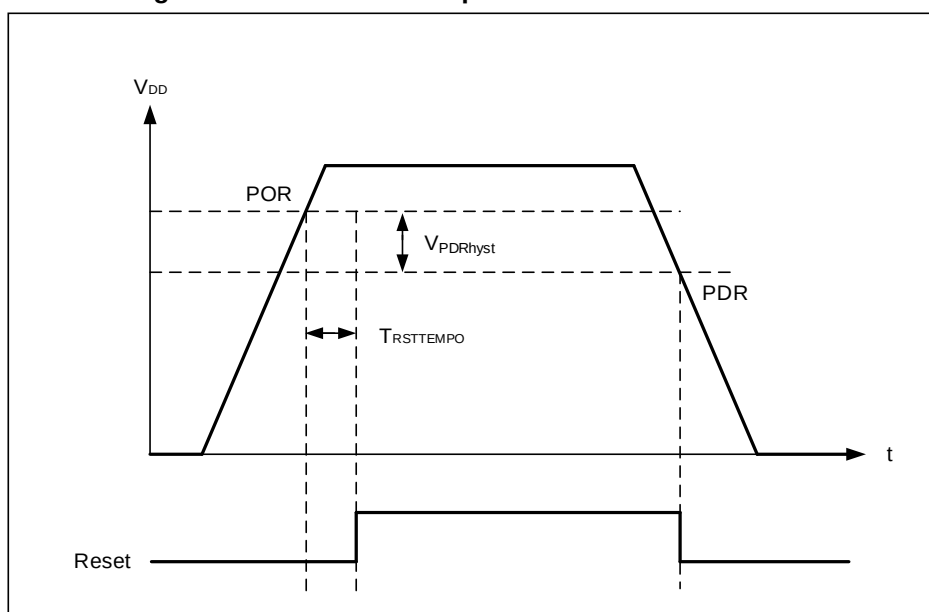
Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{PVD}	Programmable voltage detector level selection	PLS[2:0] = 001 (rising edge) ⁽¹⁾	2.19	2.28	2.37	V
		PLS[2:0] = 001 (falling edge) ⁽¹⁾	2.09	2.18	2.27	V
		PLS[2:0] = 010 (rising edge) ⁽¹⁾	2.28	2.38	2.48	V
		PLS[2:0] = 010 (falling edge)	2.18	2.28	2.38	V
		PLS[2:0] = 011 (rising edge)	2.38	2.48	2.58	V
		PLS[2:0] = 011 (falling edge)	2.28	2.38	2.48	V
		PLS[2:0] = 100 (rising edge)	2.47	2.58	2.69	V
		PLS[2:0] = 100 (falling edge)	2.37	2.48	2.59	V
		PLS[2:0] = 101 (rising edge)	2.57	2.68	2.79	V
		PLS[2:0] = 101 (falling edge)	2.47	2.58	2.69	V
		PLS[2:0] = 110 (rising edge)	2.66	2.78	2.9	V
		PLS[2:0] = 110 (falling edge)	2.56	2.68	2.8	V
		PLS[2:0] = 111 (rising edge)	2.76	2.88	3	V
		PLS[2:0] = 111 (falling edge)	2.66	2.78	2.9	V
$V_{PVDhyst}^{(2)}$	PVD hysteresis	-	-	100	-	mV
$V_{POR/PDR}^{(3)}$	Power on/power down reset threshold	Falling edge	1.85	2.15	2.35	V
		Rising edge	2.05	2.3	2.5	V
$V_{PDRhyst}^{(2)}$	PDR hysteresis	-	-	180	-	mV
$T_{RSTTEMPO}^{(2)}$	Reset temporization: CPU starts execution after V_{DD} keeps higher than V_{POR} for $T_{RSTTEMPO}$	-	-	600	-	μs

(1) PLS[2:0] = 001, 010 may be not available for its voltage detector level may be lower than $V_{POR/PDR}$.

(2) Guaranteed by design, not tested in production.

(3) The product behavior is guaranteed by design down to the minimum $V_{POR/PDR}$ value.

Figure 12. Power on reset/power down reset waveform



5.3.4 Embedded reference voltage

The parameters given in the table below are derived from tests performed under ambient temperature and V_{DD} supply voltage conditions summarized in [Table 9](#).

Table 12. Embedded internal reference voltage

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{REFINT}	Internal reference voltage	-	1.16	1.20	1.24	V
$T_{S_vrefint}^{(1)}$	ADC sampling time when reading the internal reference voltage	-	-	5.1	17.1 ⁽²⁾	μs
$T_{Coeff}^{(2)}$	Temperature coefficient	-	-	-	100	ppm/°C

(1) Shortest sampling time can be determined in the application by multiple iterations.

(2) Guaranteed by design, not tested in production.

5.3.5 Supply current characteristics

The current consumption is a function of several parameters and factors such as the operating voltage, ambient temperature, I/O pin loading, device software configuration, operating frequencies, I/O pin switching rate, and executed binary code.

The current consumption is measured as described in [Figure 11](#).

Typical current consumption

The MCU is placed under the following conditions:

- All I/O pins are in analog mode
- The Flash memory access time is adjusted to the f_{HCLK} frequency (0 wait state from 0 to 32 MHz, 1 wait state from 33 to 64 MHz, 2 wait state from 65 to 96 MHz, 3 wait state from 97 to 128 MHz, and 4 wait states above)
- Prefetch in ON (reminder: this bit must be set before clock setting and bus prescaling)
- Ambient temperature and V_{DD} supply voltage conditions summarized in [Table 9](#).
- When the peripherals are enabled:
 - $f_{PCLK1} = f_{HCLK}/2$, $f_{PCLK2} = f_{HCLK}/2$, $f_{ADCCLK} = f_{PCLK2}/4$ if $f_{HCLK} > 72$ MHz
 - $f_{PCLK1} = f_{HCLK}$, $f_{PCLK2} = f_{HCLK}$, $f_{ADCCLK} = f_{PCLK2}/4$ if $f_{HCLK} \leq 72$ MHz

Table 13. Typical current consumption in Run mode

Symbol	Parameter	Conditions	f_{HCLK}	Typ ⁽¹⁾		Unit
				All peripherals enabled	All peripherals disabled	
I_{DD}	Supply current in Run mode	External clock ⁽²⁾	150 MHz	43.5	20.1	mA
			120 MHz	36.2	17.6	
			108 MHz	32.1	15.3	
			72 MHz	24.6	11.4	
			48 MHz	17.6	8.8	
			36 MHz	13.1	6.54	
			24 MHz	9.62	5.24	
			16 MHz	6.98	4.06	
			8 MHz	4.13	2.79	
			4 MHz	2.98	2.32	
			2 MHz	2.41	2.09	
			1 MHz	2.13	1.97	
			500 kHz	1.99	1.91	
			125 kHz	1.88	1.87	
		Running on high speed internal RC (HSI)	150 MHz	43.5	20.0	mA
			120 MHz	35.5	16.7	
			108 MHz	32.1	15.2	
			72 MHz	24.0	10.8	
			48 MHz	16.9	8.06	
			36 MHz	13.0	6.44	
			24 MHz	9.52	5.13	
			16 MHz	6.88	3.96	
			8 MHz	3.84	2.49	
			4 MHz	2.68	2.02	
			2 MHz	2.11	1.79	
			1 MHz	1.83	1.67	
		Running on high speed internal RC (HSI)	500 kHz	1.69	1.61	mA
			125 kHz	1.59	1.57	

(1) Typical values are measured at $T_A = 25^\circ\text{C}$, $V_{DD} = 3.3\text{ V}$.

(2) External clock is 8 MHz and PLL is on when $f_{HCLK} > 8\text{ MHz}$.

Table 14. Typical current consumption in Sleep mode

Symbol	Parameter	Conditions	f_{HCLK}	Typ ⁽¹⁾		Unit
				All peripherals enabled	All peripherals disabled	
I_{DD}	Supply current in Sleep mode	External clock ⁽²⁾	200 MHz	64.0	15.6	mA
			144 MHz	46.7	11.8	
			100 MHz	33.0	8.74	
			72 MHz	24.7	7.24	
			48 MHz	17.1	5.46	
			36 MHz	13.3	4.58	
			24 MHz	9.46	3.71	
			16 MHz	6.95	3.13	
			8 MHz	3.97	2.08	
			4 MHz	2.81	1.86	
			2 MHz	2.23	1.76	
			1 MHz	1.94	1.70	
			500 kHz	1.79	1.67	
			125 kHz	1.68	1.65	
		Running on high speed internal RC (HSI)	200 MHz	63.9	15.3	mA
			144 MHz	46.6	11.5	
			100 MHz	32.8	8.51	
			72 MHz	24.5	6.99	
			48 MHz	16.9	5.21	
			36 MHz	13.0	4.33	
			24 MHz	9.22	3.45	
			16 MHz	6.70	2.87	
			8 MHz	3.72	1.82	
			4 MHz	2.55	1.60	
			2 MHz	1.97	1.50	
			1 MHz	1.68	1.44	
			500 kHz	1.53	1.42	
			125 kHz	1.42	1.40	

(1) Typical values are measured at $T_A = 25\text{ }^{\circ}\text{C}$, $V_{DD} = 3.3\text{ V}$.

(2) External clock is 8 MHz and PLL is on when $f_{HCLK} > 8\text{ MHz}$.

Maximum current consumption

The MCU is placed under the following conditions:

- All I/O pins are in analog mode
- The Flash memory access time is adjusted to the f_{HCLK} frequency (0 wait state from 0 to 32 MHz, 1 wait state from 33 to 64 MHz, 2 wait state from 65 to 96 MHz, 3 wait state from 97 to 128 MHz, and 4 wait states above)
- Prefetch in ON (reminder: this bit must be set before clock setting and bus prescaling)
- When the peripherals are enabled:
 - $f_{PCLK1} = f_{HCLK}/2$, $f_{PCLK2} = f_{HCLK}/2$ if $f_{HCLK} > 72$ MHz
 - $f_{PCLK1} = f_{HCLK}$, $f_{PCLK2} = f_{HCLK}$ if $f_{HCLK} \leq 72$ MHz

The parameters given in [Table 15](#) and [Table 16](#) are derived from tests performed under ambient temperature and V_{DD} supply voltage conditions summarized in [Table 9](#).

Table 15. Maximum current consumption in Run mode

Symbol	Parameter	Conditions	f_{HCLK}	Max ⁽¹⁾	Unit
				$T_A = 105^\circ\text{C}$	
I_{DD}	Supply current in Run mode	External clock ⁽²⁾ , all peripherals enabled	150 MHz	55.6	mA
			120 MHz	48.4	
			108 MHz	44.0	
			72 MHz	36.1	
			48 MHz	28.8	
			36 MHz	24.1	
			24 MHz	20.5	
			16 MHz	17.7	
			8 MHz	14.7	
		External clock ⁽²⁾ , all peripherals disabled	150 MHz	31.1	mA
			120 MHz	28.7	
			108 MHz	26.3	
			72 MHz	22.3	
			48 MHz	19.5	
			36 MHz	17.2	
			24 MHz	15.8	
			16 MHz	14.6	
			8 MHz	13.4	

(1) Guaranteed by characterization results, not tested in production.

(2) External clock is 8 MHz and PLL is on when $f_{HCLK} > 8$ MHz.

Table 16. Maximum current consumption in Sleep mode

Symbol	Parameter	Conditions	f_{HCLK}	Max ⁽¹⁾	Unit
				$T_A = 105\text{ }^{\circ}\text{C}$	
I_{DD}	Supply current in Sleep mode	External clock ⁽²⁾ , all peripherals enabled	150 MHz	46.1	mA
			120 MHz	39.7	
			108 MHz	37.0	
			72 MHz	30.9	
			48 MHz	24.9	
			36 MHz	21.7	
			24 MHz	18.8	
			16 MHz	16.5	
			8 MHz	13.8	
		External clock ⁽²⁾ , all peripherals disabled	150 MHz	16.5	mA
			120 MHz	16.0	
			108 MHz	15.6	
			72 MHz	14.6	
			48 MHz	14.1	
			36 MHz	13.5	
			24 MHz	13.4	
			16 MHz	12.9	
			8 MHz	12.1	

(1) Guaranteed by characterization results, not tested in production.

(2) External clock is 8 MHz and PLL is on when $f_{HCLK} > 8\text{ MHz}$.

Table 17. Typical and maximum current consumptions in Stop and Standby modes

Symbol	Parameter	Conditions	Typ ⁽¹⁾		Max ⁽²⁾		Unit
			V_{DD}/V_{BAT} = 2.6 V	V_{DD}/V_{BAT} = 3.3 V	$T_A =$ 85 °C	$T_A =$ 105 °C	
I_{DD}	Supply current in Stop mode	Regulator in run mode, low-speed and high-speed internal RC oscillators and high-speed oscillator OFF (no independent watchdog)	735	740	4000	6600	μA
		Regulator in low-power mode, low-speed and high-speed internal RC oscillators and high-speed oscillator OFF (no independent watchdog)	675	680	3480	6000	
	Supply current in Standby mode	Low-speed oscillator and ERTC OFF	2.5	3.6	7.0	10.3	
		Low-speed oscillator and ERTC ON	4.3	6.6	10.0	13.7	

(1) Typical values are measured at $T_A = 25\text{ }^{\circ}\text{C}$.

(2) Guaranteed by characterization results, not tested in production.

Figure 13. Typical current consumption in Stop mode with regulator in run mode vs. temperature at different V_{DD}

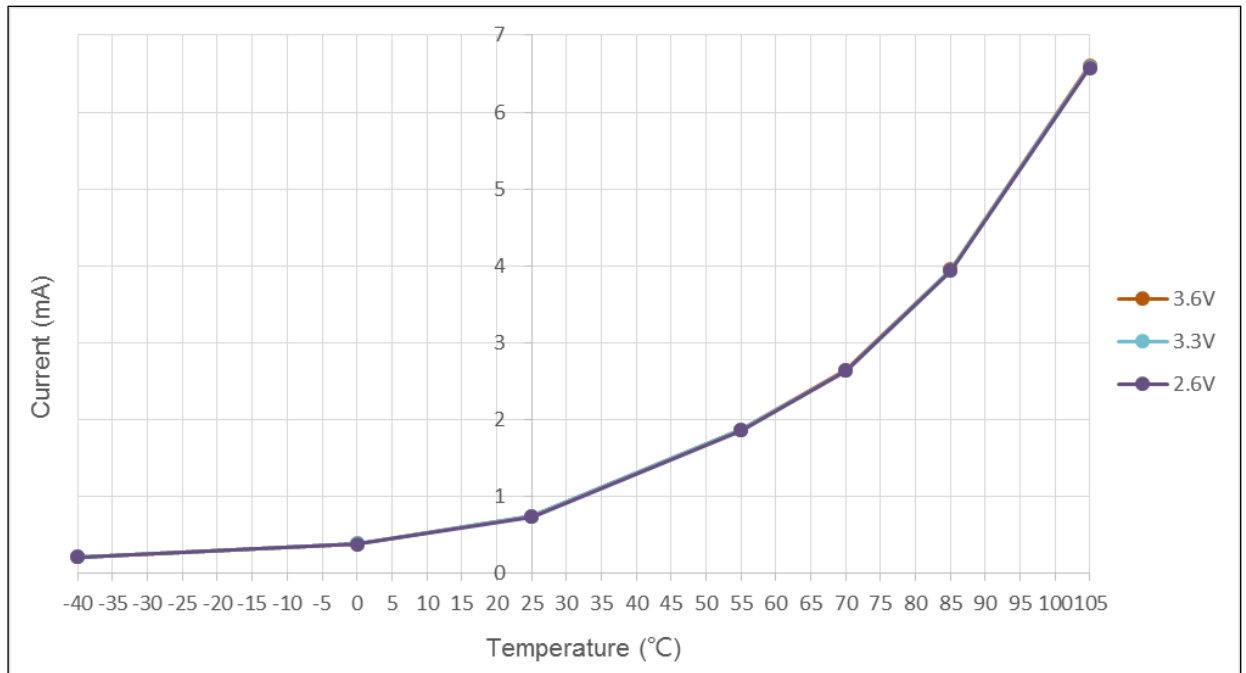


Figure 14. Typical current consumption in Stop mode with regulator with regulator in low-power mode vs. temperature at different V_{DD}

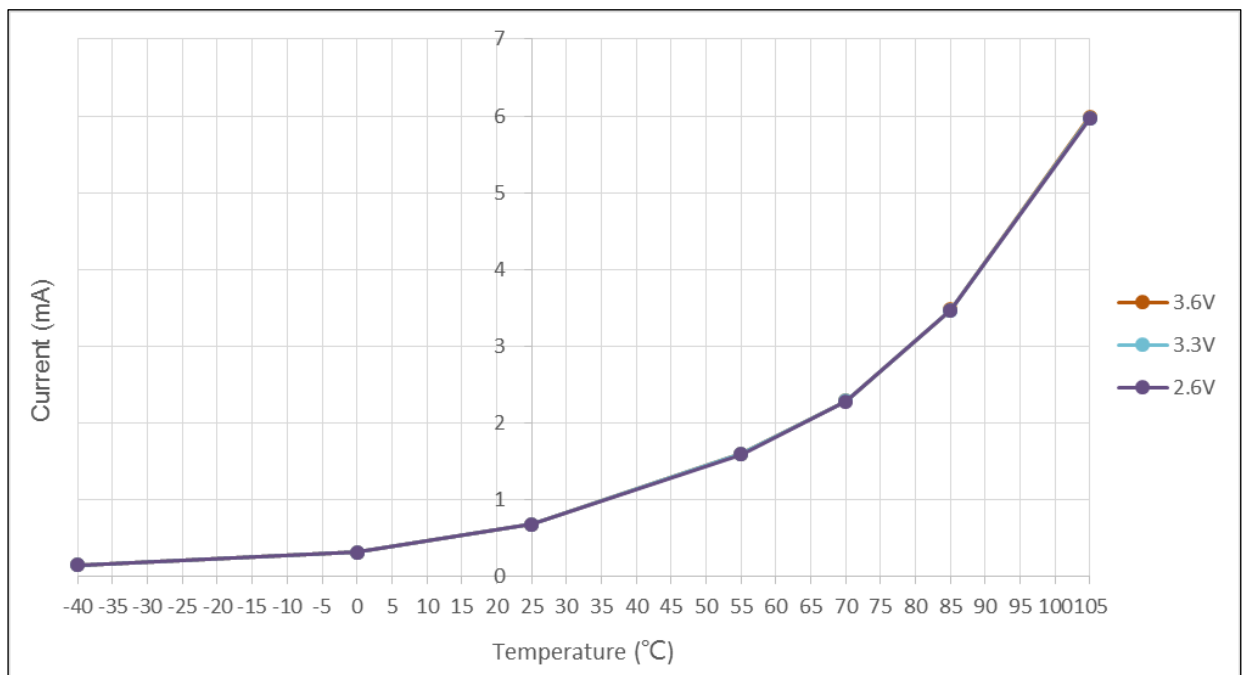


Figure 15. Typical current consumption in Standby mode vs. temperature at different V_{DD}

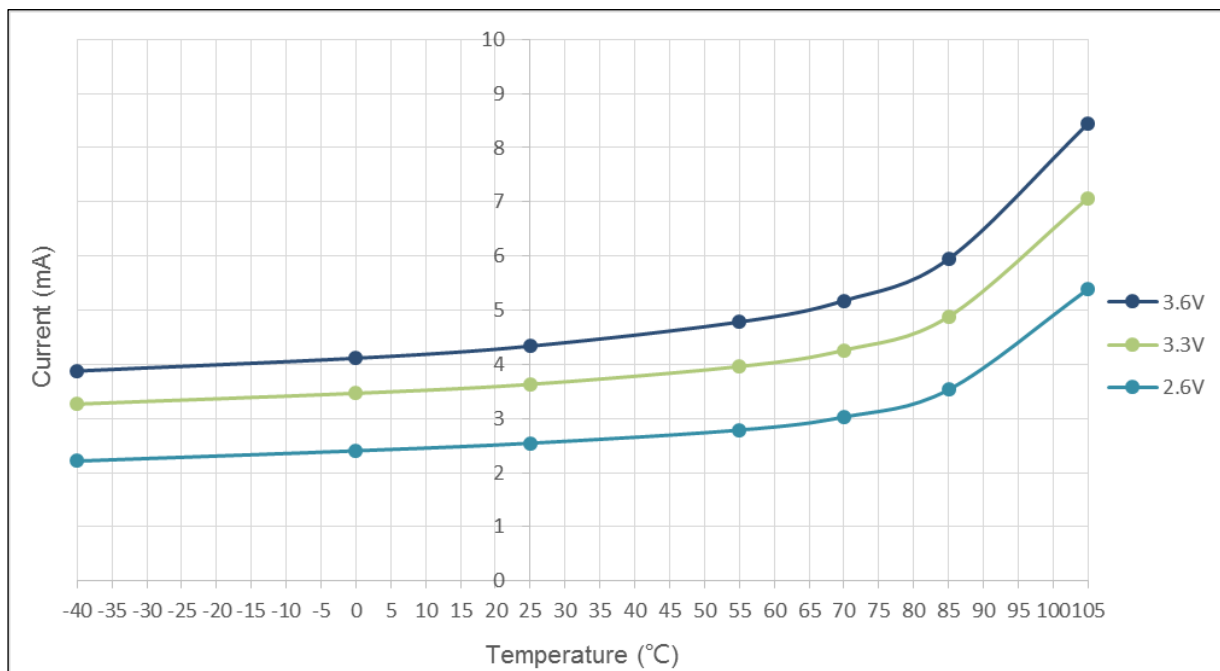


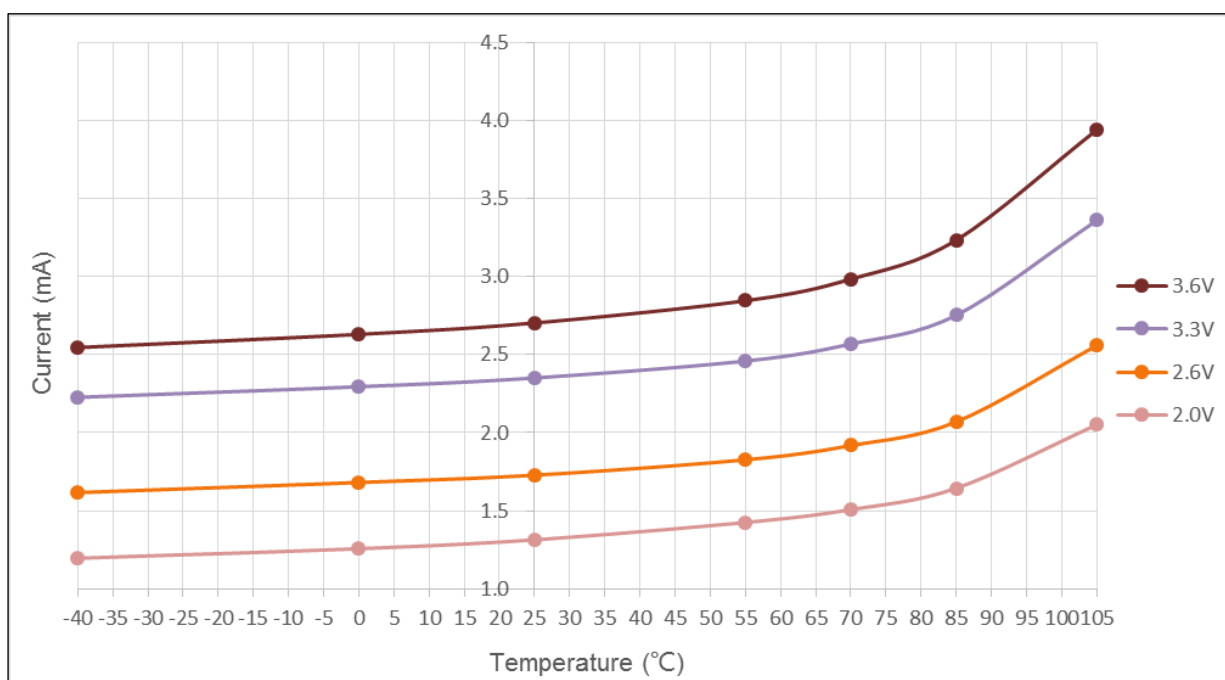
Table 18. Typical and maximum current consumptions on V_{BAT} with LSE and ERTC on

Symbol	Parameter	Conditions	Typ ⁽¹⁾			Max ⁽²⁾		Unit
			$V_{BAT} = 2.0\text{ V}$	$V_{BAT} = 2.6\text{ V}$	$V_{BAT} = 3.3\text{ V}$	$T_A = 85\text{ °C}$	$T_A = 105\text{ °C}$	
I_{DD_VBAT}	Backup domain supply current	Low-speed oscillator and RTC ON, $V_{DD} < V_{PDR}$	1.3	1.7	2.4	3.7	4.6	μA

(1) Typical values are measured at $T_A = 25\text{ °C}$.

(2) Guaranteed by characterization results, not tested in production.

Figure 16. Typical current consumption on V_{BAT} with LSE and ERTC on vs. temperature at different V_{BAT} values



On-chip peripheral current consumption

The current consumption of the on-chip peripherals is given in [Table 19](#). The MCU is placed under the following conditions:

- All I/O pins are in analog mode
- The given value is calculated by measuring the current consumption
 - with all peripherals clocked off
 - with only one peripheral clocked on
- Ambient temperature and V_{DD} supply voltage conditions summarized in [Table 9](#).

Table 19. Peripheral current consumption

Peripheral		Typ	Unit
AHB (up to 150 MHz)	DMA1	9.32	$\mu\text{A}/\text{MHz}$
	DMA2	9.41	
	GPIOA	1.25	
	GPIOB	1.33	
	GPIOC	1.27	
	GIOD	1.23	
	GPIOF	1.24	
	CRC	1.64	
	SDIO	19.3	
	USB OTG FS	46.3	
APB1 (up to 75 MHz)	TMR2	8.96	
	TMR3	6.76	
	TMR4	6.73	
	TMR5	8.97	
	SPI2/I ² S2	2.84	
	USART2	2.40	
	USART3	2.53	
	UART4	2.46	
	UART5	2.68	
	I ² C1	2.66	
	I ² C2	2.53	
	CAN	3.56	
	WWDG	0.45	
	PWR	0.38	
	COMP	0.81	
APB2 (up to 75 MHz)	AFIO	2.53	$\mu\text{A}/\text{MHz}$
	SPI1/I ² S1	2.75	
	USART1	2.48	
	TMR1	8.74	
	TMR9	4.03	
	TMR10	2.56	
	TMR11	2.60	

Peripheral		Typ	Unit
APB2 (up to 75 MHz)	ADC1	6.92	$\mu\text{A}/\text{MHz}$
	ACC	0.99	

5.3.6 External clock source characteristics

High-speed external user clock generated from an external source

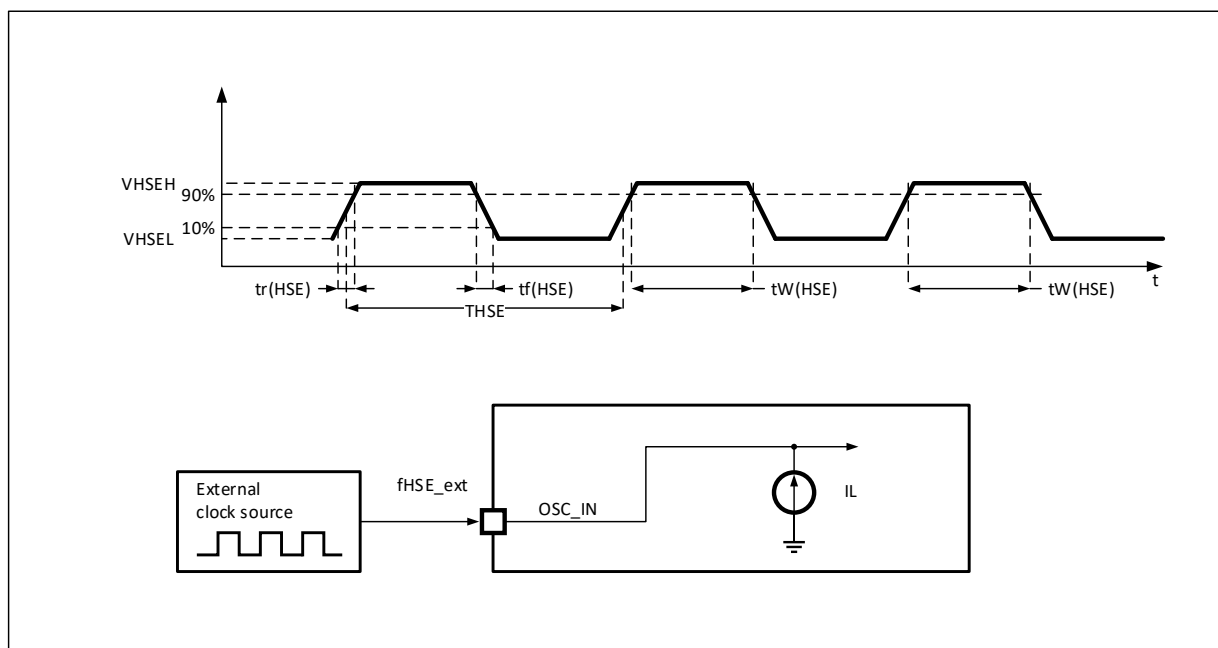
The characteristics given in the table below result from tests performed using a high-speed external clock source, and under ambient temperature and supply voltage conditions summarized in [Table 9](#).

Table 20. High-speed external user clock characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$f_{\text{HSE_ext}}$	User external clock source frequency ⁽¹⁾	-	1	8	25	MHz
V_{HSEH}	OSC_IN input pin high level voltage		$0.7V_{\text{DD}}$	-	V_{DD}	V
V_{HSEL}	OSC_IN input pin low level voltage		V_{SS}	-	$0.3V_{\text{DD}}$	
$t_{\text{w(HSE)}}$ $t_{\text{w(HSE)}}$	OSC_IN high or low time ⁽¹⁾		5	-	-	ns
$t_{\text{r(HSE)}}$ $t_{\text{f(HSE)}}$	OSC_IN rise or fall time ⁽¹⁾		-	-	20	
$C_{\text{in(HSE)}}$	OSC_IN input capacitance ⁽¹⁾	-	-	5	-	pF
$\text{DuCy}_{(\text{HSE})}$	Duty cycle	-	45	-	55	%
I_{L}	OSC_IN Input leakage current	$V_{\text{SS}} \leq V_{\text{IN}} \leq V_{\text{DD}}$	-	-	± 1	μA

(1) Guaranteed by design, not tested in production.

Figure 17. High-speed external clock source AC timing diagram



Low-speed external user clock generated from an external source

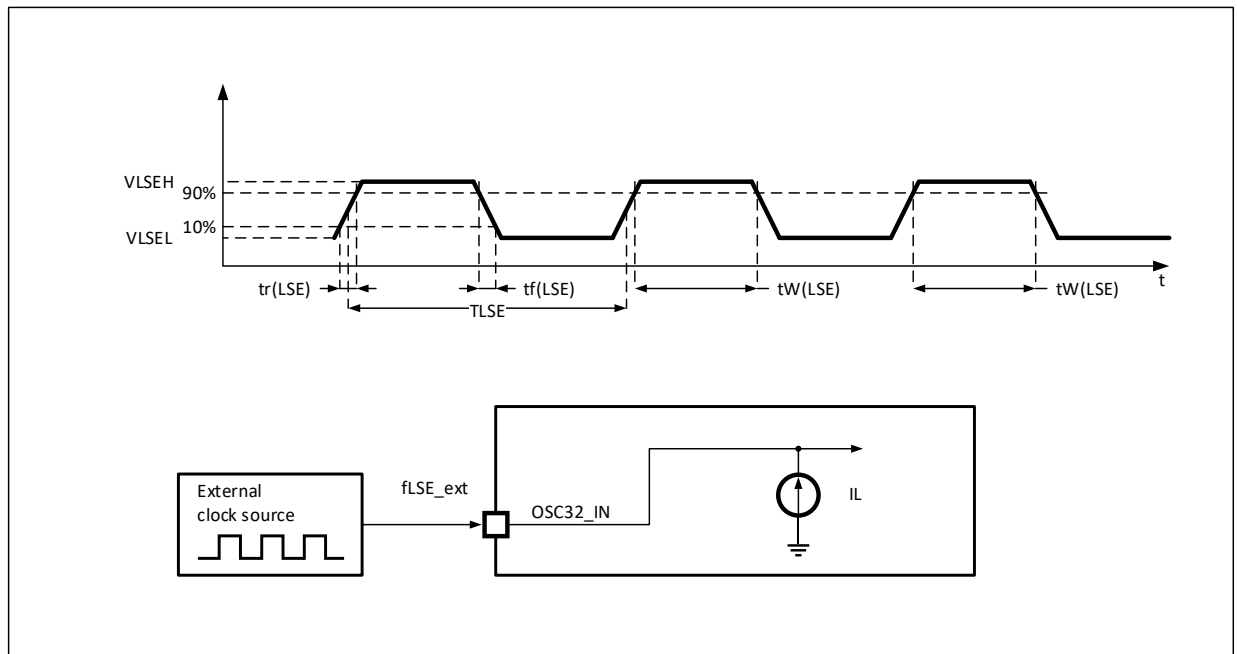
The characteristics given in the table below result from tests performed using a low-speed external clock source, and under ambient temperature and supply voltage conditions summarized in [Table 9](#).

Table 21. Low-speed external user clock characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{LSE_ext}	User External clock source frequency ⁽¹⁾		-	32.768	1000	kHz
V_{LSEH}	OSC32_IN input pin high level voltage	-	$0.7V_{DD}$	-	V_{DD}	V
V_{LSEL}	OSC32_IN input pin low level voltage		V_{SS}	-	$0.3V_{DD}$	
$t_{w(LSE)}$ $t_{w(LSE)}$	OSC32_IN high or low time ⁽¹⁾		450	-	-	ns
$t_{r(LSE)}$ $t_{f(LSE)}$	OSC32_IN rise or fall time ⁽¹⁾		-	-	50	
$C_{in(LSE)}$	OSC32_IN input capacitance ⁽¹⁾	-	-	5	-	pF
$DuCy_{(LSE)}$	Duty cycle	-	30	-	70	%
I_L	OSC32_IN input leakage current	$V_{SS} \leq V_{IN} \leq V_{DD}$	-	-	± 1	μA

(1) Guaranteed by design, not tested in production.

Figure 18. Low-speed external clock source AC timing diagram



High-speed external clock generated from a crystal/ceramic resonator

The high-speed external (HSE) clock can be supplied with a 4 to 25 MHz crystal/ceramic resonator oscillator. All the information given in this paragraph are based on characterization results obtained with typical external components specified in the table below. In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics (frequency, package, accuracy).

Table 22. HSE 4 to 25 MHz oscillator characteristics⁽¹⁾⁽²⁾

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f _{OSC_IN}	Oscillator frequency	-	4	8	25	MHz
t _{SU(HSE)} ⁽³⁾	Startup time	V _{DD} is stabilized	-	800	-	μs

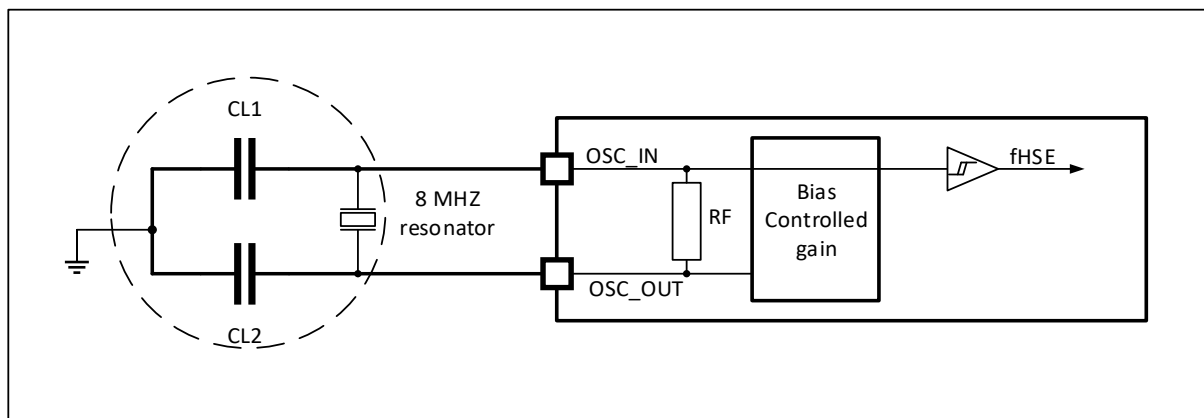
(1) Resonator characteristics given by the crystal/ceramic resonator manufacturer.

(2) Guaranteed by characterization results, not tested in production.

(3) t_{SU(HSE)} is the startup time measured from the moment it is enabled (by software) to a stabilized 8 MHz oscillation is reached. This value is measured for a standard crystal resonator and it can vary significantly with the crystal manufacturer

For C_{L1}和C_{L2}, it is recommended to use high-quality external ceramic capacitors in the 5 pF to 25 pF range (typ.), designed for high-frequency applications, and selected to match the requirements of the crystal or resonator. C_{L1} and C_{L2} are usually the same size. The crystal manufacturer typically specifies a load capacitance which is the series combination of C_{L1} and C_{L2}. PCB and MCU pin capacitance must be included (10 pF can be used as a rough estimate of the combined pin and board capacitance) when sizing C_{L1} and C_{L2}.

Figure 19. Typical application with an 8 MHz crystal



Low-speed external clock generated from a crystal/ceramic resonator

The low-speed external (LSE) clock can be supplied with a 32.768 kHz crystal/ceramic resonator oscillator. All the information given in this paragraph are based on characterization results obtained with typical external components specified in the table below. In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics (frequency, package, accuracy).

Table 23. LSE oscillator characteristics ($f_{LSE} = 32.768 \text{ kHz}$)⁽¹⁾

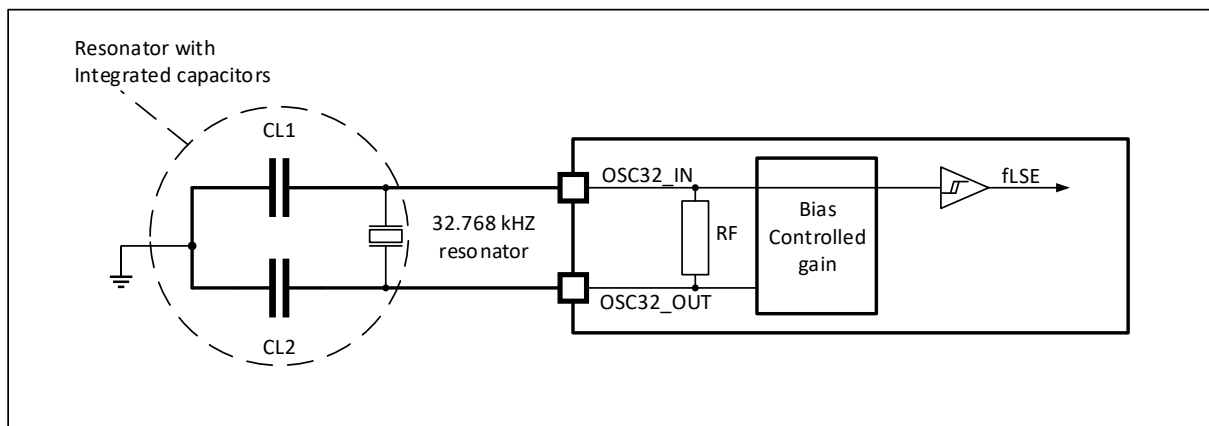
Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$t_{SU(LSE)}$	Startup time	V_{DD} is stabilized	-	200	-	ms

(1) Guaranteed by characterization results, not tested in production.

For C_{L1} and C_{L2} , it is recommended to use high-quality ceramic capacitors in the 5 pF to 15 pF range selected to match the requirements of the crystal or resonator. C_{L1} and C_{L2} , are usually the same size. The crystal manufacturer typically specifies a load capacitance which is the series combination of C_{L1} and C_{L2} .

Load capacitance C_L has the following formula: $C_L = C_{L1} \times C_{L2} / (C_{L1} + C_{L2}) + C_{stray}$ where C_{stray} is the pin capacitance and board or trace PCB-related capacitance. Typically, it is between 2 pF and 7 pF.

Figure 20. Typical application with a 32.768 kHz crystal



5.3.7 Internal clock source characteristics

The parameters given in the table below are derived from tests performed under ambient temperature and V_{DD} supply voltage conditions summarized in [Table 9](#).

High-speed internal (HSI) RC oscillator

Table 24. HSI oscillator characteristics⁽¹⁾

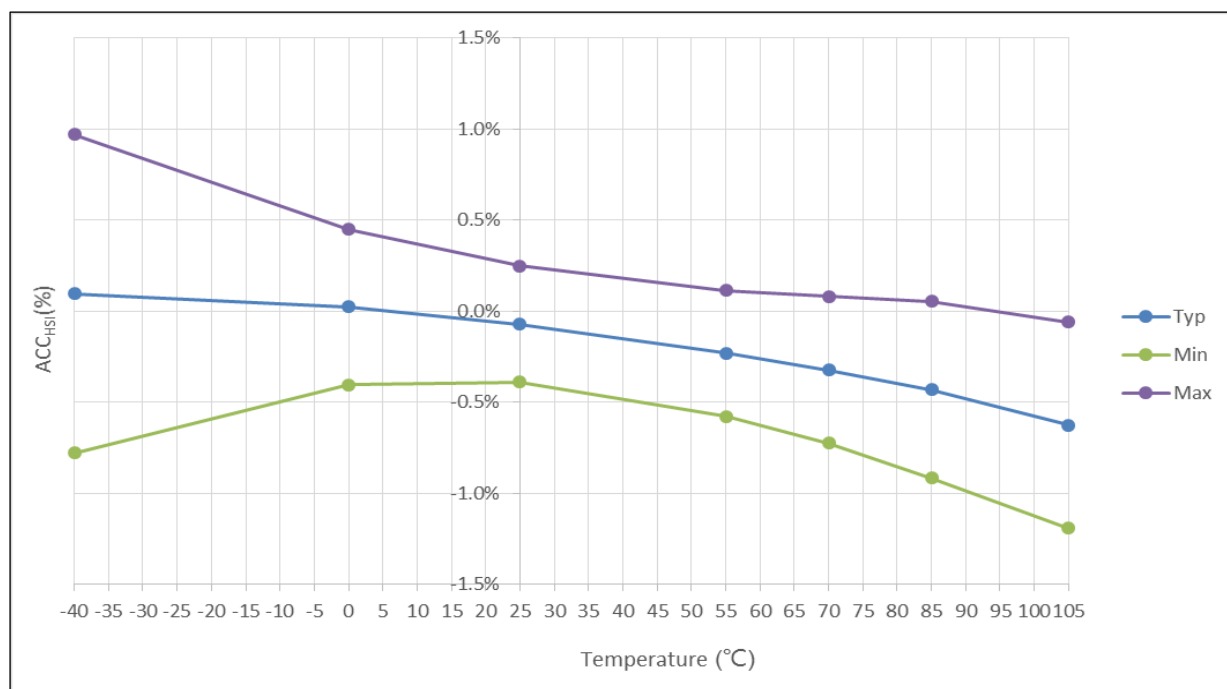
Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{HSI}	Frequency	-	-	48	-	MHz
$DuCy_{(HSI)}$	Duty cycle	-	45	-	55	%
ACC_{HSI}	Accuracy of the HSI oscillator	User-trimmed with the RCC_CTRL register	-	-	1 ⁽²⁾	%
		User-trimmed with ACC	-	-	0.25 ⁽²⁾	
		Factory-calibrated ⁽³⁾	$T_A = -40 \sim 105\text{ }^{\circ}\text{C}$	-2	1.5	%
			$T_A = -40 \sim 85\text{ }^{\circ}\text{C}$	-1.5	1.5	%
			$T_A = 25\text{ }^{\circ}\text{C}$	-1	1	%
$t_{SU(HSI)}^{(3)}$	HSI oscillator startup time	-	-	-	10	μs
$I_{DD(HSI)}^{(3)}$	HSI oscillator power consumption	-	-	200	215	μA

(1) $V_{DD} = 3.3\text{ V}$, $T_A = -40 \sim 105\text{ }^{\circ}\text{C}$, unless otherwise specified.

(2) Guaranteed by design, not tested in production.

(3) Guaranteed by characterization results, not tested in production.

Figure 21. HSI oscillator frequency accuracy vs. temperature



Low-speed internal (LSI) RC oscillator

Table 25. LSI oscillator characteristics⁽¹⁾

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$f_{LSI}^{(2)}$	Frequency	-	30	40	60	kHz

(1) $V_{DD} = 3.3\text{ V}$, $T_A = -40 \sim 105\text{ }^{\circ}\text{C}$, unless otherwise specified.

(2) Guaranteed by characterization results, not tested in production.

5.3.8 Wakeup time from low-power mode

The wakeup times given in the table below is measured on a wakeup phase with an 8 MHz HSI RC oscillator. The clock source used to wake up the device depends from the current operating mode:

- Stop or Standby mode: the clock source is the HSI RC oscillator
- Sleep mode: the clock source is the clock that was set before entering Sleep mode

All timings are derived from tests performed under ambient temperature and V_{DD} supply voltage conditions summarized in [Table 9](#).

Table 26. Low-power mode wakeup timings

Symbol	Parameter	Typ	Unit
$t_{WUSLEEP}^{(1)}$	Wakeup from Sleep mode	4.2	μs
$t_{WUSTOP}^{(1)}$	Wakeup from Stop mode (regulator in run mode)	300	μs
	Wakeup from Stop mode (regulator in low-power mode)	360	
$t_{WUSTDBY}^{(1)}$	Wakeup from Standby mode	600	μs

(1) The wakeup times are measured from the wakeup event to the point in which the user application code reads the first instruction.

5.3.9 PLL characteristics

The parameters given in the table below are derived from tests performed under ambient temperature and V_{DD} supply voltage conditions summarized in [Table 9](#).

Table 27. PLL characteristics

Symbol	Parameter	Min	Typ	Max ⁽¹⁾	Unit
f_{PLL_IN}	PLL input clock ⁽²⁾	2	8	16	MHz
	PLL input clock duty cycle	40	-	60	%
f_{PLL_OUT}	PLL multiplier output clock	16	-	200	MHz
t_{LOCK}	PLL lock time	-	-	200	μs
Jitter	Cycle-to-cycle jitter	-	-	300	ps

(1) Guaranteed by characterization results, not tested in production.

(2) Take care of using the appropriate multiplier factors so as to have PLL input clock values compatible with the range defined by f_{PLL_OUT} .

5.3.10 Memory characteristics

The characteristics in [Table 28](#) are given at $T_A = -40 \sim 105\text{ }^{\circ}\text{C}$.

Table 28. Flash memory characteristics

Symbol	Parameter	Conditions	Min	Typ	Max ⁽¹⁾	Unit
T_{PROG}	Programming time	$T_A = -40 \sim 105\text{ }^{\circ}\text{C}$	40	-	42	μs
t_{ERASE}	Page erase time	$T_A = -40 \sim 105\text{ }^{\circ}\text{C}$	6.4	-	8	ms
t_{ME}	Mass erase time	$T_A = -40 \sim 105\text{ }^{\circ}\text{C}$	8	-	10	ms
I_{DD}	Supply current	Programming mode, $V_{DD} = 3.3\text{ V}$, $T_A = 25\text{ }^{\circ}\text{C}$	-	1.69	-	mA
		Erase mode, $V_{DD} = 3.3\text{ V}$, $T_A = 25\text{ }^{\circ}\text{C}$	-	1.82	-	

Table 29. Flash memory endurance and data retention

Symbol	Parameter	Conditions	Min ⁽¹⁾	Typ	Max	Unit
N_{END}	Endurance	$T_A = -40 \sim 105\text{ }^{\circ}\text{C}$	100	-	-	kcycles
t_{RET}	Data retention	$T_A = 105\text{ }^{\circ}\text{C}$	10	-	-	years

(1) Guaranteed by design, not tested in production.

5.3.11 EMC characteristics

Susceptibility tests are performed on a sample basis during device characterization.

Functional EMS (electromagnetic susceptibility)

- **FTB:** A Burst of Fast Transient voltage (positive and negative) is applied to V_{DD} and V_{SS} through a 100 pF capacitor, until a functional disturbance occurs. This test is compliant with the IEC 61000-4-4 standard.

Table 30. EMS characteristics

Symb	Parameter	Conditions	Level/Class
V_{EFTB}	Fast transient voltage burst limits to be applied through 100 pF on V_{DD} and V_{SS} pins to induce a functional disturbance	$V_{DD} = 3.3\text{ V}$, LQFP64, $T_A = +25\text{ }^{\circ}\text{C}$, $f_{HCLK} = 150\text{ MHz}$, conforms to IEC 61000-4-4	4A (4kV)

Designing hardened software to avoid noise problems

EMC characterization and optimization are performed at component level with a typical application environment and simplified MCU software. It should be noted that good EMC performance is highly dependent on the user application and the software in particular.

Therefore it is recommended that the user applies EMC software optimization and prequalification tests in relation with the EMC level requested for his application.

Software recommendations

The software flowchart must include the management of runaway conditions such as:

- Corrupted program counter
- Unexpected reset
- Critical Data corruption (control registers...)

Prequalification trials

Most of the common failures (unexpected reset and program counter corruption) can be reproduced by manually forcing a low state on the NRST pin or the Oscillator pins for 1 second.

5.3.12 Absolute maximum ratings (electrical sensitivity)

Based on three different tests (ESD, LU) using specific measurement methods, the device is stressed in order to determine its performance in terms of electrical sensitivity.

Electrostatic discharge (ESD)

Electrostatic discharges (a positive then a negative pulse separated by 1 second) are applied to the pins of each sample according to each pin combination. The sample size depends on the number of supply pins in the device (3 parts $\times$ (n+1) supply pins). This test conforms to the JS-001-2017/JS-002-2014 standard.

Table 31. ESD absolute maximum ratings

Symbol	Parameter	Conditions	Class	Max ⁽¹⁾	Unit
V _{ESD(HBM)}	Electrostatic discharge voltage (human body model)	T _A = +25 °C, conforming to JS-001-2017	3A	5000	V
V _{ESD(CDM)}	Electrostatic discharge voltage (charge device model)	T _A = +25 °C, conforming to JS-002-2014	III	1000	

(1) Guaranteed by characterization results, not tested in production.

Static latch-up

Two complementary static tests are required on 6 parts to assess the latch-up performance:

- A supply overvoltage is applied to each power supply pin
- A current injection is applied to each input, output and configurable I/O pin

These tests are compliant with EIA/JESD78E IC latch-up standard.

Table 32. Electrical sensitivities

Symbol	Parameter	Conditions	Level/Class
LU	Static latch-up class	T _A = +105 °C, conforming to EIA/JESD78E	II level A (200 mA)

5.3.13 I/O port characteristics

General input/output characteristics

Unless otherwise specified, the parameters given in the table below are derived from tests performed under the conditions summarized in [Table 9](#). All I/Os are CMOS and TTL compliant.

Table 33. I/O static characteristics

Symb	Parameter	Conditions	Min	Typ	Max	Unit
V _{IL}	I/O ⁽¹⁾ input low level voltage	-	-0.3	-	0.28 * V _{DD} + 0.1	V
V _{IH}	Standard I/O input high level voltage	-	0.31 * V _{DD} + 0.8	-	V _{DD} + 0.3	V
	I/O FT ⁽¹⁾ input high level voltage			-	5.5	V
V _{hys}	Standard I/O Schmitt trigger voltage hysteresis ⁽²⁾	-	200	-	-	mV
	I/O FT Schmitt trigger voltage hysteresis ⁽²⁾		5% V _{DD}	-	-	mV
I _{lkg}	Input leakage current ⁽³⁾	V _{SS} ≤ V _{IN} ≤ V _{DD} Standard I/Os ⁽⁵⁾	-	-	±1	μA
		V _{SS} ≤ V _{IN} ≤ 5.5V I/O FT	-	-	±10	
R _{PU}	Weak pull-up equivalent resistor	V _{IN} = V _{SS}	60	75	110	kΩ
R _{PD}	Weak pull-down equivalent resistor ⁽⁴⁾	V _{IN} = V _{DD}	60	80	120	kΩ
C _{IO}	I/O pin capacitance	-	-	5	-	pF

(1) FT = Five-volt tolerant. In order to sustain a voltage higher than V_{DD} + 0.3 the internal pull-up/pull-down resistors must be disabled.

(2) Hysteresis voltage between Schmitt trigger switching levels. Guaranteed by characterization results.

(3) Leakage could be higher than max if negative current is injected on adjacent pins.

(4) The pull-down resistor of BOOT0 exists permanently.

All I/Os are CMOS and TTL compliant (no software configuration required). Their characteristics cover more than the strict CMOS-technology or TTL parameters.

Output driving current

In the user application, the number of I/O pins which can drive current must be limited to respect the absolute maximum rating specified in [Section 0](#):

- The sum of the currents sourced by all I/Os on V_{DD} , plus the maximum Run consumption of the MCU sourced on V_{DD} , cannot exceed the absolute maximum rating I_{VDD} (see [Table 7](#)).
- The sum of the currents sunk by all I/Os on V_{SS} , plus the maximum Run consumption of the MCU sunk on V_{SS} , cannot exceed the absolute maximum rating I_{VSS} (see [Table 7](#)).

Output voltage levels

Unless otherwise specified, the parameters given in the table below are derived from tests performed under ambient temperature and V_{DD} supply voltage conditions summarized in [Table 9](#). All I/Os are CMOS and TTL compliant.

Table 34. Output voltage characteristics

Symbol	Parameter	Conditions	Min	Max	Unit
MDEx[1:0] = 11 (maximum sourcing/sinking stregh)					
V _{OL}	Output low level voltage	CMOS standard, I _{IO} = 15 mA	-	0.4	V
V _{OH}	Output high level voltage		V _{DD} -0.4	-	
V _{OL}	Output low level voltage	TTL standard, I _{IO} = 6 mA	-	0.4	V
V _{OH}	Output high level voltage		2.4	-	
V _{OL} ⁽¹⁾	Output low level voltage	I _{IO} = 36 mA	-	1.3	V
V _{OH} ⁽¹⁾	Output high level voltage		V _{DD} -1.3	-	
MDEx[1:0] = 01 (large sourcing/sinking stregh)					
V _{OL}	Output low level voltage	CMOS standard, I _{IO} = 6 mA	-	0.4	V
V _{OH}	Output high level voltage		V _{DD} -0.4	-	
V _{OL}	Output low level voltage	TTL standard, I _{IO} = 3 mA	-	0.4	V
V _{OH}	Output high level voltage		2.4	-	
V _{OL} ⁽¹⁾	Output low level voltage	I _{IO} = 18 mA	-	1.3	V
V _{OH} ⁽¹⁾	Output high level voltage		V _{DD} -1.3	-	
MDEx[1:0] = 10 (normal sourcing/sinking stregh)					
V _{OL}	Output low level voltage	CMOS standard, I _{IO} = 4 mA	-	0.4	V
V _{OH}	Output high level voltage		V _{DD} -0.4	-	
V _{OL}	Output low level voltage	TTL standard, I _{IO} = 2 mA	-	0.4	V
V _{OH}	Output high level voltage		2.4	-	
V _{OL} ⁽¹⁾	Output low level voltage	I _{IO} = 9 mA	-	1.3	V
V _{OH} ⁽¹⁾	Output high level voltage		V _{DD} -1.3	-	

(1) Guaranteed by characterization results.

Input AC characteristics

The definition and values of input AC characteristics are given as follows.

Unless otherwise specified, the parameters given below are derived from tests performed under the ambient temperature and V_{DD} supply voltage conditions summarized in [Table 9](#).

Table 35. Input AC characteristics

Symbol	Parameter	Min	Max	Unit
$t_{EXTI\text{pw}}$	Pulse width of external signals detected by the EXTI controller	10	-	ns

5.3.14 NRST pin characteristics

The NRST pin input driver uses CMOS technology. It is connected to a permanent pull-up resistor, R_{PU} (see the table below).

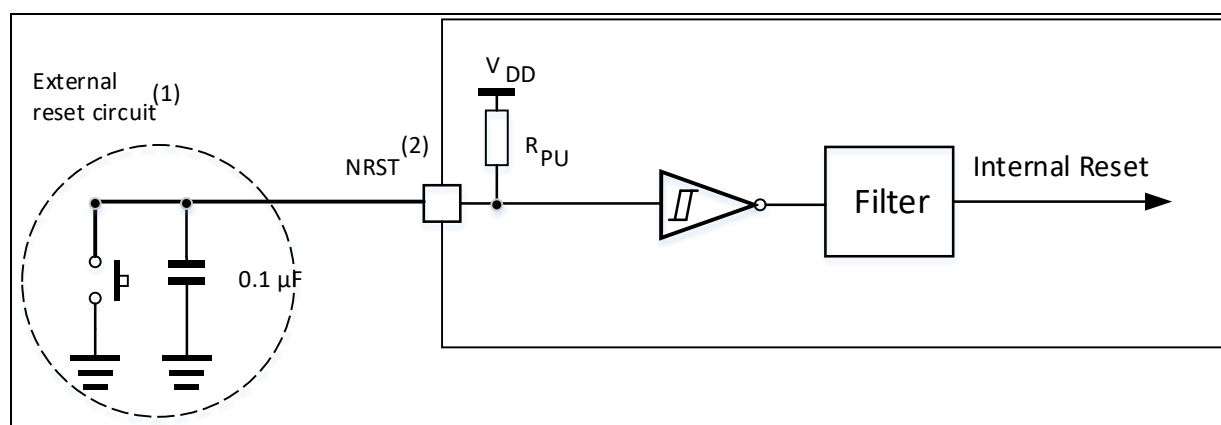
Unless otherwise specified, the parameters given in the table below are derived from tests performed under ambient temperature and V_{DD} supply voltage conditions summarized in [Table 9](#).

Table 36. NRST pin characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$V_{IL(NRST)}^{(1)}$	NRST input low level voltage	-	-0.5	-	0.8	V
$V_{IH(NRST)}^{(1)}$	NRST input high level voltage	-	2	-	$V_{DD} + 0.5$	
$V_{hys(NRST)}$	NRST Schmitt trigger voltage hysteresis	-	-	400	-	mV
R_{PU}	Weak pull-up equivalent resistor	$V_{IN} = V_{SS}$	30	40	50	k Ω
$V_{F(NRST)}^{(1)}$	NRST input filtered pulse	-	-	28	33.3	μ s
$V_{NF(NRST)}^{(1)}$	NRST input not filtered pulse	-	66.7	52	-	μ s

(1) Guaranteed by design.

Figure 22. Recommended NRST pin protection



- (1) The reset network protects the device against parasitic resets.
- (2) The user must ensure that the level on the NRST pin can go below the $V_{IL(NRST)}$ max level specified in unless otherwise specified, the parameters given in the table below are derived from tests performed under ambient temperature and V_{DD} supply voltage conditions summarized in [Table 36](#). Otherwise the reset will not be taken into account by the device.

5.3.15 TMR timer characteristics

The parameters given in the table below are guaranteed by design.

Refer to [5.3.13 I/O port characteristics](#) for details on the input/output alternate function characteristics (output compare, input capture, external clock, PWM output).

Table 37. TMRx⁽¹⁾ characteristics

Symbol	Parameter	Conditions	Min	Max	Unit
$t_{res(TMR)}$	Timer resolution time	-	1	-	$t_{TMRxCLK}$
		$f_{TMRxCLK} = 150 \text{ MHz}$	6.7	-	ns
f_{EXT}	Timer external clock frequency on CH1 to CH4	-	0	$f_{TMRxCLK}/2$	MHz
				50	MHz

(1) TMRx is used as a general term to refer to the TMR1 through TMR7 and TRM9 through TMR11.

5.3.16 Communications interfaces

I²C interface characteristics

The AT32F415 I²C interface meets the requirements of the standard I²C communication protocol with the following restrictions: the I/O pins SDA and SCL mapped to are not "true" open-drain. When configured as open-drain, the PMOS connected between the I/O pin and V_{DD} is disabled, but is still present.

The I²C characteristics are described in the table below. Refer also to [5.3.13 I/O port characteristics](#) for more details on the input/output alternate function characteristics (SDA and SCL).

Table 38. I²C characteristics

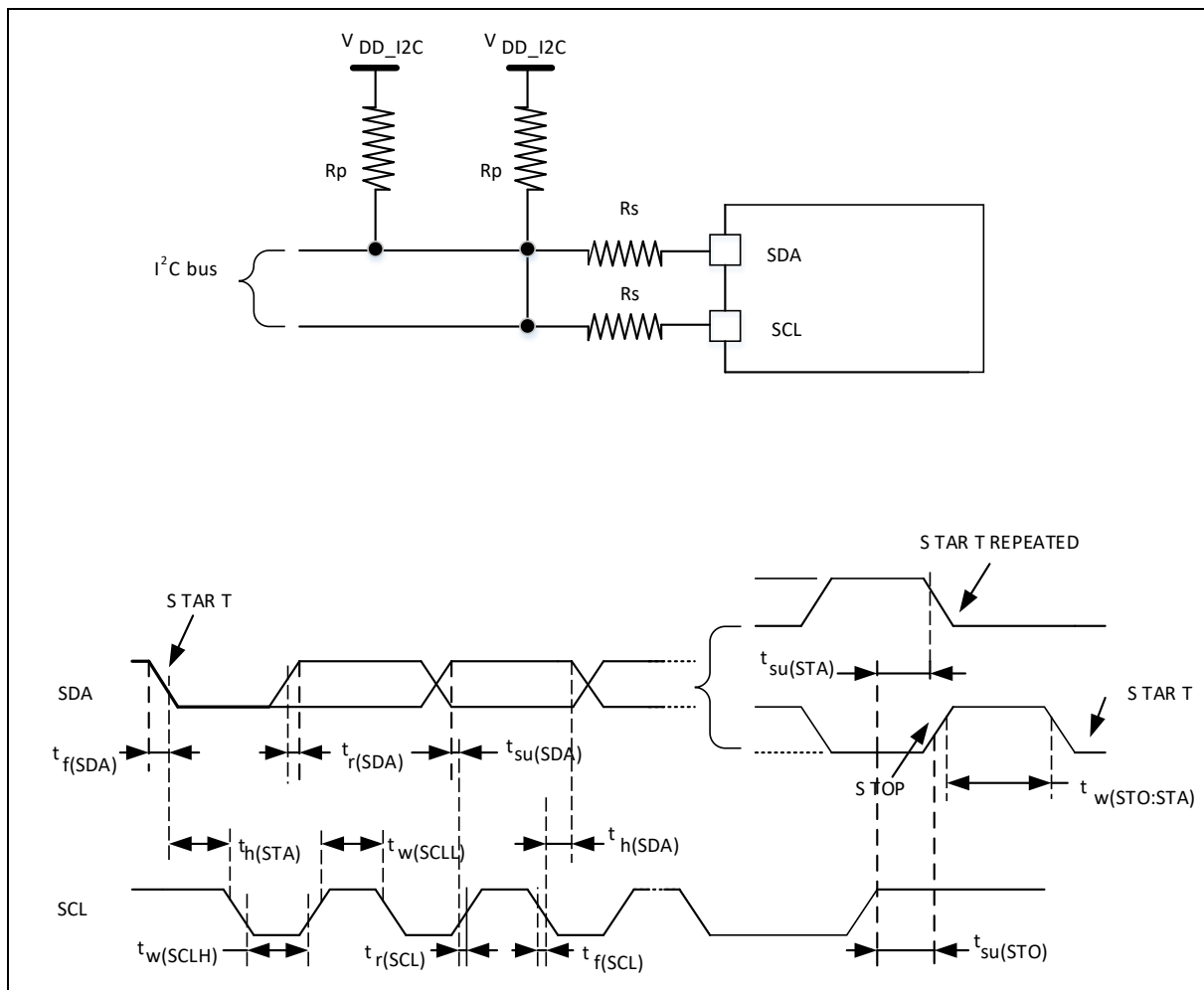
Symbol	Parameter	Standard mode I ² C ⁽¹⁾⁽²⁾		Fast mode I ² C ⁽¹⁾⁽²⁾		Unit
		Min	Max	Min	Max	
t _w (SCLL)	SCL clock low time	4.7	-	1.3	-	μs
t _w (SCLH)	SCL clock high time	4.0	-	0.6	-	
t _{su} (SDA)	SDA setup time	250	-	100	-	ns
t _h (SDA)	SDA data hold time	-	3450 ⁽³⁾	-	900 ⁽³⁾	
t _r (SDA) t _r (SCL)	SDA and SCL rise time	-	1000	-	300	
t _f (SDA) t _f (SCL)	SDA and SCL fall time	-	300	-	300	
t _h (STA)	Start condition hold time	4.0	-	0.6	-	μs
t _{su} (STA)	Repeated Start condition setup time	4.7	-	0.6	-	
t _{su} (STO)	Stop condition setup time	4.0	-	0.6	-	μs
t _w (STO:STA)	Stop to Start condition time (bus free)	4.7	-	1.3	-	μs
C _b	Capacitive load for each bus line	-	400	-	400	pF

(1) Guaranteed by design, not tested in production.

(2) f_{PCLK1} must be at least 2 MHz to achieve standard mode I²C frequencies. It must be at least 4 MHz to achieve the fast mode I²C frequencies.

(3) The device must internally provide a hold time of at least 300ns for the SDA signal in order to bridge the undefined region on the falling edge of SCL.

Figure 23. I²C bus AC waveforms and measurement circuit⁽¹⁾



(1) Measurement points are done at CMOS levels: $0.3V_{DD}$ and $0.7V_{DD}$.

Table 39. SCL frequency ($f_{PCLK1} = 36 \text{ MHz}$, $V_{DD} = 3.3 \text{ V}$)⁽¹⁾⁽²⁾

$f_{SCL}(\text{kHz})$	I2C_CLKCTRL value
	$R_P = 4.7 \text{ k}\Omega$
400	0x801E
300	0x8028
200	0x803C
100	0x00B4
50	0x0168
20	0x0384

(1) R_P = External pull-up resistance, $f_{SCL} = I^2C$ speed.

(2) For speeds around 200 kHz, the tolerance on the achieved speed is of $\pm 5\%$. For other speed ranges, the tolerance on the achieved speed $\pm 2\%$. These variations depend on the accuracy of the external components used to design the application.

SPI-I²S characteristics

Unless otherwise specified, the parameters given in [Table 40](#) for SPI or in [Table 41](#) for I²S are derived from tests performed under ambient temperature, f_{PCLKx} frequency and V_{DD} supply voltage conditions summarized in [Table 9](#).

Refer to [5.3.13 I/O port characteristics](#) for more details on the input/output alternate function characteristics (NSS, SCK, MOSI, MISO for SPI and WS, CK, SD for I²S).

Table 40. SPI and SPIM characteristics

Symbol	Parameter	Conditions	Min	Max	Unit
f_{SCK}	SPI clock frequency	SPI1~2 master mode	-	37.5	MHz
$1/t_c(SCK)$		SPI1~2 slave mode	-	$f_{PCLK}/2$	
$t_r(SCK)$ $t_f(SCK)$	SPI clock rise and fall time	Capacitive load: C = 30 pF	-	8	ns
$t_{su(NSS)}^{(1)}$	NSS setup time	Slave mode	$4t_{PCLK}$	-	ns
$t_h(NSS)^{(1)}$	NSS hold time	Slave mode	$2t_{PCLK}$	-	ns
$t_{w(SCKH)}^{(1)}$ $t_{w(SCKL)}^{(1)}$	SCK high and low time	Master mode, $f_{PCLK} = 75$ MHz, prescaler = 4	36	53	ns
$t_{su(MI)}^{(1)}$	Data input setup time	Master mode	5	-	ns
$t_{su(SI)}^{(1)}$		Slave mode	5	-	
$t_h(MI)^{(1)}$	Data input setup time	Master mode	5	-	ns
$t_h(SI)^{(1)}$		Slave mode	4	-	
$t_{a(SO)}^{(1)(2)}$	Data output access time	Slave mode, $f_{PCLK} = 20$ MHz	0	$3t_{PCLK}$	ns
$t_{dis(SO)}^{(1)(3)}$	Data output disable time	Slave mode	2	10	ns
$t_v(SO)^{(1)}$	Data output valid time	Slave mode (after enable edge)	-	25	ns
$t_v(MO)^{(1)}$	Data output valid time	Master mode (after enable edge)	-	5	ns
$t_h(SO)^{(1)}$	Data output hold time	Slave mode (after enable edge)	15	-	ns
$t_h(MO)^{(1)}$		Master mode (after enable edge)	2	-	

(1) Guaranteed by characterization results, not tested in production.

(2) Min time is for the minimum time to drive the output and the max time is for the maximum time to validate the data.

(3) Min time is for the minimum time to invalidate the output and the max time is for the maximum time to put the data in Hi-Z.

Figure 24. SPI timing diagram - slave mode and CPHA = 0

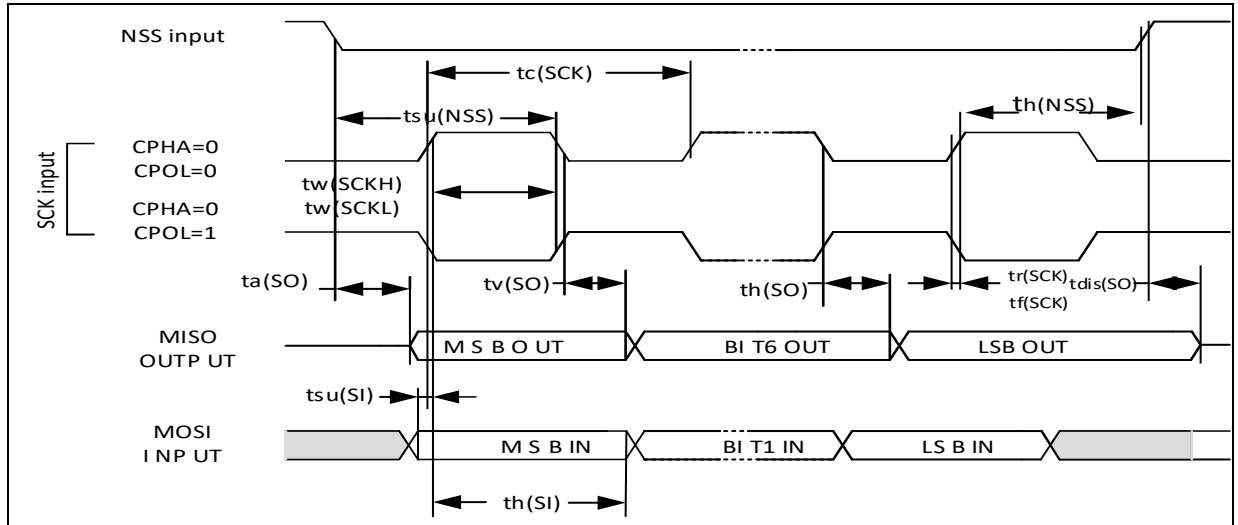
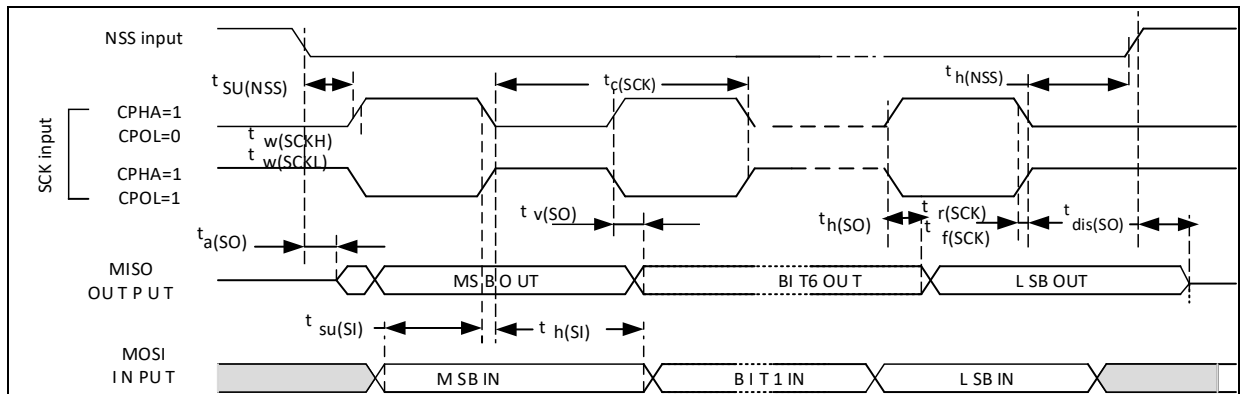
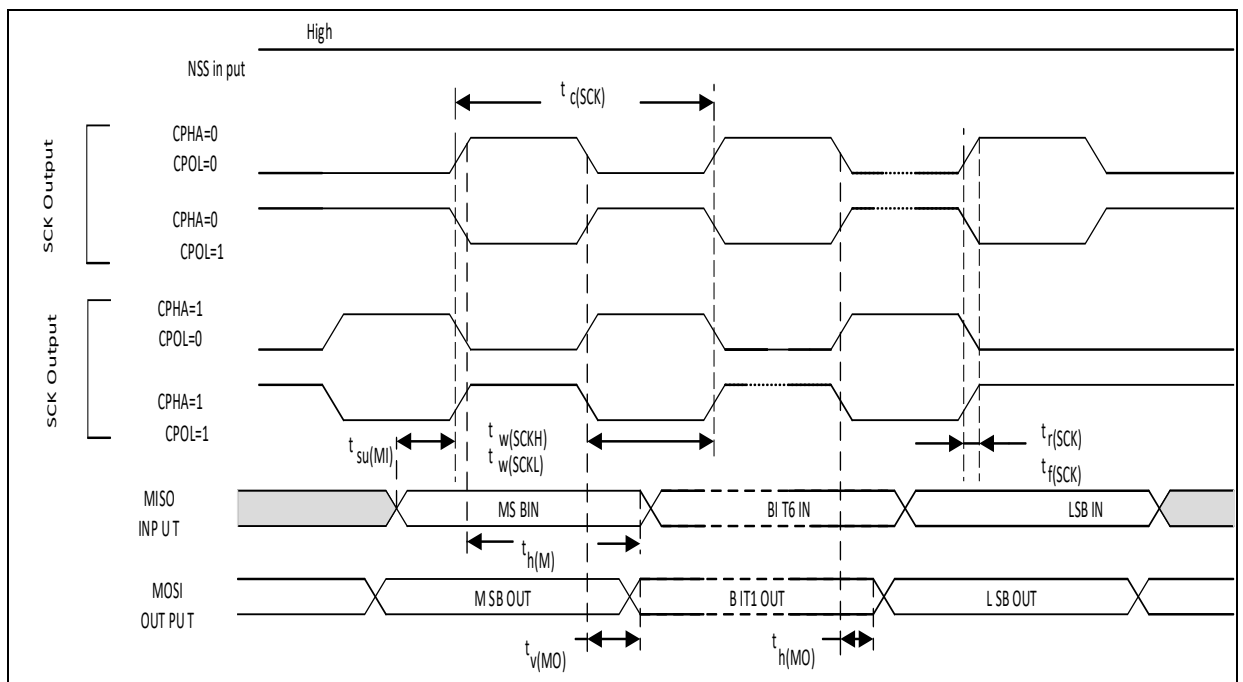


Figure 25. SPI timing diagram - slave mode and CPHA = 1⁽¹⁾



(1) Measurement points are done at CMOS levels: 0.3V_{DD} and 0.7V_{DD}.

Figure 26. SPI timing diagram - master mode⁽¹⁾



(1) Measurement points are done at CMOS levels: 0.3V_{DD} and 0.7V_{DD}.

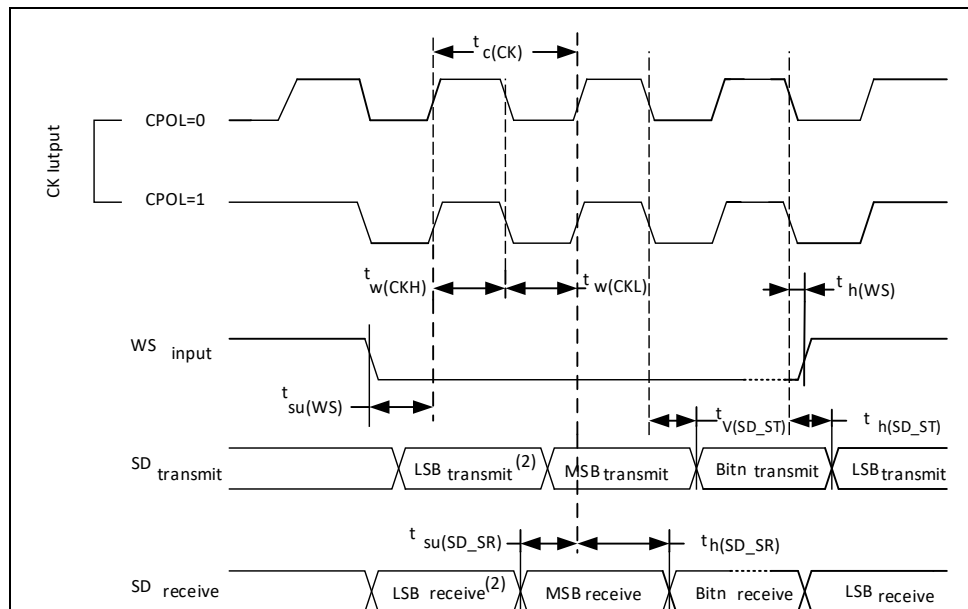
Table 41. I²S characteristics

Symbol	Parameter	Conditions	Min	Max	Unit
f_{CK} $1/t_{c(CK)}$	I ² S clock frequency	Master mode (data: 16 bits, audio frequency = 48 kHz)	1.522	1.525	MHz
		Slave mode	0	6.5	
$t_{r(CK)}$ $t_{f(CK)}$	I ² S clock rise and fall time	Capacitive load: C = 50 pF	-	8	ns
$t_{v(WS)}^{(1)}$	WS valid time	Master mode	3	-	
$t_{h(WS)}^{(1)}$	WS hold time	Master mode	2	-	
$t_{su(WS)}^{(1)}$	WS setup time	Slave mode	4	-	
$t_{h(WS)}^{(1)}$	WS hold time	Slave mode	0	-	
$t_{w(CKH)}^{(1)}$	CK high and low time	Master $f_{PCLK} = 16$ MHz, audio frequency = 48 kHz	312.5	-	
$t_{w(CKL)}^{(1)}$			345	-	
$t_{su(SD_MR)}^{(1)}$	Data input setup time	Master receiver	6.5	-	
$t_{su(SD_SR)}^{(1)}$		Slave receiver	1.5	-	
$t_{h(SD_MR)}^{(1)(2)}$	Data input hold time	Master receiver	0	-	
$t_{h(SD_SR)}^{(1)(2)}$		Slave receiver	0.5	-	
$t_{v(SD_ST)}^{(1)(2)}$	Data output valid time	Slave transmitter (after enable edge)	-	18	
$t_{h(SD_ST)}^{(1)}$	Data output hold time	Slave transmitter (after enable edge)	11	-	
$t_{v(SD_MT)}^{(1)(2)}$	Data output valid time	Master transmitter (after enable edge)	-	3	
$t_{h(SD_MT)}^{(1)}$	Data output hold time	Master transmitter (after enable edge)	0	-	

(1) Guaranteed by design and/or characterization results.

(2) Depends on f_{PCLK} . For example, if $f_{PCLK}=8$ MHz, then $T_{PCLK} = 1/f_{PCLK} = 125$ ns.

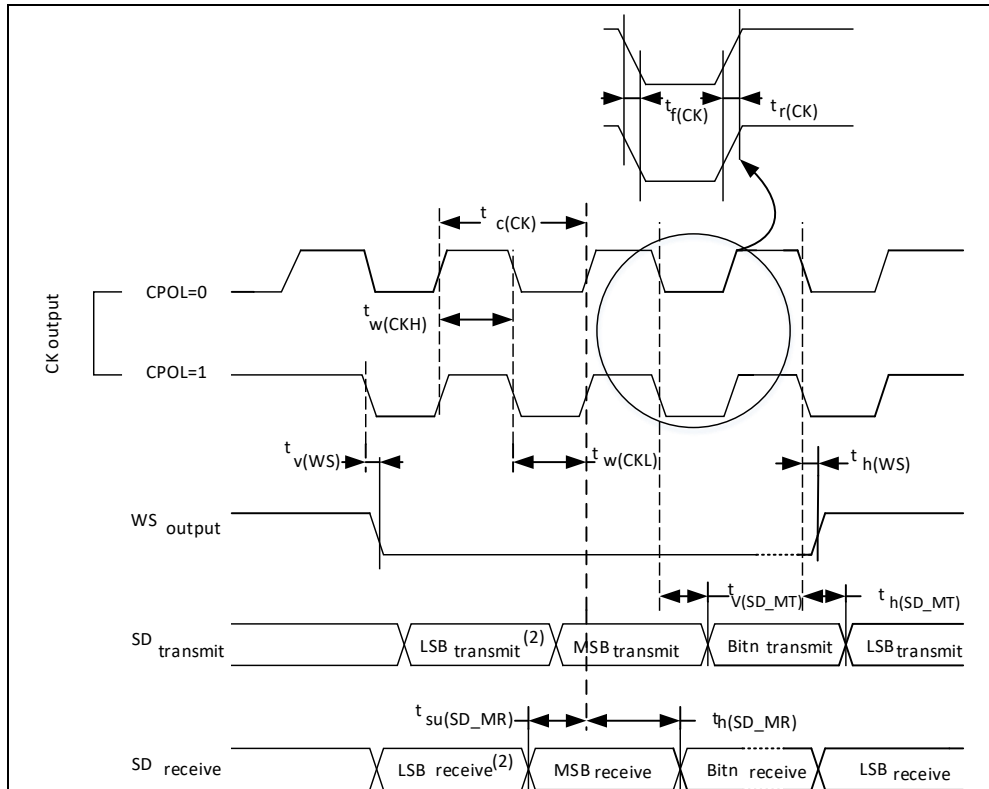
Figure 27. I²S slave timing diagram (Philips protocol)⁽¹⁾



(1) Measurement points are done at CMOS levels: 0.3V_{DD} and 0.7V_{DD}.

(2) LSB transmit/receive of the previously transmitted byte. No LSB transmit/receive is sent before the first byte.

Figure 28. I²S master timing diagram (Philips protocol)⁽¹⁾



(1) Measurement points are done at CMOS levels: $0.3V_{DD}$ and $0.7V_{DD}$.

(2) LSB transmit/receive of the previously transmitted byte. No LSB transmit/receive is sent before the first byte.

SD/SDIO MMC card host interface (SDIO) characteristics

Unless otherwise specified, the parameters given in the table below are derived from tests performed under ambient temperature, f_{PCLKx} frequency and V_{DD} supply voltage conditions summarized in [Table 9](#).

Refer to [5.3.13 I/O port characteristics](#) for more details on the input/output alternate function characteristics (D[7:0], CMD, CK).

Figure 29. SDIO high-speed mode

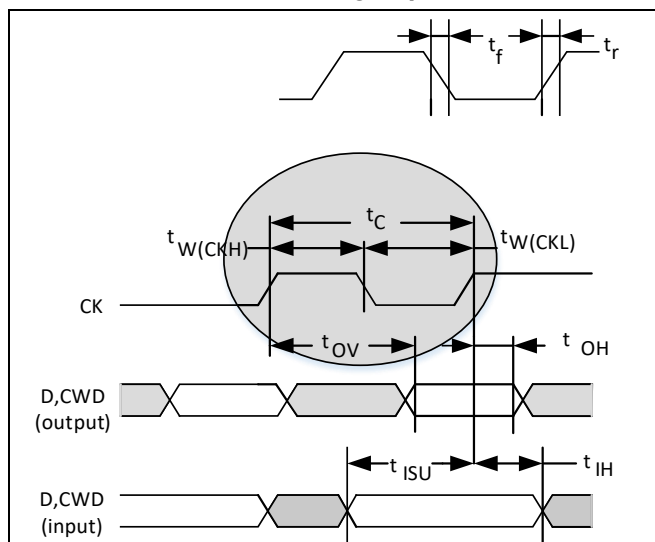


Figure 30. SD default mode

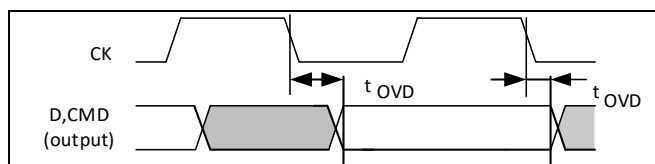


Table 42. SD / MMC characteristics

Symbol	Parameter	Conditions	Min	Max	Unit
f _{PP}	Clock frequency in data transfer mode	C _L ≤ 30 pF	0	48	MHz
t _{w(CKL)}	Clock low time, f _{PP} = 16 MHz	C _L ≤ 30 pF	32	-	ns
t _{w(CKH)}	Clock high time, f _{PP} = 16 MHz	C _L ≤ 30 pF	30	-	
t _r	Clock rise time	C _L ≤ 30 pF	-	4	
t _f	Clock fall time	C _L ≤ 30 pF	-	5	
CMD, D inputs (referenced to CK)					
t _{ISU}	Input setup time	C _L ≤ 30 pF	2	-	ns
t _{IH}	Input hold time	C _L ≤ 30 pF	0	-	
CMD, D outputs (referenced to CK) in MMC and SD HS mode					
t _{OV}	Output valid time	C _L ≤ 30 pF	-	6	ns
t _{OH}	Output hold time	C _L ≤ 30 pF	0	-	
CMD, D outputs (referenced to CK) in SD default mode ⁽¹⁾					
t _{OVD}	Output valid default time	C _L ≤ 30 pF	-	7	ns
t _{OHD}	Output hold default time	C _L ≤ 30 pF	0.5	-	

(1) Refer to SDIO_CLKCTRL, the SDIO clock control register to control the CK output.

USB OTG full-speed characteristics
Table 43. USB OTG FS startup time

Symbol	Parameter	Max	Unit
$t_{\text{STARTUP}}^{(1)}$	USB OTG transceiver startup time	1	μs

(1) Guaranteed by design, not tested in production.

Table 44. USB OTG FS DC electrical characteristics

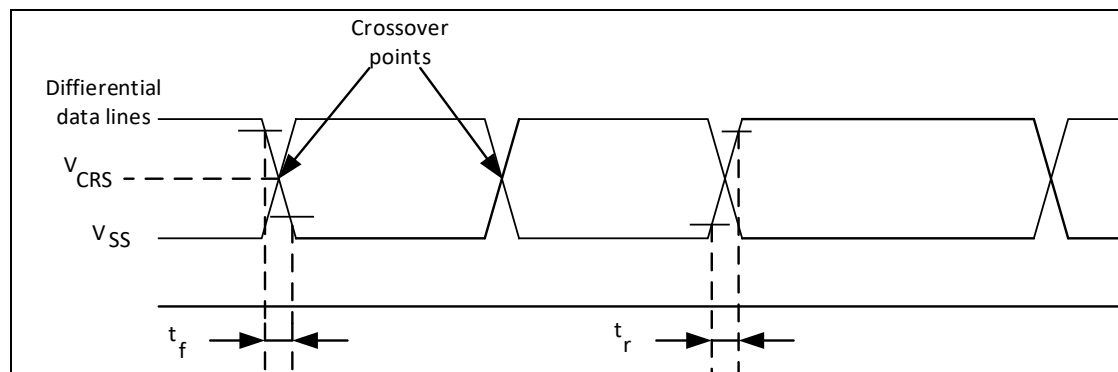
Symbol	Parameter	Conditions	Min ⁽¹⁾	Typ	Max ⁽¹⁾	Unit
Input levels	V_{DD}	USB OTG operating voltage	-	3.0 ⁽²⁾	3.6	V
	$V_{\text{DI}}^{(3)}$	Differential input sensitivity	I (OTG_FS_DP/DM)	0.2	-	V
	$V_{\text{CM}}^{(3)}$	Differential common mode range	Includes V_{DI} range	0.8	2.5	
	$V_{\text{SE}}^{(3)}$	Single ended receiver threshold	-	1.3	2.0	
Output levels	V_{OL}	Static output level low	R_{L} of intanal 1.24 k Ω to 3.6 V ⁽⁴⁾	-	0.3	V
	V_{OH}	Static output level high	R_{L} of 15 k Ω to $V_{\text{SS}}^{(4)}$	2.8	3.6	
R_{PU}	OTG_FS_DP internal pull-up	$V_{\text{IN}} = V_{\text{SS}}$	0.97	1.24	1.58	k Ω
R_{PD}	OTG_FS_DP/DM internal pull-down	$V_{\text{IN}} = V_{\text{DD}}$	15	19	25	k Ω

(1) All the voltages are measured from the local ground potential.

(2) The AT32F415 USB functionality is ensured down to 2.7 V but not the full USB electrical characteristics which are degraded in the 2.7 to 3.0 V V_{DD} voltage range.

(3) Guaranteed by characterization results, not tested in production.

(4) R_{L} is the load connected on the USB OTG FS drivers.

Figure 31. USB OTG FS timings: definition of data signal rise and fall time

Table 45. USB OTG FS electrical characteristics

Symbol	Parameter	Conditions	Min ⁽¹⁾	Max ⁽¹⁾	Unit
t_r	Rise time ⁽²⁾	$C_{\text{L}} \leq 50 \text{ pF}$	4	20	ns
t_f	Fall Time ⁽²⁾	$C_{\text{L}} \leq 50 \text{ pF}$	4	20	ns
t_{rfm}	Rise/ fall time matching	t_r/t_f	90	110	%
V_{CRS}	Output signal crossover voltage	-	1.3	2.0	V

(1) Guaranteed by design, not tested in production.

(2) Measured from 10% to 90% of the data signal. For more detailed information, please refer to USB Specification - Chapter 7 (version 2.0).

5.3.17 CAN (controller area network) interface

Refer to [5.3.13 I/O port characteristics](#) for more details on the input/output alternate function characteristics (CAN_TX and CAN_RX).

5.3.18 12-bit ADC characteristics

Unless otherwise specified, the parameters given in the table below are preliminary values derived from tests performed under ambient temperature, f_{PCLK2} frequency and V_{DDA} supply voltage conditions summarized in [Table 9](#).

Note: *It is recommended to perform a calibration after each power-up.*

Table 46. ADC characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{DDA}	Power supply	-	2.6	-	3.6	V
I_{DDA}	Current on the V_{DDA} input pin	-	-	560 ⁽¹⁾	660	μA
f_{ADC}	ADC clock frequency	-	0.6	-	28	MHz
$f_S^{(2)}$	Sampling rate	-	0.05	-	2	MHz
$f_{TRIG}^{(2)}$	External trigger frequency	$f_{ADC} = 28 \text{ MHz}$	-	-	1.65	MHz
		-	-	-	17	$1/f_{ADC}$
V_{AIN}	Conversion voltage range ⁽³⁾	-	0 (V_{REF-} internal tied to ground))		V_{REF+}	V
$R_{AIN}^{(2)}$	External input impedance	-	See Table 47 and Table 48 for details			Ω
$C_{ADC}^{(2)}$	Internal sample and hold capacitor	-	-	15	-	pF
$t_{CAL}^{(2)}$	Calibration time	$f_{ADC} = 28 \text{ MHz}$	6.14			μs
		-	172			$1/f_{ADC}$
$t_{lat}^{(2)}$	Injection trigger conversion latency	$f_{ADC} = 28 \text{ MHz}$	-	-	107	ns
		-	-	-	3 ⁽⁴⁾	$1/f_{ADC}$
$t_{latr}^{(2)}$	Regular trigger conversion latency	$f_{ADC} = 28 \text{ MHz}$	-	-	71.4	μs
		-	-	-	2 ⁽⁴⁾	$1/f_{ADC}$
$t_S^{(2)}$	Sampling time	$f_{ADC} = 28 \text{ MHz}$	0.053	-	8.55	μs
		-	1.5	-	239.5	$1/f_{ADC}$
$t_{STAB}^{(2)}$	Power-up time	-	42			$1/f_{ADC}$
$t_{CONV}^{(2)}$	Total conversion time (including sampling time)	$f_{ADC} = 28 \text{ MHz}$	0.5	-	9	μs
		-	14 to 252 (t_S for sampling + 12.5 for successive approximation)			$1/f_{ADC}$

(1) Guaranteed by characterization results, not tested in production.

(2) Guaranteed by design, not tested in production.

(3) V_{REF+} is internally connected to V_{DDA} and V_{REF-} to V_{SSA} .

(4) For external triggers, a delay of $1/f_{PCLK2}$ must be added to the latency specified in [Table 46](#).

Table 47 and Table 48 are used to determine the maximum external impedance allowed for an error below 1/4 of LSB.

Table 47. R_{AIN} max for $f_{ADC} = 14 \text{ MHz}^{(1)}$

T_s (Cycle)	t_s (μs)	R_{AIN} max ($k\Omega$)
1.5	0.11	0.25
7.5	0.54	1.3
13.5	0.96	2.5
28.5	2.04	5.0
41.5	2.96	8.0
55.5	3.96	10.5
71.5	5.11	13.5
239.5	17.11	40

(1) Guaranteed by design.

Table 48. R_{AIN} max for $f_{ADC} = 28 \text{ MHz}^{(1)}$

T_s (Cycle)	t_s (μs)	R_{AIN} max ($k\Omega$)
1.5	0.05	0.1
7.5	0.27	0.6
13.5	0.48	1.2
28.5	1.02	2.5
41.5	1.48	4.0
55.5	1.98	5.2
71.5	2.55	7.0
239.5	8.55	20

(1) Guaranteed by design.

Table 49. ADC accuracy ($V_{DDA} = 3.0$ to 3.6 V, $V_{REF+} = V_{DDA}$, $T_A = 25$ °C)⁽¹⁾⁽²⁾

Symbol	Parameter	Test Conditions	Typ	Max ⁽³⁾	Unit
ET	Total unadjusted error	$f_{PCLK2} = 56$ MHz, $f_{ADC} = 28$ MHz, $R_{AIN} < 10$ k Ω , $V_{DDA} = 3.0$ to 3.6 V, $T_A = 25$ °C Measurements made after ADC calibration $V_{REF+} = V_{DDA}$	± 2	± 3	LSB
EO	Offset error		± 1	± 1.6	
EG	Gain error		± 1.5	± 3	
ED	Differential linearity error		± 0.6	± 1	
EL	Integral linearity error		± 1	± 2	

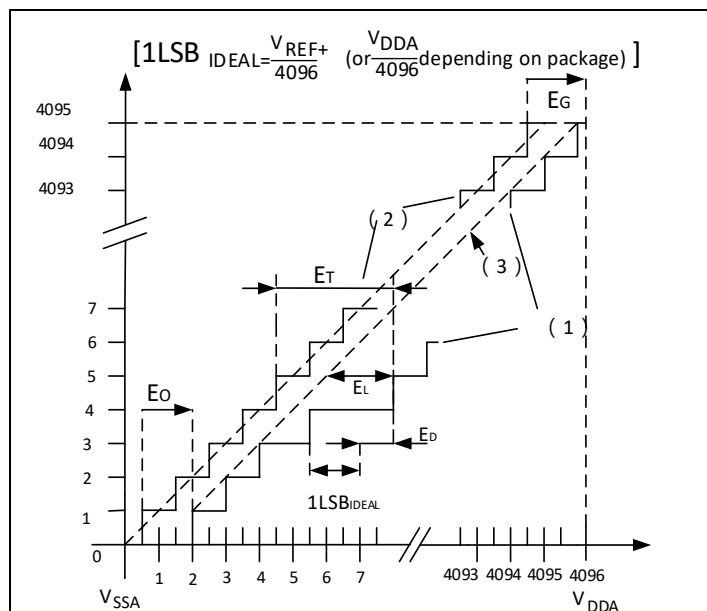
- (1) ADC DC accuracy values are measured after internal calibration.
(2) ADC Accuracy vs. Negative Injection Current: Injecting negative current on any analog input pins should be avoided as this significantly reduces the accuracy of the conversion being performed on another analog input. It is recommended to add a Schottky diode (pin to ground) to analog pins which may potentially inject negative current.
(3) Guaranteed by characterization results, not tested in production.

Table 50. ADC accuracy ($V_{DDA} = 2.6$ to 3.6 V, $T_A = -40$ to $+105$ °C)⁽¹⁾⁽²⁾

Symbol	Parameter	Test Conditions	Typ	Max ⁽³⁾	Unit
ET	Total unadjusted error	$f_{PCLK2} = 56$ MHz, $f_{ADC} = 28$ MHz, $R_{AIN} < 10$ k Ω , $V_{DDA} = 2.6$ to 3.6 V Measurements made after ADC calibration	± 2	± 4	LSB
EO	Offset error		± 1	± 2	
EG	Gain error		± 1.5	± 3.5	
ED	Differential linearity error		± 0.6	$+1.5/-1$	
EL	Integral linearity error		± 1	± 2	

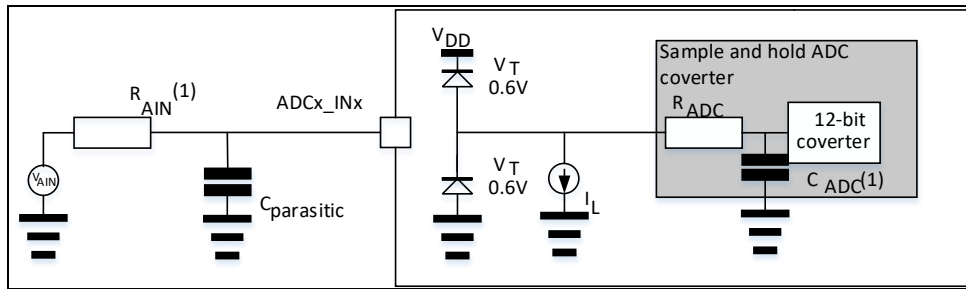
- (1) ADC DC accuracy values are measured after internal calibration.
(2) ADC Accuracy vs. Negative Injection Current: Injecting negative current on any analog input pins should be avoided as this significantly reduces the accuracy of the conversion being performed on another analog input. It is recommended to add a Schottky diode (pin to ground) to analog pins which may potentially inject negative current.
(3) Guaranteed by characterization results, not tested in production.

Figure 32. ADC accuracy characteristics



- (1) Example of an actual transfer curve.
(2) Ideal transfer curve.
(3) End point correlation line.
(4) ET = Maximum deviation between the actual and the ideal transfer curves.
EO = Deviation between the first actual transition and the first ideal one.
EG = Deviation between the last ideal transition and the last actual one.
ED = Maximum deviation between actual steps and the ideal one.
EL = Maximum deviation between any actual transition and the end point correlation line.

Figure 33. Typical connection diagram using the ADC

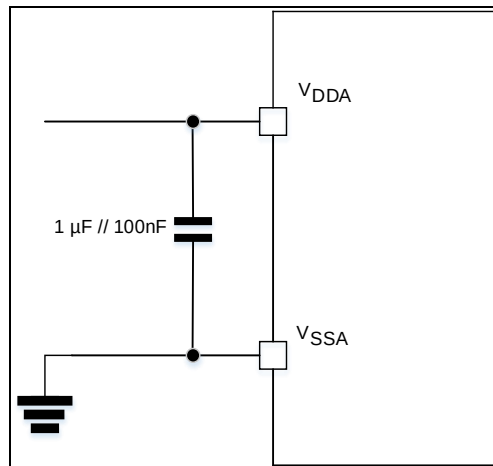


- (1) Refer to [Table 46](#) for the values of R_{AIN} and C_{ADC} .
- (2) $C_{parasitic}$ represents the capacitance of the PCB (dependent on soldering and PCB layout quality) plus the pad capacitance (roughly 7 pF). A high $C_{parasitic}$ value will downgrade conversion accuracy. To remedy this, f_{ADC} should be reduced.

General PCB design guidelines

Power supply decoupling should be performed as shown in [Figure 34](#). They should be placed them as close as possible to the chip.

Figure 34. Power supply and reference decoupling



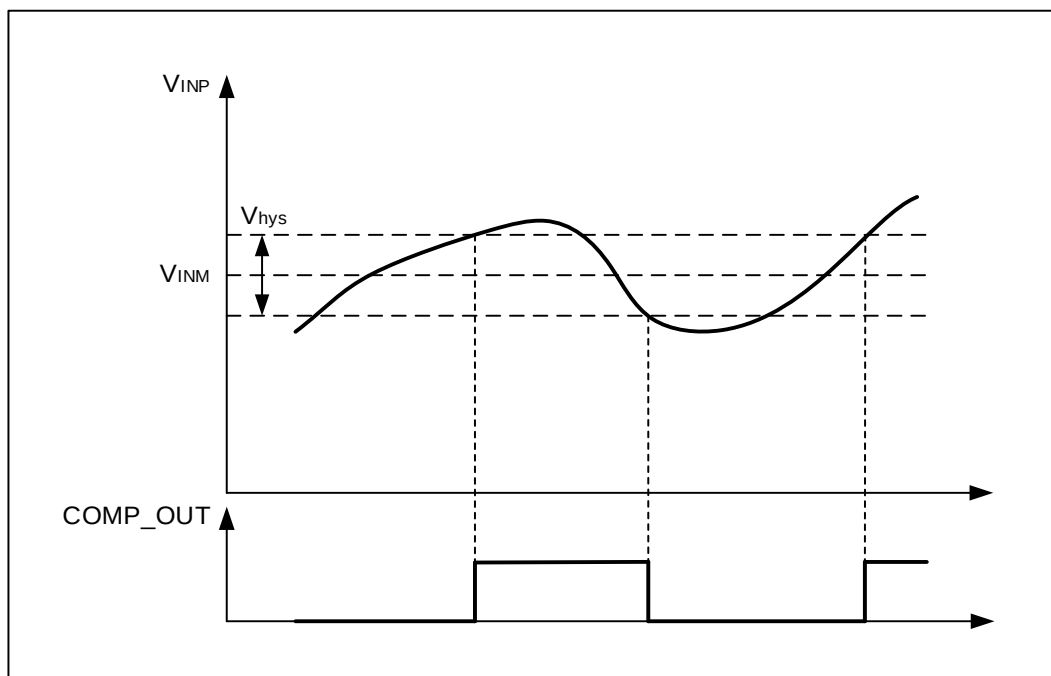
5.3.19 Comparator characteristics

Table 51. Comparator characteristics

Symbol	Parameter	Conditions	Min ⁽¹⁾	Typ	Max ⁽¹⁾	単位
V_{DDA}	Analog supply voltage	-	2.6	-	3.6	V
V_{IN}	Input voltage range	-	0	-	V_{DDA}	V
t_{START}	Startup time	High speed mode	-	2.0	3.2	μs
		Low power mode	-	3.6	5.5	
t_D	Propagation delay for 200 mV step with 100 mV overdrive	High speed mode	-	105	320	ns
		Low power mode	-	1.2	3	μs
V_{offset}	Offset voltage	-	-	± 3	± 10	mV
V_{hys}	Hysteresis	No hysteresis	-	0	-	mV
		High speed mode	Low hysteresis	40	65	
			Medium hysteresis	120	180	
			High hysteresis	200	320	
		Low power mode	Low hysteresis	15	25	
			Medium hysteresis	50	70	
			High hysteresis	90	120	
I_{DDA}	Current consumption	High speed mode	-	120	165	μA
		Low power mode	-	1.9	3.5	

(1) Guaranteed by characterization results, not tested in production.

Figure 35. Comparator hysteresis



5.3.20 Temperature sensor characteristics

Table 52. Temperature sensor characteristics

Symbol	Parameter	Min	Typ	Max	Unit
$T_L^{(1)}$	V_{SENSE} linearity with temperature	-	± 2	± 5	$^{\circ}\text{C}$
Avg_Slope ⁽¹⁾⁽²⁾	Average slope	-4.13	-4.34	-4.54	mV/ $^{\circ}\text{C}$
$V_{25}^{(1)(2)}$	Voltage at 25 $^{\circ}\text{C}$	1.26	1.32	1.38	V
$t_{START}^{(3)}$	Startup time	-	-	100	μs
$T_{S_temp}^{(3)(4)}$	ADC sampling time when reading the temperature	-	8.6	17.1	μs

(1) Guaranteed by characterization results, not tested in production.

(2) The temperature sensor output voltage changes linearly with temperature. The offset of this line varies from chip to chip due to process variation (up to 30 $^{\circ}\text{C}$ from one chip to another). The internal temperature sensor is more suited to applications that detect temperature variations instead of absolute temperatures. If accurate temperature readings are needed, an external temperature sensor part should be used.

(3) Guaranteed by design, not tested in production.

(4) Shortest sampling time can be determined in the application by multiple iterations.

Obtain the temperature using the following formula:

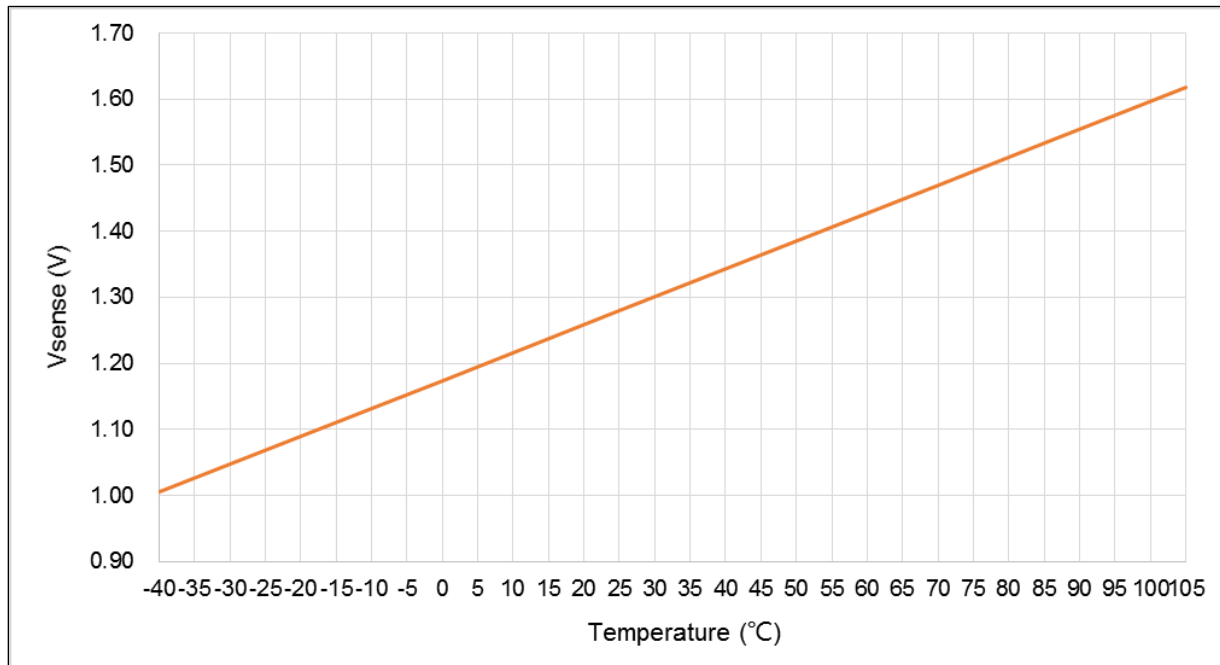
$$\text{Temperature (in } ^{\circ}\text{C)} = \{(V_{25} - V_{SENSE}) / \text{Avg_Slope}\} + 25.$$

Where,

$V_{25} = V_{SENSE}$ value for 25 $^{\circ}\text{C}$ and

Avg_Slope = Average Slope for curve between Temperature vs. V_{SENSE} (given in mV/ $^{\circ}\text{C}$).

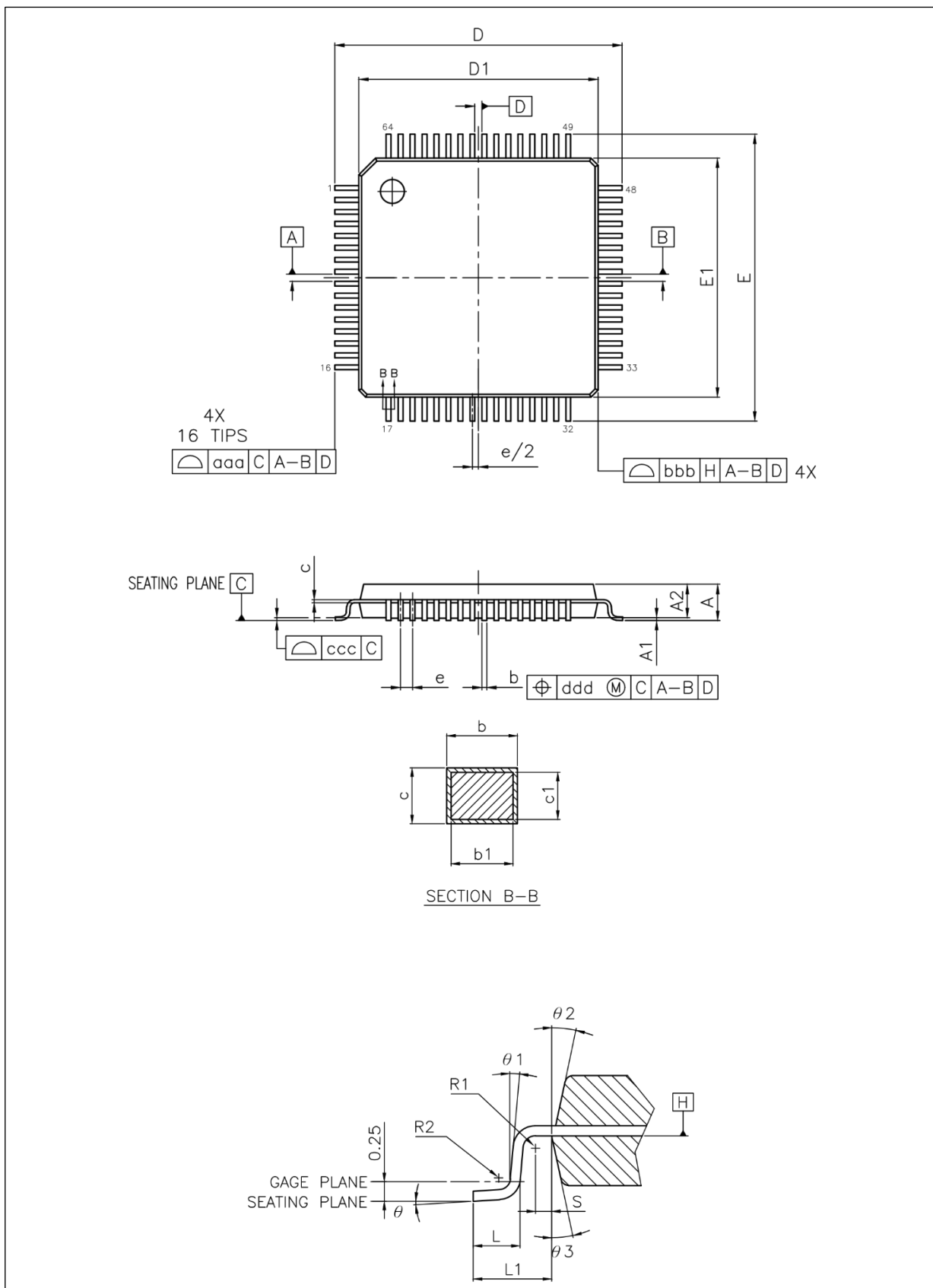
Figure 36. V_{SENSE} vs. temperature



6 Package information

6.1 LQFP64 – 10 x 10 mm package information

Figure 37. LQFP64 – 10 x 10 mm 64 pin low-profile quad flat package outline



(1) Drawing is not in scale.

Table 53. LQFP64 – 10 x 10 mm 64 pin low-profile quad flat package mechanical data

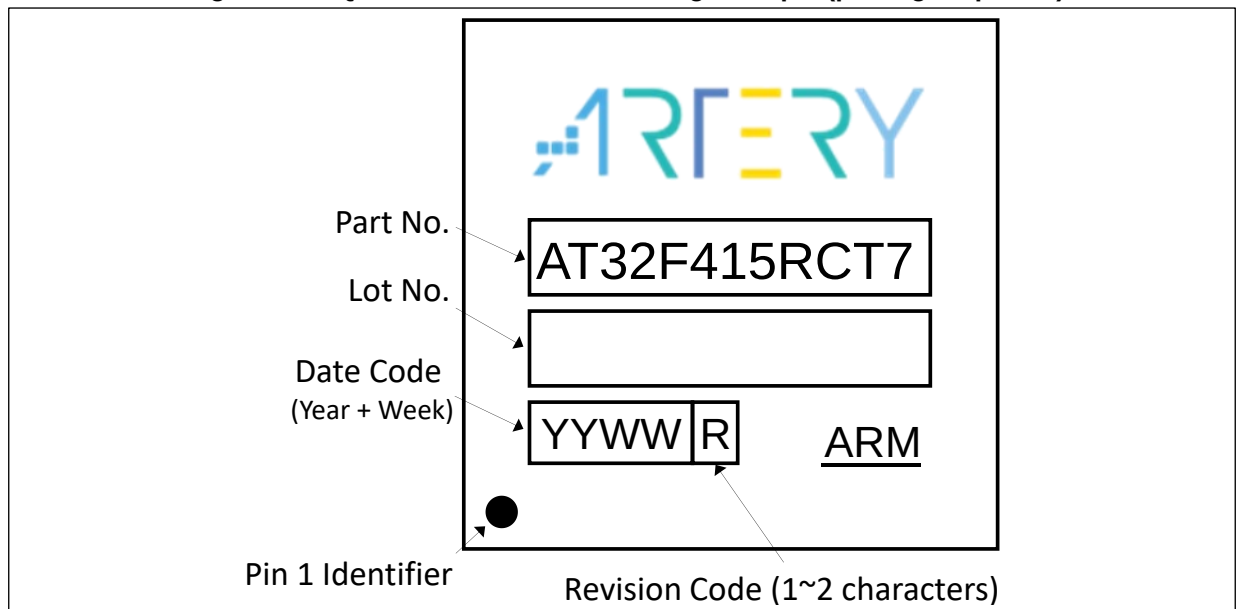
Symbol	millimeters			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
A	-	-	1.60	-	-	0.063
A1	0.05	-	0.15	0.002	-	0.006
A2	1.35	1.40	1.45	0.053	0.055	0.057
b	0.17	0.20	0.27	0.007	0.008	0.011
c	0.09	-	0.20	0.004	-	0.008
D	11.75	12.00	12.25	0.463	0.472	0.482
D1	9.90	10.00	10.10	0.390	0.394	0.398
E	11.75	12.00	12.25	0.463	0.472	0.482
E1	9.90	10.00	10.10	0.390	0.394	0.398
e	0.50 BSC.			0.020 BSC.		
Θ	3.5° REF.			3.5° REF.		
L	0.45	0.60	0.75	0.018	0.024	0.030
L1	1.00 REF.			0.039 REF.		
ccc	0.08			0.003		

(1) Values in inches are converted from mm and rounded to 3 decimal digits.

Device marking for LQFP64 – 10 x 10 mm

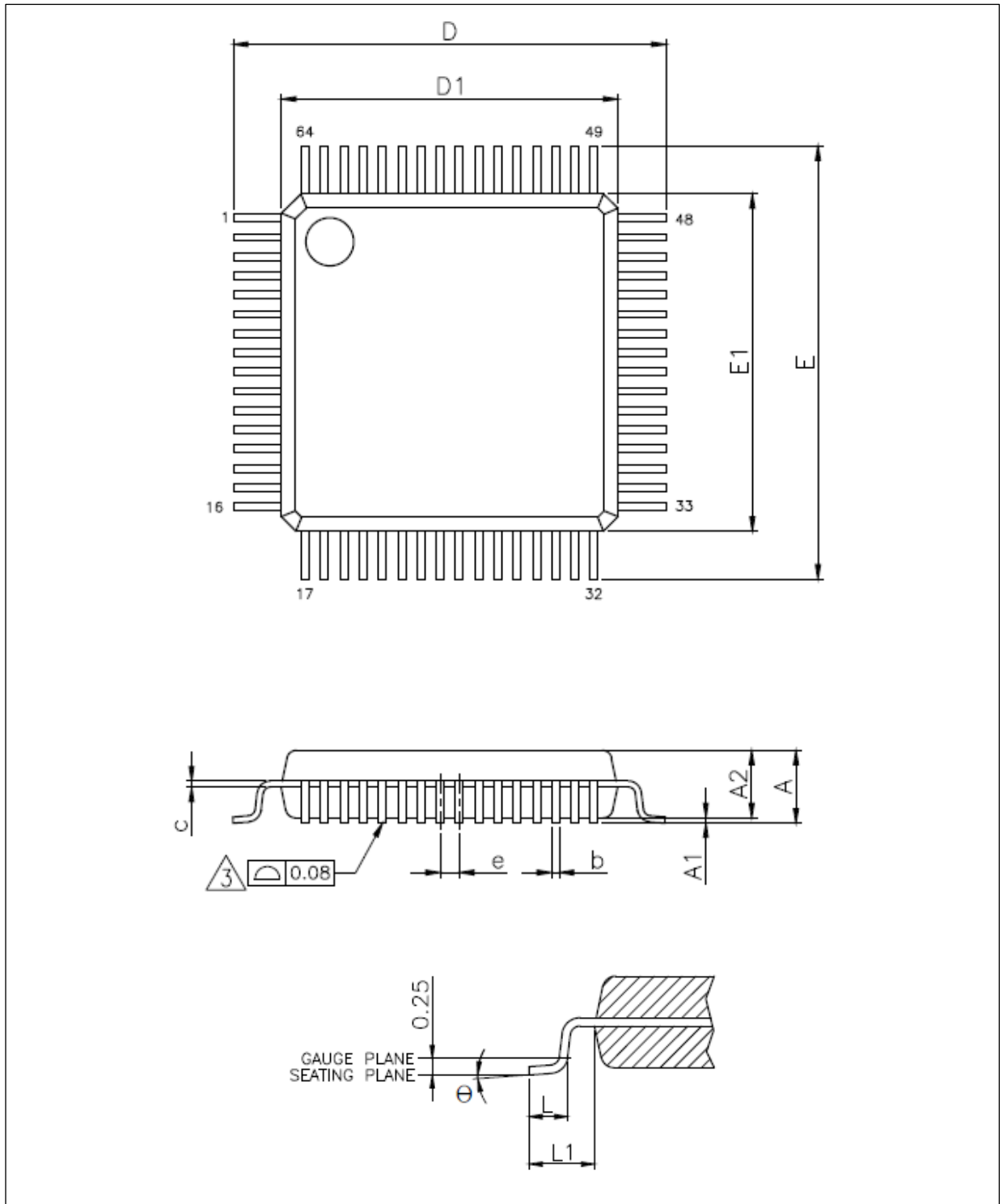
The following figure gives an example of topside marking orientation versus pin 1 identifier location.

Figure 38. LQFP64 – 10 x 10 mm marking example (package top view)



6.2 LQFP64 – 7 x 7 mm package information

Figure 39. LQFP64 – 7 x 7 mm 64 pin low-profile quad flat package outline



(1) Drawing is not in scale.

Table 54. LQFP64 – 7 x 7 mm 64 pin low-profile quad flat package mechanical data

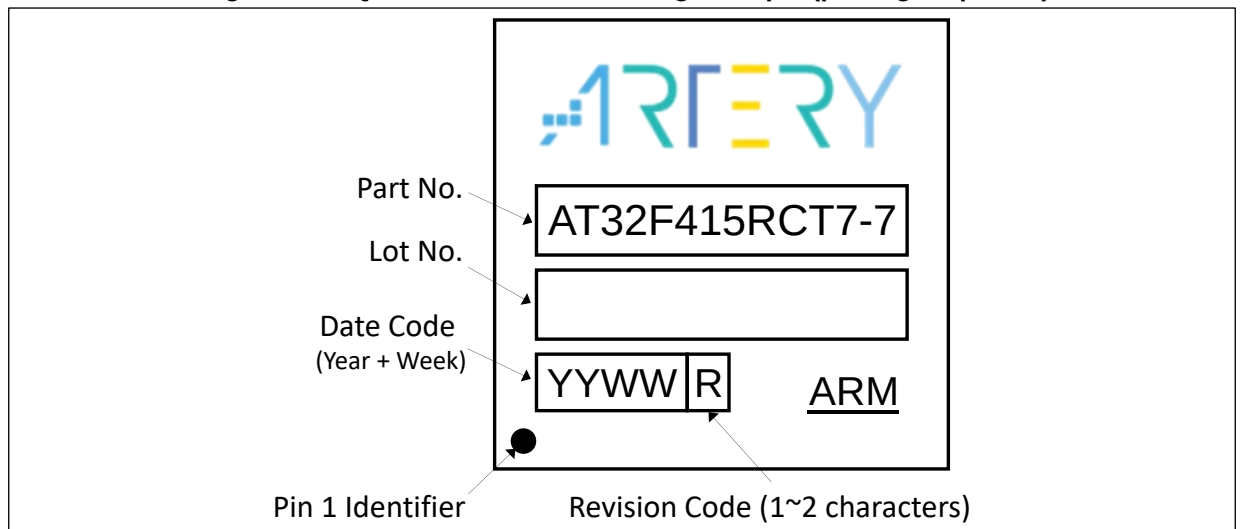
Symbol	millimeters			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
A	-	-	1.60	-	-	0.063
A1	0.05	-	0.15	0.002	-	0.006
A2	1.35	1.40	1.45	0.053	0.055	0.057
b	0.17	0.20	0.27	0.007	0.008	0.011
c	0.09	-	0.20	0.004	-	0.008
D	11.75	12.00	12.25	0.463	0.472	0.482
D1	9.90	10.00	10.10	0.390	0.394	0.398
E	11.75	12.00	12.25	0.463	0.472	0.482
E1	9.90	10.00	10.10	0.390	0.394	0.398
e	0.50 BSC.			0.020 BSC.		
Θ	3.5° REF.			3.5° REF.		
L	0.45	0.60	0.75	0.018	0.024	0.030
L1	1.00 REF.			0.039 REF.		
ccc	0.08			0.003		

(1) Values in inches are converted from mm and rounded to 3 decimal digits.

Device marking for LQFP64 – 7 x 7 mm

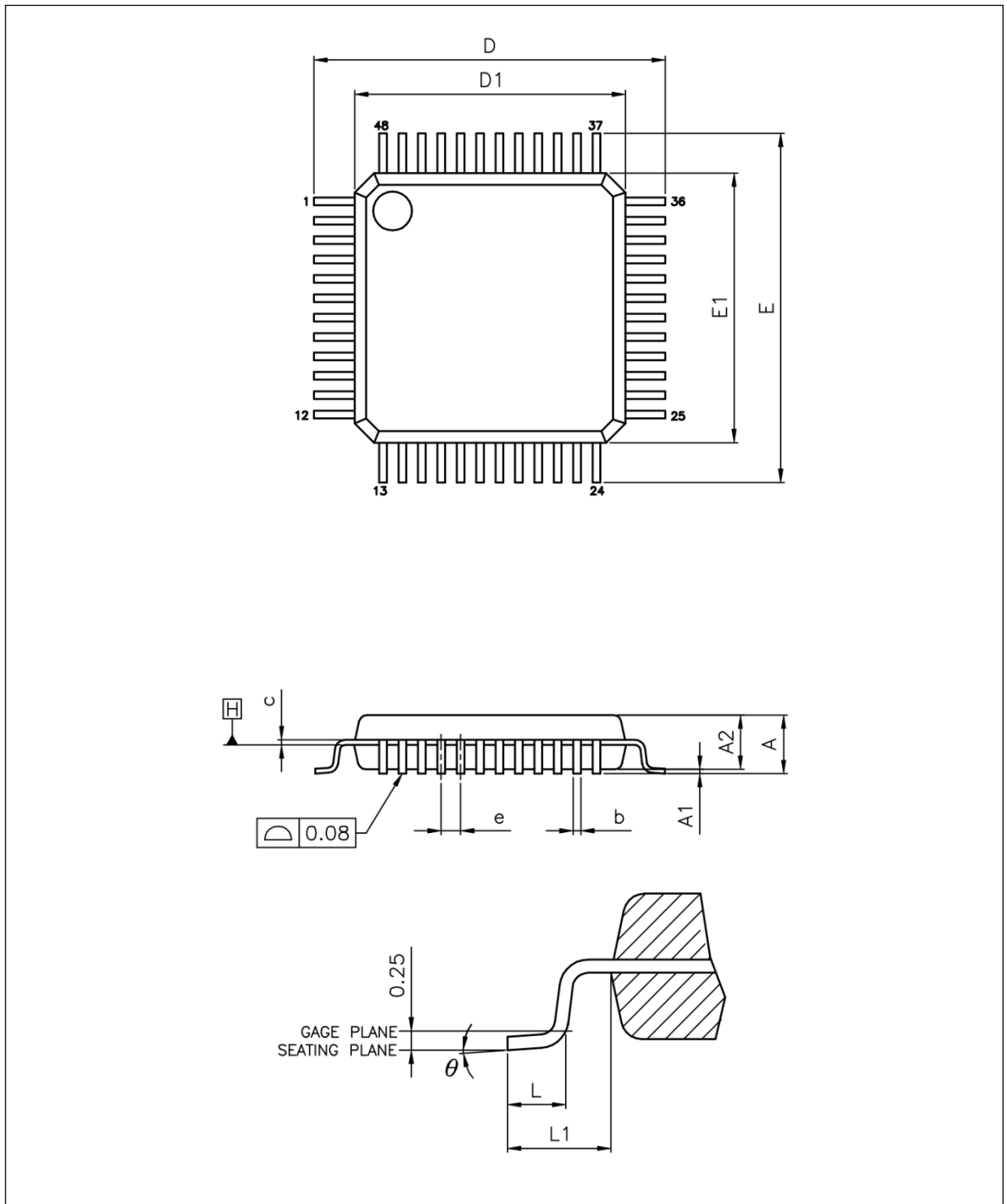
The following figure gives an example of topside marking orientation versus pin 1 identifier location.

Figure 40. LQFP64 – 7 x 7 mm marking example (package top view)



6.3 LQFP48 – 7 x 7 mm package information

Figure 41. LQFP48 – 7 x 7 mm 48 pin low-profile quad flat package outline



(1) Drawing is not in scale.

Table 55. LQFP48 – 7 x 7 mm 48 pin low-profile quad flat package mechanical data

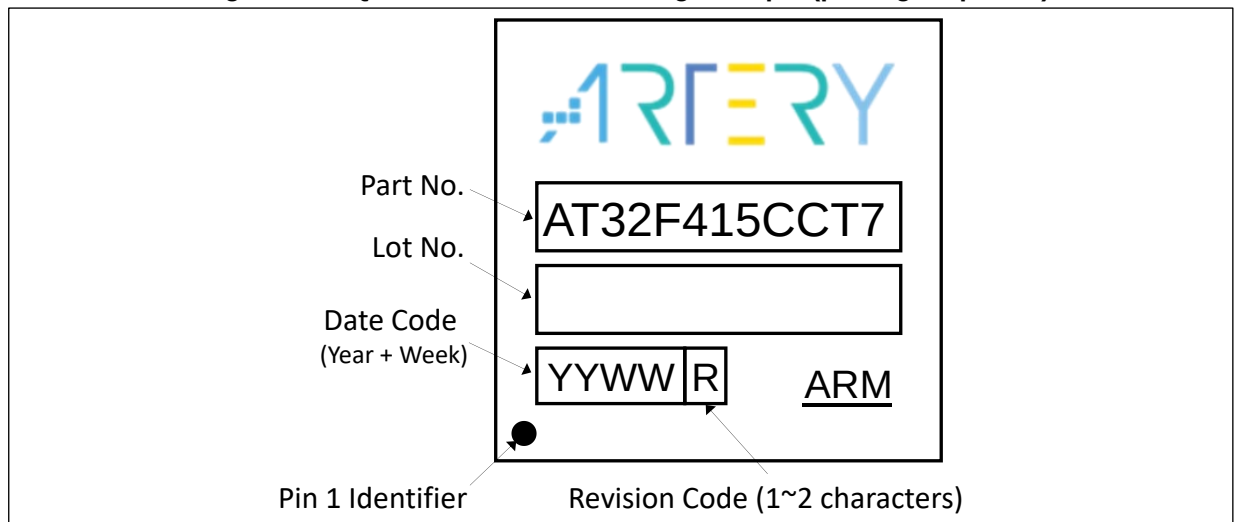
Symbol	millimeters			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
A	-	-	1.60	-	-	0.063
A1	0.05	-	0.15	0.002	-	0.006
A2	1.35	1.40	1.45	0.053	0.055	0.057
b	0.17	0.22	0.27	0.007	0.009	0.011
c	0.09	-	0.20	0.004	-	0.008
D	9.00 BSC			0.345 BSC		
D1	7.00 BSC			0.276 BSC		
E	9.00 BSC			0.345 BSC		
E1	7.00 BSC			0.276 BSC		
e	0.50 BSC.			0.020 BSC.		
Θ	0°	3.5°	7°	0°	3.5°	7°
L	0.45	0.60	0.75	0.018	0.024	0.030
L1	1.00 REF.			0.039 REF.		

(1) Values in inches are converted from mm and rounded to 3 decimal digits.

Device marking for LQFP48

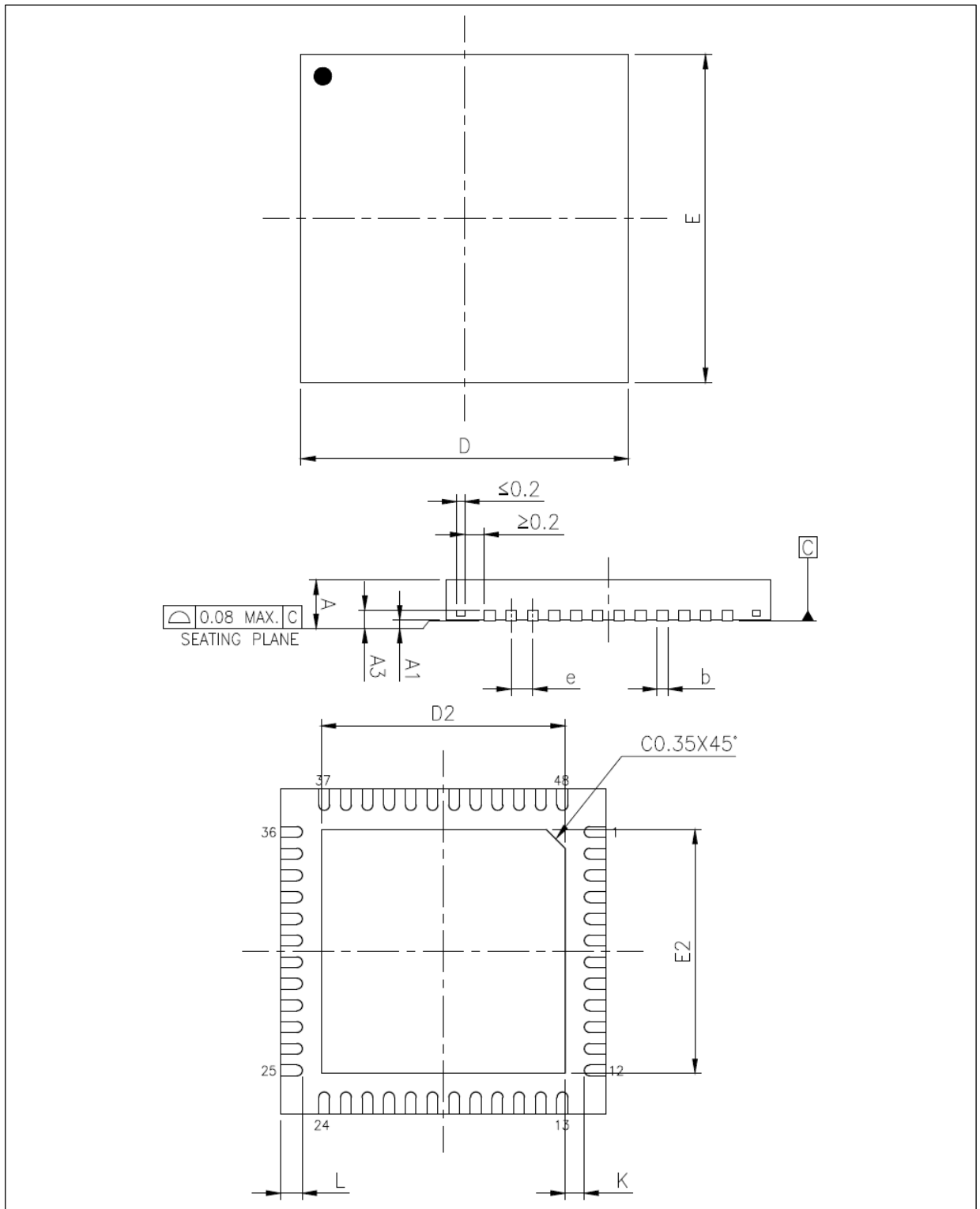
The following figure gives an example of topside marking orientation versus pin 1 identifier location.

Figure 42. LQFP48 – 7 x 7 mm marking example (package top view)



6.4 QFN48 – 6 x 6 mm package information

Figure 43. QFN48 – 6 x 6 mm 48 pin fine-pitch quad flat package outline



(1) Drawing is not in scale.

Table 56. QFN48 – 6 x 6 mm 48 pin fine-pitch quad flat package mechanical data

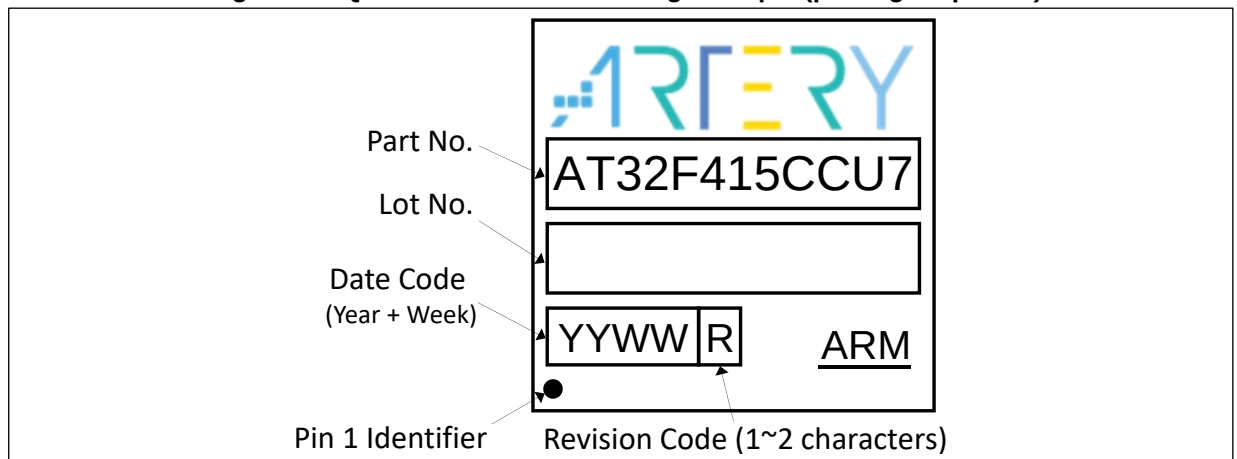
Symbol	millimeters			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
A	0.80	0.85	0.90	0.031	0.033	0.035
A1	0.00	0.02	0.05	0.000	0.001	0.002
A3	0.203 REF.			0.008 REF.		
b	0.15	0.20	0.25	0.006	0.008	0.010
D	6.00 BSC.			0.236 BSC.		
D2	4.45	4.50	4.55	0.175	0.177	0.179
E	6.00 BSC.			0.236 BSC.		
E2	4.45	4.50	4.55	0.175	0.177	0.179
e	0.40 BSC.			0.016 BSC.		
K	0.20	-	-	0.008	-	-
L	0.35	0.40	0.45	0.014	0.016	0.018

(1) Values in inches are converted from mm and rounded to 3 decimal digits.

Device marking for QFN48

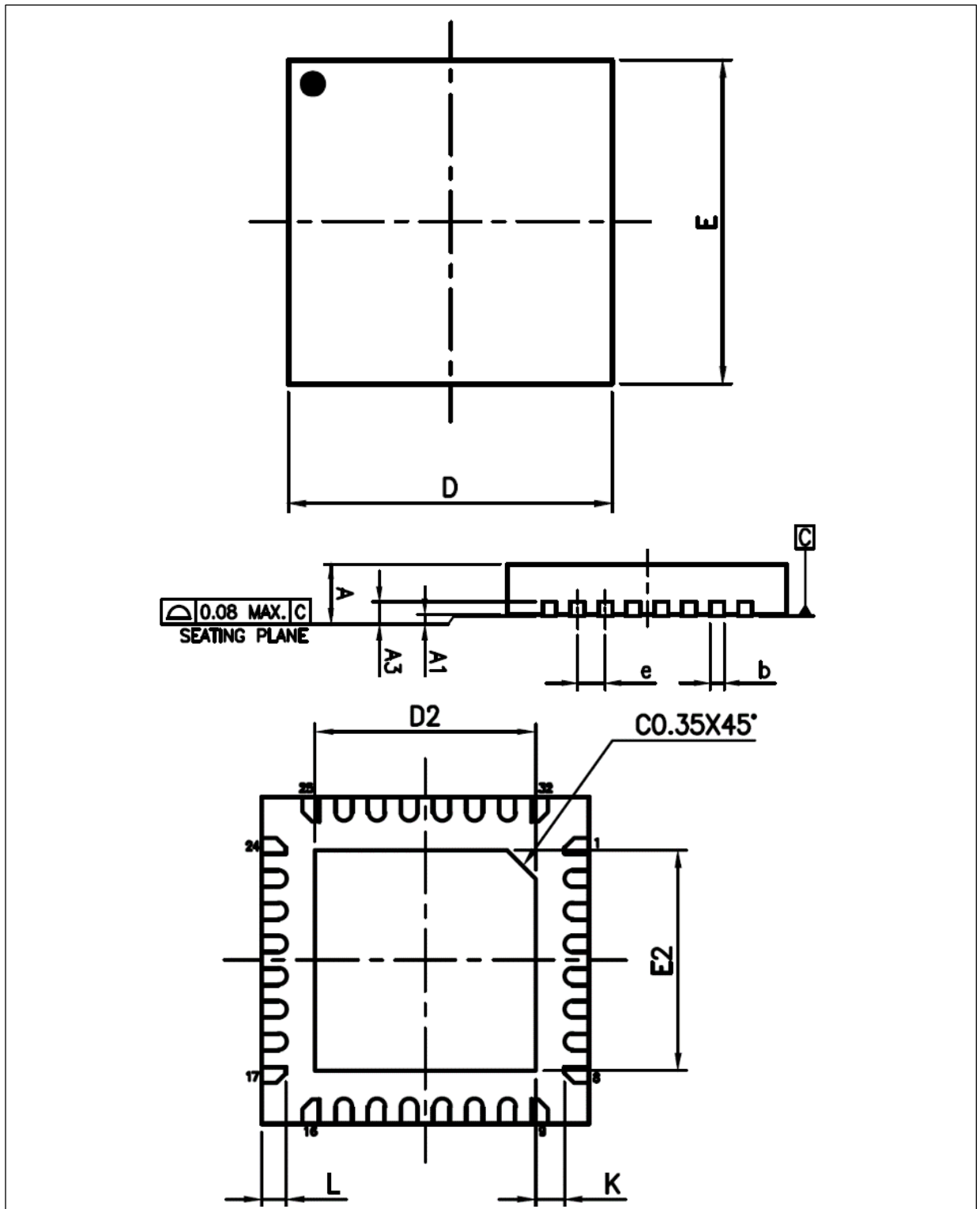
The following figure gives an example of topside marking orientation versus pin 1 identifier location.

Figure 44. QFN48 – 6 x 6 mm marking example (package top view)



6.5 QFN32 – 4 x 4 mm package information

Figure 45. QFN32 – 4 x 4 mm 32 pin fine-pitch quad flat package outline



(1) Drawing is not in scale.

Table 57. QFN32 – 4 x 4 mm 32 pin fine-pitch quad flat package mechanical data

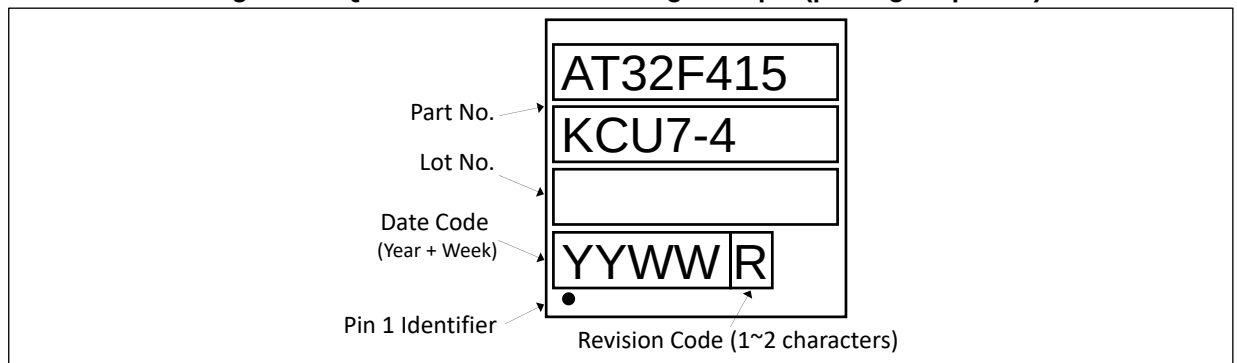
Symbol	millimeters			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
A	0.80	0.85	0.90	0.031	0.033	0.035
A1	0.00	0.02	0.05	0.000	0.001	0.002
A3	0.203 REF.			0.008 REF.		
b	0.15	0.20	0.25	0.006	0.008	0.010
D	4.00 BSC.			0.157 BSC.		
D2	2.65	2.70	2.75	0.104	0.106	0.108
E	4.00 BSC.			0.157 BSC.		
E2	2.65	2.70	2.75	0.104	0.106	0.108
e	0.40 BSC.			0.016 BSC.		
K	0.20	-	-	0.008	-	-
L	0.25	0.30	0.35	0.010	0.012	0.014

(1) Values in inches are converted from mm and rounded to 3 decimal digits.

Device marking for QFN32

The following figure gives an example of topside marking orientation versus pin 1 identifier location.

Figure 46. QFN32 – 4 x 4 mm marking example (package top view)



6.6 Thermal characteristics

The maximum chip junction temperature ($T_J \text{ max}$) must never exceed the values given in [Table 9](#).

The maximum chip-junction temperature, $T_J \text{ max}$, in degrees Celsius, may be calculated using the following equation:

$$T_J \text{ max} = T_a \text{ max} + (P_d \text{ max} \times \Theta_{JA})$$

Where:

- $T_a \text{ max}$ is the maximum ambient temperature in °C,
- Θ_{JA} is the package junction-to-ambient thermal resistance, in °C/W,
- $P_d \text{ max}$ is the sum of $P_{INT} \text{ max}$ and $P_{I/O} \text{ max}$ ($P_d \text{ max} = P_{INT} \text{ max} + P_{I/O} \text{ max}$),
- $P_{INT} \text{ max}$ is the product of I_{DD} and V_{DD} , expressed in Watts. This is the maximum chip internal power.

$P_{I/O} \text{ max}$ represents the maximum power dissipation on output pins where:

$$P_{I/O} \text{ max} = \Sigma(V_{OL} \times I_{OL}) + \Sigma((V_{DD} - V_{OH}) \times I_{OH}),$$

taking into account the actual V_{OL} / I_{OL} and V_{OH} / I_{OH} of the I/Os at low and high level in the application.

Table 58. Package thermal characteristics

Symbol	Parameter	Value	Unit
Θ_{JA}	Thermal resistance junction-ambient LQFP64 – 10 × 10 mm / 0.5 mm pitch	75.3	°C/W
	Thermal resistance junction-ambient LQFP64 – 7 × 7 mm / 0.4 mm pitch	80.4	
	Thermal resistance junction-ambient LQFP48 – 7 × 7 mm / 0.5 mm pitch	76.8	
	Thermal resistance junction-ambient QFN48 – 6 × 6 mm / 0.4 mm pitch	38.8	
	Thermal resistance junction-ambient QFN32 – 4 × 4 mm / 0.4 mm pitch	59.7	

7 Part numbering

Table 59. AT32F415 ordering information scheme

Example:	AT32	F	4	1	5	R	C	T	7	-7
Product family										
AT32 = ARM-based 32-bit microcontroller										
Product type										
F = General-purpose										
Core										
4 = Cortex®-M4										
Product series										
1 = Value line										
Product application										
5 = USB OTG series										
Pin count										
K = 32 pins										
C = 48 pins										
R = 64 pins										
Internal Flash memory size										
8 = 64 KBytes of the internal Flash memory										
B = 128 KBytes of the internal Flash memory										
C = 256 KBytes of the internal Flash memory										
Package										
T = LQFP										
U = QFN										
Temperature range										
7 = -40 °C to +105 °C										
Package details										
-7 = LQFP64 - 7 x 7 mm										
-4 = QFN32 - 4 x 4 mm										
Blank = Other packages										

For a list of available options (speed, package, etc.) or for further information on any aspect of this device, please contact your nearest Artery sales office.

8 Revision history

Table 60. Document revision history

Date	Version	Change
2019.8.1	1.00	Initial release.
2019.10.11	1.01	1. Modified DMA2 as 7 channels 2. Modified USART/UART maximum communication rate 3. Added Table 35 4. Added AT32F415CCU7 and AT32F415CBU7
2020.3.10	1.02	1. Separated “wakeup time from low-power mode” as 5.3.8 2. Added thorough description of $T_{RSTTEMPO}$ in Table 11 3. Modified the note of Figure 2 of the maximum CPU frequency when USB clock is direct form HSI 48 MHz 4. Modified the maximum accessible speed of I/O control registers as f_{AHB} on the cover page 5. Added the note (8) of Table 5 to describe the usage limitation of PA9
2020.6.5	1.03	Corrected a typo of PA8 in the note (8) of Table 5 . It should be PA9

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